PrepFusion

Power Electronics

Detailed Solutions

GATE Electrical Engineering

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Previous Year Questions Solutions

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SOLUTIONS

I

Power Semiconductor Devices

Topics

1. Power Diode
2. Power Transistor
3. Static V-I Characteristics of Thyristor (SCR)
4. Static V-I Characteristics of MOSFET and IGBT
5. Firing and Gating Circuits for Thyristor, MOSFET and IGBT
6. Protection Schemes, Series and Parallel Operation of SCR
7. V-I Characteristics of GTO, TRIAC and Switch Modules

Q1.26.1 MCQ | 2026 | 1M | Ans: C

Method 1

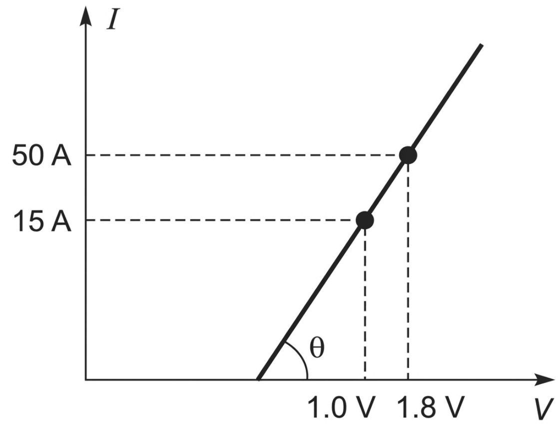

Fig. Solution Q1.26.1

Based on the given current-voltage (I–V) characteristic curve of the diode, the cut-in voltage can be obtained from the straight-line characteristic.

$$\tan\theta = \frac{50}{1.8-1} = \frac{15}{x-1}$$

$$50(x-1) = 15\times 0.8$$

$$50x - 50 = 12$$

$$x = \frac{62}{50} = 1.24\,\text{V}$$

For a diode current of

$$I = 15\,\text{A},$$

the voltage drop is

$$V_{\text{drop}} = 1.24\,\text{V}.$$

Hence, the power loss is

$$P_{\text{loss}} = V_{\text{drop}} I$$

$$P_{\text{loss}} = 1.24\times 15 = 18.6\,\text{W}.$$

Method 2

Using the standard piecewise conduction-loss formula,

$$P_{\text{loss}} = V_D I_{\text{Davg}} + I_{\text{Drms}}^2 R_{\text{on}}.$$

For constant current,

$$I_{\text{Davg}} = I_{\text{Drms}} = 15\,\text{A}.$$

Also,

$$R_{\text{on}} = \frac{0.8}{50} = 0.016\,\Omega.$$

Therefore,

$$P_{\text{loss}} = 1(15) + (15)^2(0.016) = 15 + 3.6 = 18.6\,\text{W}.$$

$\boxed{C}$

Key Points

- The piecewise linear diode model uses a threshold voltage and an ON-state resistance.
- For constant current, $I_{\text{avg}} = I_{\text{rms}} = I$.
- Semiconductor conduction loss is evaluated using the voltage drop and current.

Mistakes to be avoided

- Do not use only VI if the ON-state resistance is separately considered.
- Do not forget that the current is continuous DC in this question.

Q1.26.2 MCQ 2026 1M Ans: B ● ○ ○

From $t = 0$ to $t = t_1$, during the turn-ON condition of the MOSFET, current flows through the inductor, and the inductor charges. After t_1, during the MOSFET turn-OFF process, the inductor tries to discharge through the diode (D) and the resistor (R_f).

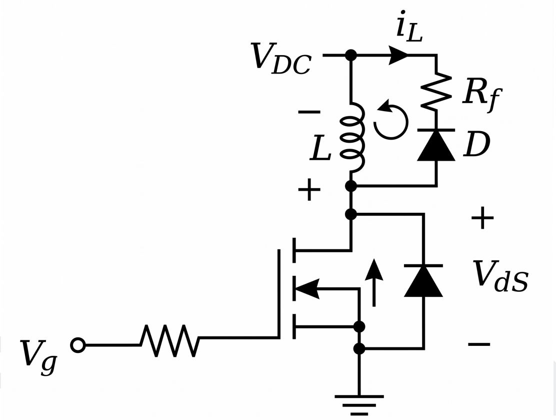

Fig. Solution Q1.26.2

Assuming the diode is ideal, the voltage across the inductor can be written as

$$V_L = i_L R_f$$

where the discharging current is given by

$$i_L = I_o e^{-t/\tau}$$

Using KVL, the voltage across the MOSFET during the turn-OFF condition (V_{ds}) can be calculated as

$$V_{ds} = V_{DC} + V_L$$

$$V_{ds} = V_{DC} + i_L R_f$$

As the resistance R_f increases, the peak voltage across the MOSFET V_{ds} increases. Based on the given relationship and circuit behavior,

$\boxed{B}$

Key Points

- During turn-off of a switch in an inductive circuit, the stored energy in the inductor causes a voltage spike ($L\frac{di}{dt}$).
- The peak voltage across the switch is the sum of the supply voltage and the transient voltage across the discharge resistor.
- The time constant $\tau = \frac{L}{R}$ dictates the rate of discharge; a higher resistance leads to a faster discharge but a higher peak voltage.

Mistakes to be avoided

- Neglecting the polarity of V_L during discharging; it adds to V_{DC} across the switch.
- Forgetting that V_{ds} reaches its maximum value at the instant of turn-off when i_L is maximum.

Q1.24.1 MCQ 2024 1M Ans: D

Among the given power semiconductor devices:

- SCR has the slowest switching speed due to latching and turn-off limitations.
- GTO is faster than SCR but requires a large negative gate current for turn-off.
- IGBT has moderate switching speed due to minority carrier conduction.
- Power MOSFET is a majority carrier device and has the highest switching speed.

Hence, the fastest switching device among those with similar power ratings is the **Power MOSFET**.

$$\boxed{D}$$

Q1.23.1 MSQ 2023 1M Ans: A,D

From the required switch characteristics, the switch must block voltage of only one polarity during the OFF state and conduct current in both directions during the ON state.

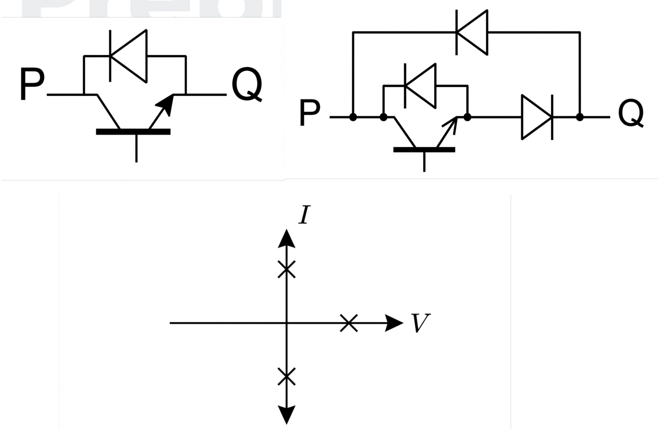

Fig. Solution Q1.23.1

The switch configurations in options (A) and (D) satisfy bidirectional current conduction with one-polarity voltage blocking.

$$\boxed{A, D}$$

Key Points

- Bidirectional current conduction requires current paths in both directions.
- One-polarity voltage blocking means only one OFF-state voltage polarity needs to be blocked.

Mistakes to be avoided

- Do not confuse bidirectional current conduction with bidirectional voltage blocking.

Q1.23.2 NAT 2023 1M Ans: 7.33

Initially, the capacitor is charged to the supply voltage:

$$V_C = V_s = 100\,\text{V}.$$

When thyristor T is triggered, the capacitor discharges through the resonant LC loop. The resonant conduction time for one half-cycle is

$$t_1 = \pi\sqrt{LC}.$$

The peak resonant current is

$$I_p = V_s\sqrt{\frac{C}{L}} = 100\sqrt{\frac{1\,\mu\text{F}}{4\,\mu\text{H}}} = 50\,\text{A}.$$

The load current is

$$I_0 = \frac{V_s}{R} = \frac{100}{4} = 25\,\text{A}.$$

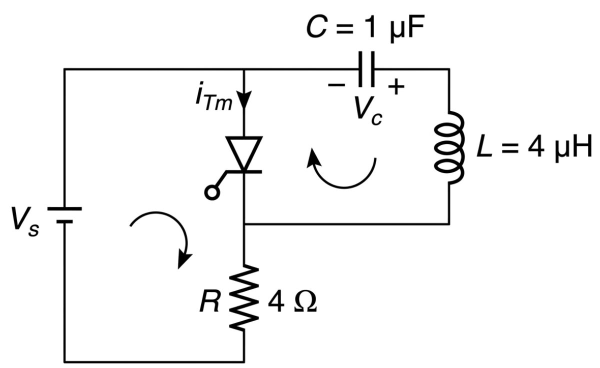

Fig. Solution Q1.23.2

The thyristor turns OFF when the total current becomes zero:

$$0 = \frac{V_s}{R} - I_p\sin(\omega_0 t_2).$$

$$\omega_0 t_2 = \sin^{-1}\left(\frac{I_0}{I_p}\right) = \sin^{-1}\left(\frac{25}{50}\right) = \frac{\pi}{6}.$$

Thus,

$$t_2 = \frac{\pi}{6}\sqrt{LC}.$$

Total thyristor conduction time is

$$t_{\text{total}} = t_1 + t_2 = \pi\sqrt{LC} + \frac{\pi}{6}\sqrt{LC} = \frac{7\pi}{6}\sqrt{LC}.$$

Substituting the given values,

$$t_{\text{total}} = \frac{7\pi}{6}\sqrt{(1\times10^{-6})(4\times10^{-6})} = \frac{7\pi}{3}\times10^{-6}\text{ s.}$$

$$t_{\text{total}} \approx 7.33\,\mu\text{s.}$$

$$\boxed{7.33}$$

Key Points

- Resonant discharge through an LC circuit gives a half-cycle conduction time of $\pi\sqrt{LC}$.
- Total thyristor conduction time is obtained by adding the resonant interval and current decay interval.

Mistakes to be avoided

- Do not stop the calculation at only $\pi\sqrt{LC}$ when load current is also present.
- Keep the final NAT answer numerical only.

Q1.22.1 MCQ 2022 1M Ans: C ● ○ ○

Among IGBT, thyristor, MOSFET and BJT, the MOSFET is best suited for high switching frequency operation such as 200 kHz because it is a majority carrier device.

$$\boxed{C}$$

Key Points

- MOSFETs are preferred for high-frequency low-to-medium power converters.
- Thyristors and IGBTs are relatively slower due to minority carrier effects.

Mistakes to be avoided

- Do not select thyristor for high-frequency laptop charger applications.

Q1.22.2 MCQ 2022 1M Ans: D ● ○ ○

For an ideal MOSFET operating in saturation, the gate current is zero. Hence, the small-signal current gain is:

$$A_i = \left|\frac{i_o}{i_i}\right| = \left|\frac{i_D}{i_G}\right|$$

Since $i_G = 0$,

$$A_i = \infty$$

Thus, the magnitude of the small-signal current gain of a common-drain (source follower) amplifier is

$$\boxed{D}$$

Key Points

- An ideal MOSFET has infinite input resistance.
- Gate current is zero under small-signal operation.
- Current gain is theoretically infinite because the input current is zero.

Q1.20.1 NAT 2020 1M Ans: 3 ● ○ ○

The IGBT current is

$$I_{\text{IGBT}} = -I_D + I_{\text{load}}.$$

At point 3, the diode current reaches its peak reverse recovery value:

$$I_D = -I_{D,\text{peak(rr)}}.$$

Therefore,

$$I_{\text{IGBT}} = -\left(-I_{D,\text{peak(rr)}}\right) + I_{\text{load}} = I_{D,\text{peak(rr)}} + I_{\text{load}}.$$

Hence, the IGBT experiences the highest current stress at point 3.

$$\boxed{3}$$

Key Points

- Reverse recovery current adds to the load current in the IGBT.
- Maximum IGBT current occurs at peak reverse recovery current.

Mistakes to be avoided

- Do not consider diode current magnitude alone; include load current also.

Q1.18.1 MCQ 2018 1M Ans: B ● ○ ○

Only TRIAC and MOSFET can carry current in the shown direction when gated appropriately.

$$\boxed{B}$$

Key Points

- TRIAC can conduct bidirectionally when triggered.
- MOSFET can conduct in both directions due to channel conduction and body diode behavior.

Mistakes to be avoided

- Do not assume a thyristor can conduct reverse current.

Q1.17.1 MCQ 2017 1M Ans: D ● ○ ○

MOSFET is a majority carrier device. IGBT, diode and thyristor involve minority carrier conduction.

$$\boxed{D}$$

Key Points

- MOSFET is a unipolar majority carrier device.
- IGBT, diode and thyristor are minority carrier devices.

Mistakes to be avoided

- Do not classify IGBT as a purely majority carrier device.

Q1.16.1 NAT 2016 1M Ans: 170 ● ○ ○

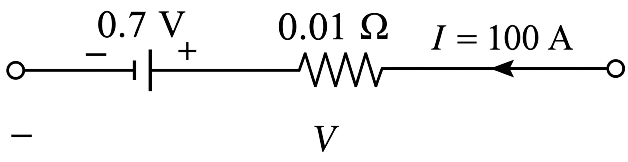

Fig. Solution Q1.16.1

No current flows through the IGBT; hence, the current flows only through the diode. The total forward voltage drop across the conducting diode is

$$V = IR + 0.7.$$

Given,

$$I = 100\,\text{A}, \qquad R = 0.01\,\Omega.$$

Therefore,

$$V = (100)(0.01) + 0.7 = 1.7\,\text{V}.$$

The conduction power loss is

$$P = VI = 1.7 \times 100 = 170\,\text{W}.$$

$$\boxed{170}$$

Key Points

- Practical diode model uses a cut-in voltage and forward resistance.
- Conduction loss is $P = VI$.

Mistakes to be avoided

- Do not include IGBT loss if the current path is only through the diode.

Q1.16.2 NAT 2016 1M Ans: 75 ● ○ ○

For linear switching transitions, energy loss is the triangular area under the v–i curve.

$$E = \frac{1}{2}VIT_1 + \frac{1}{2}VIT_2.$$

Substituting the given values,

$$E = 600\left(\frac{150}{2} \times 10^{-6}\right) + 100\left(\frac{600}{2} \times 10^{-6}\right).$$

$$E = 0.045 + 0.030 = 0.075\,\text{J}.$$

$$E = 75\,\text{mJ}.$$

$$\boxed{75}$$

Key Points

- For linear switching transitions, $E = \frac{1}{2}VIt$.
- Total switching energy is the sum of the two triangular areas.

Mistakes to be avoided

- Convert joules to millijoules correctly.

Q1.14.1 MCQ 2014 1M Ans: C ● ○ ○

For bidirectional voltage blocking in the OFF state, the switch arrangement must block voltage for both polarities across terminals a and b.

Circuit (ii) consists of a controlled switch in series with a diode. The controlled device blocks one polarity and the series diode blocks the opposite polarity.

Circuit (iii) also provides blocking capability for both polarities in the OFF state.

Therefore, the switches that can block voltages of either polarity are

$$\boxed{C}$$

Key Points

- Series diode helps in reverse voltage blocking.
- Bidirectional voltage blocking is different from bidirectional current conduction.

Mistakes to be avoided

- Do not select circuits that contain an anti-parallel diode across the switch when reverse voltage blocking is needed.

Q1.14.2 NAT 2014 2M Ans: 6060 ● ● ○

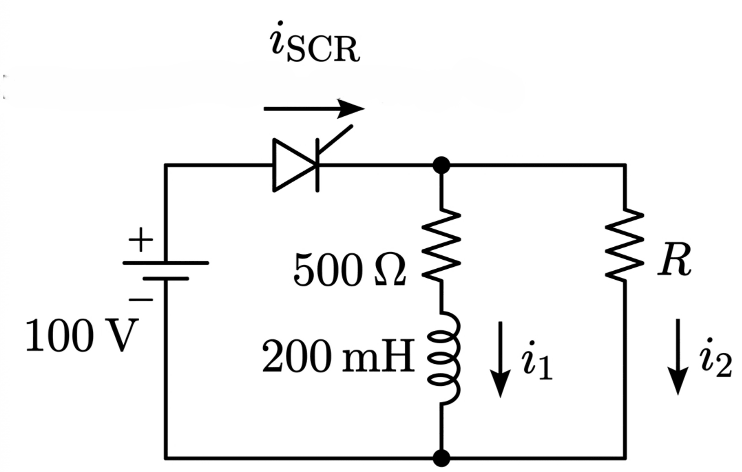

Fig. Solution Q1.14.2

Initially,

$$i_1(0^+) = 0, \qquad i_1(\infty) = \frac{100}{500} = 0.2\,\text{A}.$$

The time constant of the RL branch is

$$\tau = \frac{L}{R} = \frac{200\,\text{mH}}{500} = 0.4\,\text{ms}.$$

The current through the RL branch is

$$i_1(t) = 0.2\left(1 - e^{-t/\tau}\right).$$

Hence, the SCR current is

$$i_{\text{SCR}} = 0.2\left(1 - e^{-t/\tau}\right) + \frac{100}{R}.$$

At

$$t = 50\,\mu\text{s}, \qquad i_{\text{SCR}} = 40\,\text{mA}.$$

Therefore,

$$40 \times 10^{-3} = 0.2\left(1 - e^{-\frac{50\times10^{-6}}{0.4\times10^{-3}}}\right) + \frac{100}{R}.$$

$$R = \frac{100}{0.0165} = 6060.6\,\Omega.$$

$$\boxed{6060}$$

Key Points

- SCR latches only when anode current reaches the latching current before the gate pulse is removed.
- Current rise in an RL circuit is exponential.

Mistakes to be avoided

- Use the current at the end of the gate pulse, not the steady-state current.

Q1.14.3 MCQ 2014 2M Ans: C ● ○ ○

When thyristor T is triggered at $t = 0$, it forms a series resonant LC circuit.

The resonant angular frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

Given,

$$L = \left(\frac{100}{\pi}\right)\mu\text{H}, \qquad C = \left(\frac{100}{\pi}\right)\mu\text{F}.$$

Hence,

$$LC = \left(\frac{100}{\pi} \times 10^{-6}\right)^2 = \frac{10^{-8}}{\pi^2}.$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 10^4\pi\ \text{rad/s}.$$

The thyristor conducts until current becomes zero again:

$$\omega_0 t_c = \pi.$$

$$10^4 \pi t_c = \pi.$$

$$t_c = 10^{-4}\,\text{s} = 100\,\mu\text{s}.$$

$\boxed{C}$

Key Points

- For series LC commutation, $t_c = \pi\sqrt{LC}$.
- The thyristor remains ON until the current naturally falls to zero.

Mistakes to be avoided

- Do not confuse gate pulse width with conduction time.

Q1.12.1 MCQ 2012 1M Ans: B ● ○ ○

The typical ratio of latching current I_L to holding current I_H for a thyristor is approximately 2.

$\boxed{B}$

Key Points

- Latching current is greater than holding current.
- A typical value of I_L/I_H is nearly 2.

Mistakes to be avoided

- Do not take $I_L = I_H$.

Q1.11.1 MCQ 2011 1M Ans: C ● ○ ○

Circuit turn-OFF time t_c is the time duration provided by the external commutation circuit to keep the SCR reverse biased after its current becomes zero.

For successful turn-OFF,

$$t_c > t_q,$$

where t_q is the device turn-OFF time.

Thus, the correct option is

$\boxed{C}$

Key Points

- Circuit turn-OFF time is decided by the external circuit.
- Device turn-OFF time is a property of the SCR.
- Successful commutation requires $t_c > t_q$.

Mistakes to be avoided

- Do not confuse circuit turn-OFF time with the time for current to merely become zero.

Q1.10.1

MCQ | 2010 | 1M | Ans: C

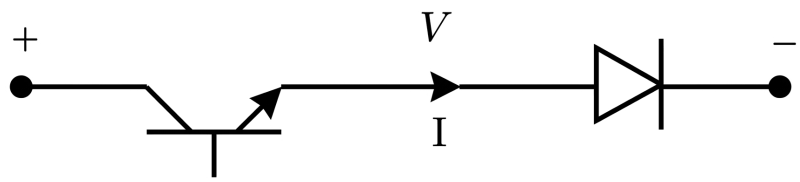

Fig. Solution Q1.10.1

The composite switch shown blocks voltages of both polarities and allows unidirectional current in the direction shown.

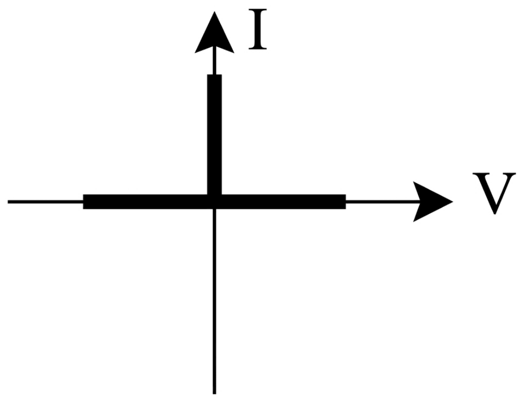

Fig. Solution Q1.10.1

Hence, the correct option is

$$\boxed{C}$$

Key Points

- Series diode with a controlled switch can modify voltage blocking capability.
- Composite switch characteristics are obtained by combining individual device characteristics.

Mistakes to be avoided

- Do not ignore the direction of the series diode.

Q1.10.2

MCQ | 2010 | 2M | Ans: D

Method 1

Initially,

$$V_C(0^-) = 100\,\text{V}, \qquad i_L(0^-) = 0.$$

Since capacitor voltage and inductor current cannot change instantaneously,

$$V_C(0^+) = 100\,\text{V}, \qquad i_L(0^+) = 0.$$

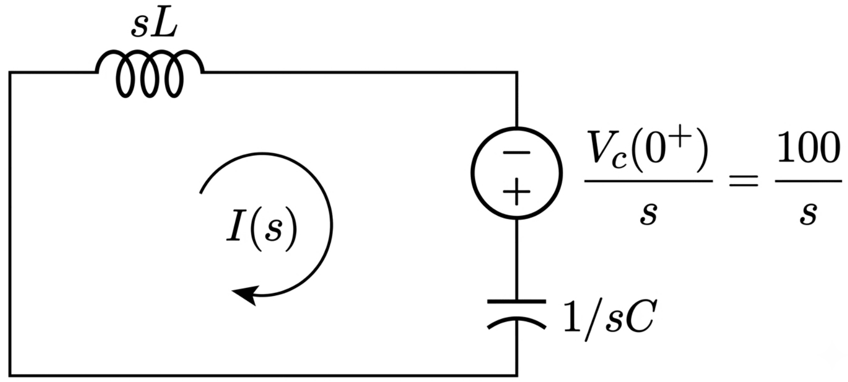

Fig. Solution Q1.10.2

The s-domain loop current is

$$I(s) = \frac{100/s}{sL + \frac{1}{sC}} = \frac{100}{L}\frac{1}{s^2 + \frac{1}{LC}}.$$

Taking inverse Laplace transform,

$$i(t) = 100\sqrt{\frac{C}{L}}\sin\left(\frac{t}{\sqrt{LC}}\right).$$

Substituting $L = 1\,\text{mH}$ and $C = 10\,\mu\text{F}$,

$$\omega_0 = \frac{1}{\sqrt{LC}} = 10^4\ \text{rad/s}.$$

Therefore,

$$i(t) = 10\sin(10^4 t)\ \text{A}.$$

Method 2

For a source-free LC oscillator with initial capacitor voltage $V_C(0^+)$,

$$i(t) = V_C(0^+)\sqrt{\frac{C}{L}}\sin(\omega_0 t).$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 10^4\ \text{rad/s}.$$

Therefore,

$$i(t) = 100\sqrt{\frac{10\times10^{-6}}{10^{-3}}}\sin(10^4 t) = 10\sin(10^4 t)\ \text{A}.$$

$\boxed{D}$

Key Points

- Inductor current and capacitor voltage cannot change instantaneously.
- Natural frequency of an LC circuit is $\omega_0 = 1/\sqrt{LC}$.

Mistakes to be avoided

- Do not forget to include initial capacitor voltage.

Q1.10.3 MCQ 2010 2M Ans: A ● ● ○

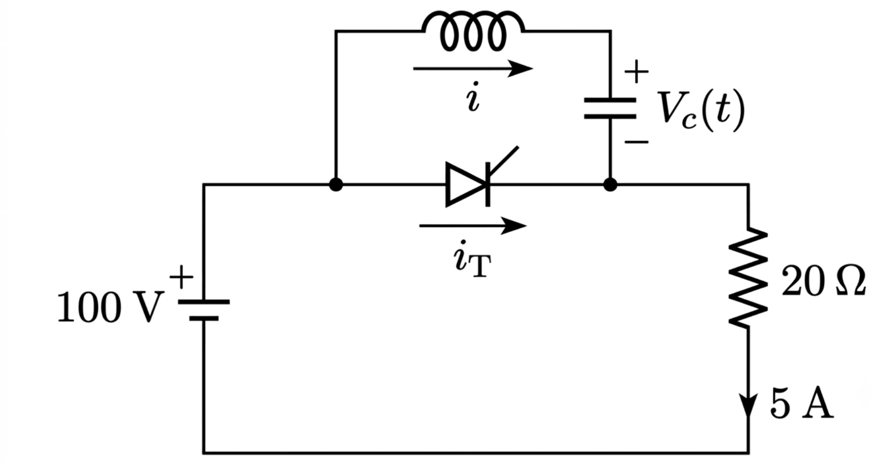

Fig. Solution Q1.10.3

When the switch is closed, oscillatory current flows through the LC commutation branch.
Given load current,

$$I_0 = 5\,\text{A}.$$

The net thyristor current is

$$i_T = I_0 - i.$$

Using the resonant current,

$$i = 10\sin(10^4 t),$$

we get

$$i_T = 5 - 10\sin(10^4 t).$$

For turn-OFF,

$$i_T = 0.$$

$$5 - 10\sin(10^4 T) = 0.$$

$$\sin(10^4 T) = 0.5.$$

$$10^4 T = \frac{\pi}{6}.$$

$$T = \frac{\pi}{6 \times 10^4} = 52.3\,\mu\text{s}.$$

$$\boxed{A}$$

Key Points

- Forced commutation occurs when reverse resonant current equals load current.
- Net thyristor current becomes zero at turn-OFF.

Mistakes to be avoided

- Use the first instant at which $i_T = 0$.

Q1.09.1 MCQ 2009 2M Ans: C

The device characteristics are matched as follows.

Device A behaves as an ideal diode:

$$V_s = 0, \quad i_s > 0$$

in ON state and

$$V_s < 0, \quad i_s = 0$$

in reverse blocking state.

Device B behaves as a thyristor with forward and reverse voltage blocking capability.

Device C behaves as an ideal transistor with unidirectional current conduction and forward voltage blocking.

Device D behaves as a transistor with anti-parallel diode, giving bidirectional current capability with forward voltage blocking.

Therefore,

$$\boxed{C}$$

Key Points

- Diode: unidirectional conduction and reverse voltage blocking.
- Thyristor: unidirectional conduction and bidirectional voltage blocking.
- Transistor with anti-parallel diode: bidirectional current conduction.

Mistakes to be avoided

- Do not confuse voltage blocking direction with current conduction direction.

Q1.07.1 MCQ 2007 2M Ans: C

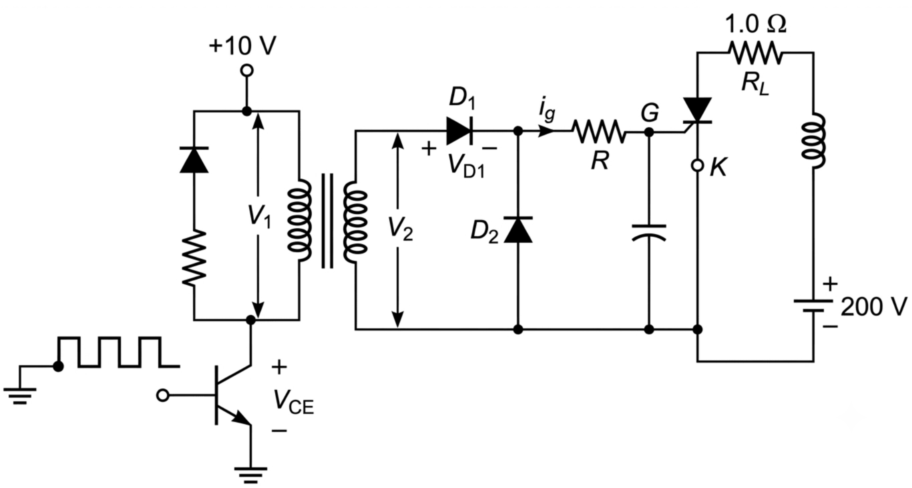

Fig. Solution Q1.07.1

When the transistor is ON, its collector-emitter voltage drop is 1 V. Hence, the voltage across the primary winding is

$$V_1 = 10 - 1 = 9\,\text{V}.$$

Since the pulse transformer has a turns ratio of 1 : 1,

$$V_2 = V_1 = 9\,\text{V}.$$

During the pulse, D_1 is forward biased and drops 1 V, and the SCR gate-cathode junction also drops 1 V. Applying KVL in the gate circuit,

$$V_R = V_2 - V_{D1} - V_{gk} = 9 - 1 - 1 = 7\,\text{V}.$$

For maximum gate current,

$$I_g = 150\,\text{mA} = 0.15\,\text{A}.$$

$$R = \frac{V_R}{I_g} = \frac{7}{0.15} = 46.67\,\Omega.$$

$\boxed{C}$

Key Points

- Pulse transformer secondary voltage equals primary voltage for a 1 : 1 transformer.
- Gate resistor is selected from allowable gate current.

Mistakes to be avoided

- Include diode and gate-cathode voltage drops in KVL.

Q1.07.2 MCQ 2007 2M Ans: A ●●○

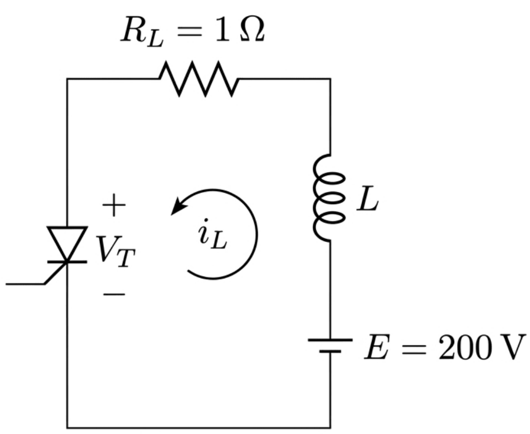

Fig. Solution Q1.07.2

Applying KVL,

$$E - L\frac{di_a}{dt} - R_L i_a - V_T = 0.$$

Substituting,

$$200 - 0.15\frac{di_a}{dt} - i_a - 1 = 0.$$

$$0.15\frac{di_a}{dt} + i_a = 199.$$

With $i_a(0) = 0$,

$$i_a(t) = 199\left(1 - e^{-t/0.15}\right).$$

The gate pulse must be present until anode current reaches the latching current,

$$I_L = 250\,\text{mA} = 0.25\,\text{A}.$$

$$0.25 = 199\left(1 - e^{-T/0.15}\right).$$

$$T = -0.15\ln\left(1 - \frac{0.25}{199}\right) \approx 188.56\,\mu\text{s}.$$

Volt-second rating is

$$VT = 10 \times 188.56\,\mu\text{s} = 1885.6\,\mu\text{V}\cdot\text{s} \approx 2000\,\mu\text{V}\cdot\text{s}.$$

$$\boxed{A}$$

Key Points

- Gate pulse must persist until SCR current reaches latching current.
- Volt-second rating is voltage multiplied by pulse width.

Mistakes to be avoided

- Do not confuse latching current with gate current.

Q1.07.3 MCQ 2007 2M Ans: C ● ● ○

The given commutation circuit is a Class-B commutation circuit, also called a current commutation circuit.
Th_M will be turned OFF after Mode-1, i.e.,

$$T_1 = \pi\sqrt{LC}$$

Given,

$$L = 25.28\,\mu\text{H}, \qquad C = 10\,\mu\text{F}$$

$$T_1 = \pi\sqrt{10 \times 10^{-6} \times 25.28 \times 10^{-6}}$$

$$T_1 = 50\,\mu\text{s}$$

Hence, Th_M is turned OFF between

$$50\,\mu\text{s} < t \leq 75\,\mu\text{s}$$

$$\boxed{C}$$

Key Points

- Class-B commutation is also called resonant pulse commutation.
- In LC commutation, the first mode duration is $\pi\sqrt{LC}$.
- The main SCR turns OFF after the commutating current reverses the current through it.

Q1.06.1 MCQ 2006 2M Ans: B

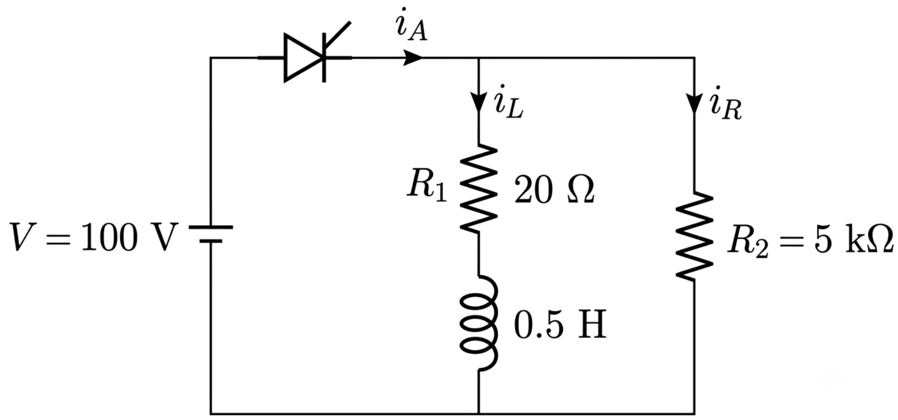

Fig. Solution Q1.06.1

The current through the 5 kΩ branch is

$$i_R = \frac{100}{5 \times 10^3} = 0.02\text{ A}.$$

The current through the R_1–L branch is

$$i_L(t) = \frac{100}{20}\left(1 - e^{-\frac{20}{0.5}t}\right) = 5\left(1 - e^{-40t}\right)\text{ A}.$$

Hence, the SCR anode current is

$$i_a(t) = 0.02 + 5\left(1 - e^{-40t}\right).$$

For successful turn-ON,

$$i_a(T) = I_L = 50\text{ mA} = 0.05\text{ A}.$$

$$0.02 + 5\left(1 - e^{-40T}\right) = 0.05.$$

$$e^{-40T} = 0.994.$$

$$T = -\frac{\ln(0.994)}{40} \approx 150\,\mu\text{s}.$$

$$\boxed{B}$$

Key Points

- SCR gate pulse width must be long enough for anode current to reach latching current.
- RL current rises exponentially.

Mistakes to be avoided

- Do not use holding current for turn-ON latching calculation.

Q1.06.2 MCQ 2006 2M Ans: A ● ○ ○

The circuit turn-OFF time is obtained by multiplying the device turn-OFF time by the safety factor.

$$T_c = RC \ln 2 = (50\,\mu\text{s}) \times 2$$

$$50 \times C \times \ln 2 = 100\,\mu\text{s}$$

$$C = \frac{2}{\ln 2}\,\mu\text{F} = 2.88\,\mu\text{F}$$

$$\boxed{A}$$

Key Points

- Circuit turn-off time must be greater than device turn-off time.
- A safety factor is used to ensure reliable commutation.
- For RC turn-off circuit, $T_c = RC \ln 2$.

Mistakes to be avoided

- Do not use device turn-off time directly without safety factor.
- Do not forget the $\ln 2$ factor in the RC turn-off expression.

Q1.05.1 MCQ 2005 1M Ans: D ● ○ ○

For the given n-channel MOSFET,

$$V_{GS} = 2\,\text{V}, \qquad V_{TH} = 1\,\text{V}.$$

Since

$$V_{GS} > V_{TH},$$

the MOSFET is ON.
Also,

$$V_{DS(\text{sat})} = V_{GS} - V_{TH} = 1\,\text{V}.$$

As the MOSFET behaves as an ideal closed switch,

$$V_{ab} = 0.$$

$$\boxed{D}$$

Key Points

- Enhancement MOSFET turns ON when $V_{GS} > V_{TH}$.
- An ideal ON switch has zero voltage drop.

Mistakes to be avoided

- Do not take threshold voltage as the output voltage.

Q1.05.2

MCQ | 2005 | 1M | Ans: A | ● ○ ○

For a power MOSFET,

$$P_{\text{cond}} = I^2 R_{\text{DS(on)}}.$$

Therefore, conduction loss varies as the square of the current. Hence, the conduction loss versus current characteristic is a parabola.

$$\boxed{(A)}$$

Key Points

- MOSFET ON-state loss is $I^2 R_{\text{DS(on)}}$.
- A square-law relation gives a parabolic characteristic.

Mistakes to be avoided

- Do not use a constant voltage-drop model for MOSFET conduction loss.

Q1.05.3

MCQ | 2005 | 2M | Ans: A | ● ● ○

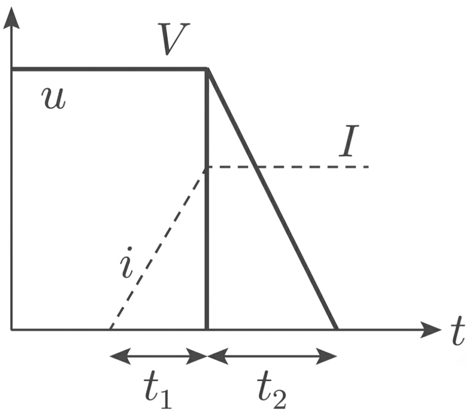

Fig. Solution Q1.05.3

During t_1, the voltage across the device is constant at V, while current rises linearly from 0 to I.

$$E_1 = V \int i\,dt = V\left(\frac{1}{2} I t_1\right) = \frac{1}{2} V I t_1.$$

During t_2, current is constant at I, while voltage falls linearly from V to 0.

$$E_2 = I \int v\,dt = I\left(\frac{1}{2} V t_2\right) = \frac{1}{2} V I t_2.$$

Hence,

$$E = E_1 + E_2 = \frac{1}{2} V I (t_1 + t_2).$$

The waveform corresponds to turn-ON transition.

$$\boxed{A}$$

Key Points

- Switching energy is the area under the instantaneous power curve.
- Linear transitions give triangular energy areas.

Mistakes to be avoided

- Do not omit the factor $\frac{1}{2}$ for linear rise/fall.

Q1.05.4 MCQ 2005 2M Ans: C

The required electronic switch must block voltage of either polarity in its OFF state and conduct current in only one direction in its ON state.

Circuit P satisfies this requirement because the thyristor can block both forward and reverse voltages in the OFF state and conduct only forward current in the ON state.

Circuit R also satisfies the requirement because the transistor blocks one polarity and the series diode blocks the opposite polarity, while conduction is allowed only in one direction.

Therefore, the valid realizations are

$$\boxed{C}$$

Key Points

- Bidirectional voltage blocking requires OFF-state blocking for both polarities.
- Unidirectional current conduction allows only one current direction in ON state.

Mistakes to be avoided

- Anti-parallel diodes prevent reverse voltage blocking.

Q1.04.1 MCQ 2004 1M Ans: D

The triggering voltage varies as

$$V_b = 12 \pm 4\,\text{V}$$

Hence,

$$(V_b)_{\text{min}} = 12 - 4 = 8\,\text{V}$$

$$(V_b)_{\text{max}} = 12 + 4 = 16\,\text{V}$$

The thyristor must turn ON even at the minimum triggering voltage. Therefore, the gate resistor is designed using the minimum value of V_b.

Given,

$$I_g = 10\,\text{mA}$$

Neglecting the gate-to-source voltage drop,

$$R = \frac{(V_b)_{\text{min}}}{I_g} = \frac{8}{10 \times 10^{-3}} = 800\,\Omega$$

$$\boxed{D}$$

Key Points

- The gate resistor is designed using the minimum triggering voltage to ensure reliable firing under worst-case conditions.
- Gate resistance is calculated as $R = \frac{V_g}{I_g}$ after neglecting the gate-cathode voltage drop (if specified).

Q1.04.2

MCQ | 2004 | 2M | Ans: C

The average power loss in any switch modeled as a resistance is

$$P_{avg} = I_{rms}^2 R$$

Given,

$$I_{rms} = \frac{10}{\sqrt{2}} \text{ A}, \qquad R = 0.15\,\Omega$$

Therefore,

$$P_{avg} = \left(\frac{10}{\sqrt{2}}\right)^2 \times 0.15$$

$$P_{avg} = \frac{100 \times 0.15}{2} = 7.5\,\text{W}$$

$$\boxed{C}$$

Key Points

- Conduction loss in a resistive switch is calculated using RMS current.
- Use $P_{avg} = I_{rms}^2 R$.

Q1.03.1

MCQ | 2003 | 1M | Ans: A

For a depletion-mode MOSFET, drain current exists even when the gate-to-source voltage is zero.
From the given characteristic,

$$I_D \neq 0 \quad \text{at} \quad V_{GS} = 0$$

Hence, the MOSFET is a depletion-mode device.
Also, the characteristic corresponds to an n-channel MOSFET.

$$\boxed{A}$$

Key Points

- In a depletion-mode MOSFET, current flows even at $V_{GS} = 0$.
- In an enhancement-mode MOSFET, current is zero at $V_{GS} = 0$.
- The given characteristic represents an n-channel depletion-mode MOSFET.

Mistakes to be avoided

- Do not select enhancement-mode device when I_D exists at $V_{GS} = 0$.
- Do not confuse depletion-mode operation with enhancement-mode operation.
- Always check the drain current at $V_{GS} = 0$ first.

Q1.03.2 MCQ 2003 1M Ans: B

The anti-parallel diode conducts in the reverse direction.
When reverse current flows through diode D, the diode behaves like a short circuit.

$$I_S < 0 \quad \text{and} \quad V_{DS} = 0$$

When the MOSFET is in the ON state,

$$I_S > 0 \quad \text{and} \quad V_{DS} = 0$$

When the MOSFET is in the OFF state,

$$I_S = 0 \quad \text{and} \quad V_{DS} > 0$$

$$\boxed{(B)}$$

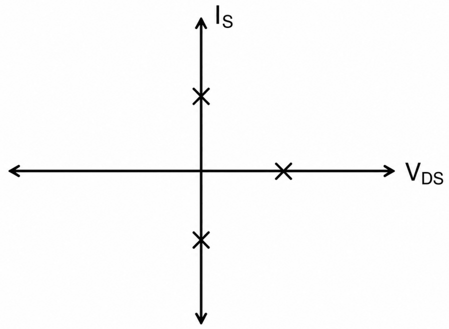

Fig. Solution Q1.03.2

Q1.03.3 MCQ 2003 1M Ans: D

In forward blocking mode, J_1 and J_3 are forward biased and J_2 is reverse biased.
In reverse blocking mode, J_1 and J_3 are reverse biased while J_2 is forward biased.
In forward conduction mode, all three junctions are forward biased.

$$\boxed{D}$$

Q1.01.1 MCQ 2001 2M Ans: B

For a JFET,

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right)$$

Given,

$$V_P = -5\,\text{V}, \qquad V_{GS} = -3\,\text{V}, \qquad g_m = 1\,\text{mA/V}$$

Therefore,

$$1 = g_{m0}\left(1 - \frac{-3}{-5}\right)$$

$$1 = g_{m0}(1 - 0.6) = 0.4g_{m0}$$

$$g_{m0} = \frac{1}{0.4} = 2.5\,\text{mA/V}$$

$\boxed{B}$

Key Points

- Maximum transconductance of JFET is g_{m0}.
- For JFET, $g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right)$.
- For an n-channel JFET, both V_{GS} and V_P are negative.

Q1.01.2 MCQ 2001 1M Ans: B ● ○ ○

A pulse transformer is connected between the triggering circuit and the thyristor gate primarily to provide galvanic (electrical) isolation.

It isolates the low-voltage control circuit from the high-voltage power circuit while transferring the gate pulse safely.

Therefore, the primary purpose of the pulse transformer is electrical isolation.

$\boxed{(B)}$

Q1.98.1 MCQ 1998 1M Ans: C ● ○ ○

An uncontrolled switch is a device whose turn-ON and turn-OFF depend only on the external circuit conditions and not on any gate or control signal.

Among the given devices,

- Thyristor (SCR) is a semi-controlled device.
- Bipolar Junction Transistor (BJT) is a controlled device.
- MOSFET is a fully controlled device.
- A diode conducts automatically when forward biased and blocks when reverse biased, requiring no control signal.

Hence, the uncontrolled electronic switch employed in power electronic converters is the diode.

$\boxed{C}$

Q1.98.2 MCQ 1998 1M Ans: A ● ○ ○

A MOSFET in its ON state can be modeled as a drain-to-source resistance because of the power loss in the switch.

$$P_{\text{loss}} = \left(I_{D,\text{rms}}\right)^2 R_{DS}$$

Hence, in the ON state, the MOSFET behaves as a resistor.

$\boxed{A}$

Q1.98.3 MCQ 1998 1M Ans: B ● ○ ○

The circuit turn-off time must always be greater than the time required by the device to turn off. Otherwise, stored charges remain in the device and may cause false triggering.

This phenomenon is known as **commutation failure**.

$$t_c > t_q$$

where,

$$t_c = \text{Circuit turn-off time}, \qquad t_q = \text{Device turn-off time}$$

$$\boxed{B}$$

Key Points

- Circuit turn-off time must always be greater than the thyristor turn-off time.
- Stored charge must be completely removed before forward voltage is reapplied.
- If $t_c < t_q$, the SCR may turn ON unintentionally, resulting in commutation failure.

Mistakes to be avoided

- Do not confuse circuit turn-off time (t_c) with device turn-off time (t_q).
- Do not assume the SCR regains forward blocking capability immediately after current becomes zero.
- Remember the condition $t_c > t_q$ for reliable commutation.

Q1.97.1 MCQ 1997 2M Ans: 0.384,221.46 ● ● ●

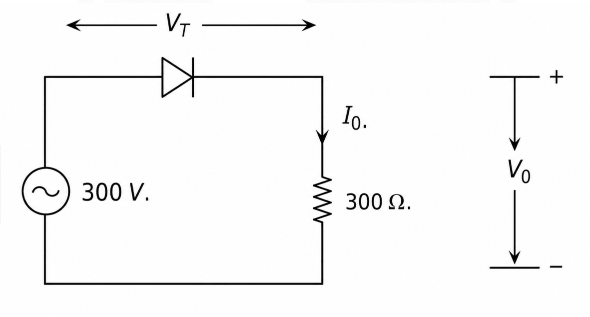

Fig. Solution Q1.97.1

Waveforms:

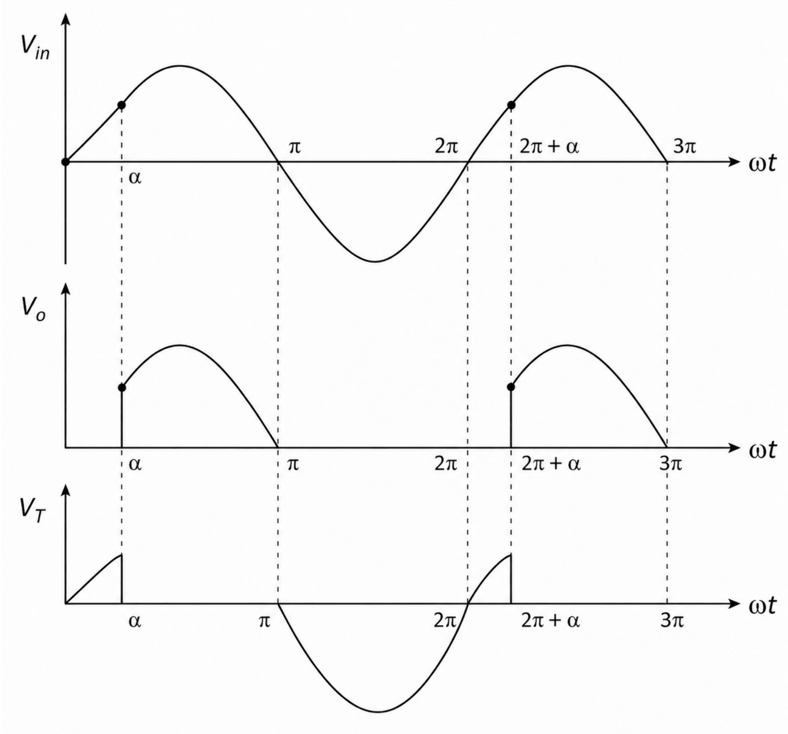

Fig. Solution Q1.97.1

(a) Reading of a moving coil ammeter

A moving coil ammeter measures the **average value** of the current.

For a half-wave rectifier circuit with an SCR and a resistive load,

$$V_{\text{avg}} = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

At

$$\alpha = 45^\circ,$$

$$V_{\text{avg}} = \frac{300\sqrt{2}}{2\pi}(1 + \cos 45^\circ) = 115.32\text{ V}$$

Hence,

$$I_{\text{avg}} = \frac{V_{\text{avg}}}{R} = \frac{115.32}{300} = 0.384\text{ A}$$

(b) Reading of a moving iron voltmeter across the SCR

A moving iron voltmeter measures the **RMS value** of the voltage across the device.

The voltage across the SCR is equal to the supply voltage when the SCR is OFF and is approximately zero when the SCR is ON.

Therefore,

$$V_{T,\text{rms}} = \sqrt{\frac{1}{2\pi}\left[\int_0^{\alpha}(V_m \sin\omega t)^2\, d(\omega t) + \int_{\pi}^{2\pi}(V_m \sin\omega t)^2\, d(\omega t)\right]}$$

Using

$$\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4},$$

we obtain

$$V_{T,\text{rms}} = V_m\sqrt{\frac{1}{4\pi}\left[\alpha - \frac{\sin 2\alpha}{2} + \pi\right]}$$

Substituting

$$V_m = 300\sqrt{2}\,\text{V}, \qquad \alpha = \frac{\pi}{4},$$

$$V_{T,\text{rms}} = 424.26\sqrt{\frac{1.25\pi - 0.5}{4\pi}}$$

$$V_{T,\text{rms}} = 221.46\,\text{V}$$

$$\boxed{I_{\text{avg}} = 0.384\,\text{A}, \qquad V_{T,\text{rms}} = 221.46\,\text{V}}$$

Key Points

- A moving coil ammeter measures the average (DC) value of current.
- A moving iron voltmeter measures the RMS value of voltage.
- For a half-wave controlled rectifier,
$$V_{\text{avg}} = \frac{V_m}{2\pi}(1 + \cos\alpha).$$
- The SCR voltage is equal to the source voltage during the blocking interval and nearly zero during conduction.

Mistakes to be avoided

- Do not use RMS value when calculating the moving coil ammeter reading.
- Do not forget that the SCR voltage is zero during its conduction interval.
- Evaluate the RMS integral over the correct OFF intervals only.

Q1.97.2 MCQ 1997 1M Ans: C ● ○ ○

When the diode turns ON, it reverse biases the thyristor. Hence, the stored charge in the thyristor is removed more quickly, making the turn-off process faster.

However, since the diode conducts during the turn-off interval, additional current flows through the diode, which increases the turn-off power loss.

$$\boxed{C}$$

Key Points

- An anti-parallel diode provides a reverse-bias path for the SCR during turn-off.
- Reverse bias accelerates carrier removal, thereby reducing the SCR turn-off time.
- Conduction through the diode during commutation increases the turn-off energy (power) loss.

Q1.96.1 MCQ 1996 1M Ans: D ● ○ ○

Power semiconductor devices are classified based on their control mechanisms.

- **SCR (Thyristor):** Current-triggered device. A gate current initiates conduction.
- **GTO (Gate Turn-Off Thyristor):** Current-controlled device. Positive gate current turns it ON, and negative gate current turns it OFF.
- **TRIAC:** Current-triggered bidirectional thyristor. A gate current triggers conduction in either direction.
- **MOSFET:** Voltage-controlled device. It is turned ON/OFF by the gate-to-source voltage with negligible steady-state gate current.

Hence, the only device that is **not current triggered** is the MOSFET.

$$\boxed{D}$$

Q1.96.2 MCQ 1996 2M Ans: D ● ○ ○

A TRIAC is a bidirectional thyristor capable of conducting current in both directions. Hence, it is suitable only for AC power control applications.

Among the given options:

- Inverter uses forced commutation devices such as MOSFETs or IGBTs.
- Rectifiers generally employ SCRs or diodes.
- Multiquadrant choppers require self-commutating devices.
- Cycloconverters use TRIACs (or anti-parallel SCRs) for direct AC-to-AC conversion.

Therefore, the TRIAC can be used only in a cycloconverter.

$$\boxed{D}$$

Q1.96.3 MCQ 1996 1M Ans: D ● ○ ○

A high rate of rise of voltage can cause false triggering due to dv/dt triggering, but it cannot cause permanent damage to an SCR. The other three factors can cause thermal damage to the SCR.

$$\boxed{D}$$

Key Points

- Excessive dv/dt may turn ON an SCR unintentionally due to junction capacitance.
- Excessive di/dt causes localized heating near the gate region.
- Overcurrent and excessive junction temperature can permanently damage the SCR.

Mistakes to be avoided

- Do not confuse false triggering due to dv/dt with permanent device failure.
- Remember that dv/dt protection is provided using an RC snubber circuit.
- Thermal damage is mainly associated with excessive current or temperature, not with dv/dt alone.

Q1.95.1 NAT | 1995 | 2M | Ans: 0.733ms

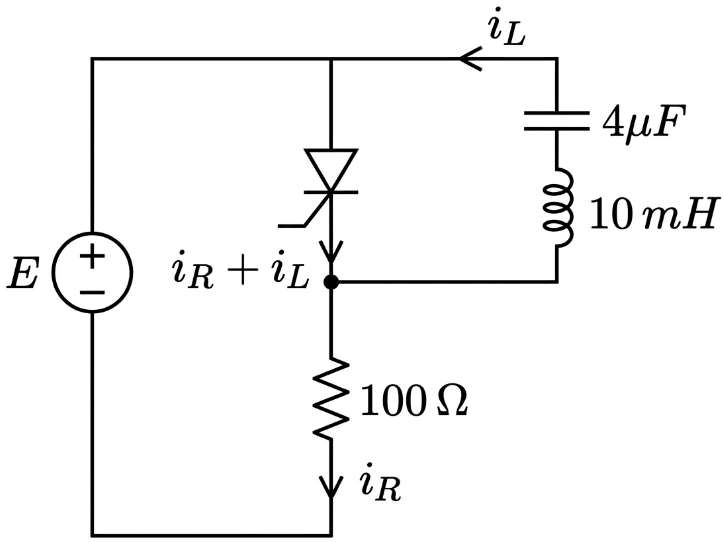

Fig. Solution Q1.95.1

When the thyristor turns OFF,

$$i_R + i_L = 0$$

The resistor current is

$$i_R = \frac{E}{100}$$

The LC branch current is

$$i_L = E\sqrt{\frac{C}{L}}\sin\omega_0 t$$

where

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

Hence,

$$\frac{E}{100} + E\sqrt{\frac{4\times10^{-6}}{10\times10^{-3}}}\sin\omega_0 t = 0$$

Cancelling E,

$$0.01 + \sqrt{\frac{4\times10^{-6}}{10\times10^{-3}}}\sin\omega_0 t = 0$$

Since

$$\sqrt{\frac{4\times10^{-6}}{10\times10^{-3}}} = 0.02,$$

we get

$$0.01 + 0.02\sin\omega_0 t = 0$$

$$\sin \omega_0 t = -\frac{1}{2}$$

Taking the first valid solution,

$$\omega_0 t = \frac{7\pi}{6}$$

Also,

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times 10^{-6} \times 10 \times 10^{-3}}} = 5000 \text{ rad/s}.$$

Therefore,

$$t_{on} = \frac{7\pi}{6\omega_0} = \frac{7\pi}{6 \times 5000} = 0.733 \times 10^{-3} \text{ s}$$

$$\boxed{0.733}$$

Key Points

- At turn-OFF, the resistor current and resonant LC current satisfy $i_R + i_L = 0$.
- The natural resonant frequency is $\omega_0 = 1/\sqrt{LC}$.
- The LC branch current amplitude is $E\sqrt{C/L}$.
- The first valid instant satisfying the current condition determines the thyristor turn-OFF time.

Q1.95.2 NAT 1995 1M Ans: 100 ● ○ ○

The voltage across parallel branches remains the same.

Assuming that the ON-state voltage drop is the same for both thyristors, ΔV_T,

$$\Delta V_T + 0.05 \times 120 = \Delta V_T + 0.06 \times I$$

$$0.05 \times 120 = 0.06I$$

$$I = \frac{0.05 \times 120}{0.06} = \frac{600}{6}$$

$$\boxed{100}$$

Key Points

- Voltage across parallel branches is same.
- For identical ON-state thyristor voltage drop, ΔV_T cancels out.
- Current sharing depends on the slope resistance of the thyristor branch.

Q1.94.1 MCQ 1994 1M Ans: B ● ○ ○

A switched mode power supply (SMPS) operating at 20 kHz to 100 kHz requires a switching device with a very high switching speed and low switching losses.

Among the given options, the MOSFET is best suited for high-frequency switching applications.

$$\boxed{B}$$

Q1.94.2 NAT 1994 1M Ans: Either positive or negative ● ○ ○

A TRIAC can conduct in both directions and can be triggered by either a positive or a negative gate pulse with respect to MT_1.

Hence, it can be triggered by a gate pulse of either polarity.

$$\boxed{\text{Either positive or negative}}$$

Key Points

- A TRIAC is a bidirectional thyristor.
- It can be triggered in all four quadrants.
- The gate pulse may be positive or negative with respect to MT_1 depending on the operating quadrant.

Q1.93.1 MCQ 1993 1M Ans: D ● ○ ○

$$\text{Thermal Resistance} = \frac{\text{Temperature Difference between Body and Ambient}}{\text{Average Power dissipated in the device}}$$

$$R_\theta = \frac{T_{\text{body}} - T_{\text{ambient}}}{P_{\text{avg}}}$$

$$\boxed{D}$$

Q1.92.1 MCQ 1992 1M Ans: A ● ● ○

Since the system is initially uncharged, when the switch is turned ON, the diode will turn ON and the current will be given by,

$$i = V\sqrt{\frac{C}{L}}\sin\omega_0 t$$

After $\omega_0 t = \pi$, the current reverses, but the thyristor does not allow reverse flow of current. Therefore, the diode turns OFF, and the circuit becomes open.

The capacitor voltage is given by

$$V_c = V(1 - \cos\omega_0 t)$$

When the diode turns OFF,

$$\omega_0 t = \pi$$

$$V_c = V(1 - \cos\pi) = 2V = 200\,\text{V}$$

$$\boxed{A}$$

Key Points

- For an initially uncharged LC circuit, current is sinusoidal during natural oscillation.
- The diode/thyristor conducts only until current tries to reverse.

- Maximum capacitor voltage occurs at $\omega_0 t = \pi$.

Q1.91.1 MCQ 1991 1M Ans: B

Operating states of a diode:

- Forward conduction
- Backward (reverse) blocking

Operating states of an SCR:

- Forward blocking
- Forward conduction
- Reverse blocking

Hence, the **forward blocking state** distinguishes an SCR from a diode.

$$\boxed{B}$$

Key Points

- A diode starts conducting immediately under forward bias.
- An SCR remains in the forward blocking state until it is triggered by a gate pulse.
- Both diode and SCR possess the reverse blocking state.

Q1.91.2 MCQ 1991 2M Ans: A:R,B:P,C:S,D:Q

(a) A reactor offers inductive reactance against sudden change in current. Hence, for di/dt protection, a series reactor is used.

(b) Switching problems arise due to the dv/dt limit. Whenever there is a switching problem, an RC snubber circuit is used.

(c) The i^2t limit is fixed to avoid runaway speeds under no-load conditions.

(d) Thyristors are mounted on heat sinks for thermal protection to maintain the junction temperature within permissible limits.

$$\boxed{(A) \rightarrow (R),\ (B) \rightarrow (P),\ (C) \rightarrow (S),\ (D) \rightarrow (Q)}$$

DEBUG INFO

Chapter I

Power Semiconductor Devices

Total Questions : 53

MCQ : 44

MSQ : 1

NAT : 8

SOLUTIONS

II

AC to DC Converters/Phase Controlled Rectifiers

Topics

1. Single-Phase Rectifier:
   (a) Half Wave Rectifier with R/RE Loads
   (b) Filters and Rectifiers with Pure Inductance(L) Load
   (c) Half Wave Rectifiers with RLE Loads
   (d) Half Wave Rectifiers with Free wheeling diode
   (e) Full Wave Rectifier with R/RE Loads
   (f) Full Wave Rectifier with L/RL/RLE Loads
   (g) Full Wave Rectifiers with Free wheeling diode
2. Three-Phase Rectifier:
   (a) Half Wave Rectifier with R/RL/RLE Loads
   (b) Full Wave Rectifier with RL/RLE Load with Phase Voltage
   (c) Full Wave Rectifier with RL/RLE Load with Line Voltage
   (d) Circuit Turn off time concept
   (e) Full wave Semi-Converter
   (f) Effect of Source Inductance in Rectifiers
   (g) Fourier Series Analysis of Supply Currents
3. Line-Commutated AC to DC Converters
4. Bidirectional AC to DC Voltage Source Converters
5. Line Current Harmonics, Power Factor and Distortion Factor

Q2.26.1 NAT 2026 1M Ans: 32.5

$$R = 10\,\Omega, \quad L = 500\,\mu\text{F}$$

First, calculating the inductive reactance (X_L) at a standard power frequency of $f = 50$ Hz:

$$X_L = \omega L = 2\pi \cdot 50 \cdot 500 \cdot 10^{-6}$$

$$X_L = 0.157\,\Omega$$

The total magnitude of the load impedance ($|Z|$) is given by:

$$|Z| = \sqrt{R^2 + X_L^2} = \sqrt{10^2 + (0.157)^2} = 10.001\,\Omega \simeq R$$

Since the inductive reactance value is extremely small compared to the resistance ($X_L \ll R$), the total impedance magnitude is dominated entirely by the resistor:

$$|Z| \simeq R$$

Therefore, we can approximate the network as a pure load resistance. Assuming a peak supply voltage of $V_M = 325$ V, the peak load current (i_{peak}) is evaluated as:

$$i_{\text{peak}} = \frac{V_M}{R} = \frac{325}{10} = 32.5\text{ A}$$

$$\boxed{32.5}$$

Q2.26.2 NAT 2026 2M Ans: 1.08

Method 1

The transient current equation

When the supply voltage V_s exceeds 50 V, the conduction starts.

$$i = \frac{V_s - 50}{10} = 10\sin\alpha - 5$$

Conduction occurs between the conduction angles where $i > 0$.
Equating $i = 0$

$$10\sin\alpha = 5 \implies \sin\alpha = \frac{1}{2}$$

This yields the lower limit $\alpha = \frac{\pi}{6}$ and the upper limit $\alpha = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

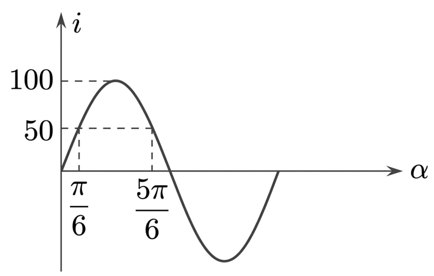

Fig. Solution Q2.26.2

Average current (I_{avg})

$$I_{avg} = \frac{1}{2\pi} \int_{\pi/6}^{5\pi/6} i \cdot d\alpha$$

$$I_{avg} = \frac{1}{2\pi} \int_{\pi/6}^{5\pi/6} (10 \sin \alpha - 5)\, d\alpha$$

$$I_{avg} = \frac{1}{2\pi} \Big[-10 \cos \alpha - 5\alpha \Big]_{\pi/6}^{5\pi/6}$$

$$I_{avg} = \frac{-1}{2\pi} \left\{ \left[10 \cos \frac{5\pi}{6} - 10 \cos \frac{\pi}{6} \right] + \left[5\left(\frac{5\pi}{6}\right) - 5\left(\frac{\pi}{6}\right) \right] \right\}$$

$$I_{avg} = \frac{1}{2\pi} [17.32 - 10.47] = \frac{6.85}{2\pi} \approx 1.08 \text{ A}$$

$$\boxed{1.08}$$

Method 2

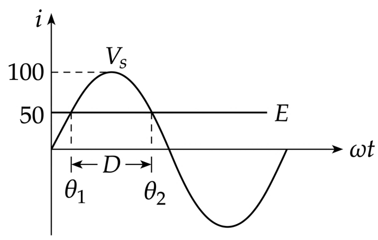

Fig. Solution Q2.26.2

Calculation of Conduction Angles

The conduction begins when the instantaneous supply voltage reaches the battery voltage level E:

$$\theta_1 = \sin^{-1}\left(\frac{E}{V_m}\right) = \sin^{-1}\left(\frac{50}{100}\right) = \frac{\pi}{6} \quad (30^\circ)$$

Due to the symmetry of the sine wave, the conduction stops when the supply falls back below E:

$$\theta_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \quad (150^\circ)$$

Integration for Average Current

The average current equation over a full cycle (2π) is given by:

$$I_D = I_0 = \frac{1}{2\pi} \cdot \int_{\theta_1}^{\theta_2} \left(\frac{V_m \sin \omega t - E}{R}\right) d(\omega t)$$

$$= \frac{1}{2\pi \cdot 10} \cdot \left[100(\cos 30^\circ - \cos 150^\circ) - 50\left(\frac{5\pi}{6} - \frac{\pi}{6}\right) \right]$$

$$= \frac{1}{20\pi} \cdot \left[100\left(\frac{\sqrt{3}}{2} - \left(-\frac{\sqrt{3}}{2}\right)\right) - 50\left(\frac{4\pi}{6}\right) \right]$$

$$= \frac{1}{20\pi} \cdot \left[100\sqrt{3} - \frac{100\pi}{3} \right]$$

$$= \frac{1}{20\pi} \cdot [173.20 - 104.72] = \frac{68.48}{20\pi} \approx 1.08 \text{ A}$$

$$\boxed{1.08}$$

Mistakes to be avoided

- Remember to distribute the negative sign carefully when simplifying terms involving $\cos(5\pi/6) = -\sqrt{3}/2$.
- Do not integrate over the entire 0 to π range; the limits must strictly reflect the actual conduction angle limits ($\frac{\pi}{6}$ to $\frac{5\pi}{6}$).

Q2.25.1 MCQ 2025 2M Ans: A

Given the sinusoidal voltage expression:

$$v = 230\sqrt{2}\sin(2\pi \cdot 50t)$$

A SCR with $\alpha = 0$ behaves like a diode.
The current expression in a purely inductive load supplied by a single-phase voltage source is given by:

$$i = \frac{1}{\omega L}\int_0^{\omega t} V_m \sin(\omega t)\, d(\omega t)$$

$$i = \frac{V_m}{\omega L}[1 - \cos\omega t]$$

The peak value of the current occurs at $\omega t = \pi$:

$$i_{\text{peak}} = \frac{2V_m}{\omega L}$$

Substituting the given values ($V_m = 230\sqrt{2}\,\text{V}$, $f = 50\,\text{Hz}$, and $L = 100 \times 10^{-3}\,\text{H}$):

$$i_{\text{peak}} = \frac{2 \times 230\sqrt{2}}{2\pi \times 50 \times 100 \times 10^{-3}} \approx 20.73\,\text{A}$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- When a purely inductive circuit is energized at the zero-crossing of a sinusoidal voltage, the resulting current contains a DC offset equal to the peak value of the AC steady-state component.
- The maximum value or peak current reaches twice the steady-state amplitude ($i_{\text{peak}} = \frac{2V_m}{\omega L}$) exactly at $\omega t = \pi$ due to this transient asymmetry.

Q2.25.2 MCQ 2025 2M Ans: A

Given that the power dissipated in the resistor is:

$$P_R = I_{\text{or}}^2 R = 64\,\text{W}$$

Substituting the resistance value $R = 1\,\Omega$:

$$I_{\text{or}}^2 \times 1 = 64$$

$$I_{\text{or}} = 8\,\text{A} = I_0 \quad (\because \text{Current is ripple-free})$$

The average output voltage V_0 for the converter circuit is given by:

$$V_0 = E + I_0 R$$

Substituting the given values into the equation:

$$400\left[1 + \frac{\cos\alpha}{3}\right] = 500 + 8 \times 1 = 508$$

Solving for $\cos\alpha$:

$$1+\frac{\cos\alpha}{3}=\frac{508}{400}=1.27$$
$$\frac{\cos\alpha}{3}=0.27$$
$$\cos\alpha=0.81$$
$$\alpha=\cos^{-1}(0.81)\approx 35.9^\circ$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- The output voltage equation for a converter feeding a highly inductive load with an EMF source is $V_0 = E + I_0R$.

Q2.25.3 MCQ 2025 1M Ans: A

The given voltage and current are

$$v(t)=300\sin(\omega t)\ \text{V}$$

$$i(t)=10\sin\left(\omega t-\frac{\pi}{6}\right)+2\sin\left(3\omega t+\frac{\pi}{6}\right)+\sin\left(5\omega t+\frac{\pi}{2}\right)$$

The fundamental, third and fifth harmonic components are

$$i_1(t)=10\sin\left(\omega t-\frac{\pi}{6}\right)$$

$$i_3(t)=2\sin\left(3\omega t+\frac{\pi}{6}\right)$$

$$i_5(t)=\sin\left(5\omega t+\frac{\pi}{2}\right)$$

Hence,

$$I_{1,\text{RMS}}=\frac{10}{\sqrt{2}},\qquad I_{3,\text{RMS}}=\frac{2}{\sqrt{2}},\qquad I_{5,\text{RMS}}=\frac{1}{\sqrt{2}}$$

The effective RMS current is

$$I_{\text{eff(RMS)}}=\sqrt{I_{1,\text{RMS}}^2+I_{3,\text{RMS}}^2+I_{5,\text{RMS}}^2}$$

$$=\sqrt{\left(\frac{10}{\sqrt{2}}\right)^2+\left(\frac{2}{\sqrt{2}}\right)^2+\left(\frac{1}{\sqrt{2}}\right)^2}=7.246\ \text{A}$$

The distortion factor is

$$\text{DF}=\frac{I_{1,\text{RMS}}}{I_{\text{eff(RMS)}}}=\frac{10/\sqrt{2}}{7.246}=0.976$$

The displacement power factor is

$$\cos\theta=\cos\frac{\pi}{6}=0.866$$

Therefore,

$$\text{Input PF} = \text{DF} \times \cos\theta = 0.976 \times 0.866 = 0.845$$

Hence, the correct option is

$$\boxed{A}$$

Q2.24.1 NAT 2024 2M Ans: 0.955

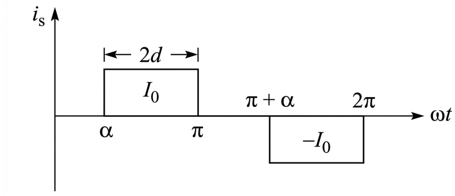

Fig. Solution Q2.24.1

For a 1-ϕ Half Controlled Bridge Rectifier operating with a firing angle $\alpha = 60°$, the source current waveform i_s is quasi-square as shown in the waveform:

$$\text{Conduction angle or pulse duration, } 2d = \pi - \alpha$$

The fundamental component RMS source current (I_{s1}) and total RMS source current (I_{sr}) are defined to determine the total distortion factor or fundamental power factor (g):

$$g = \frac{I_{s1}}{I_{sr}} = \frac{\frac{2\sqrt{2}}{\pi} I_0 \cdot \cos\frac{\alpha}{2}}{I_0\left(\frac{\pi-\alpha}{\pi}\right)^{1/2}} = \frac{2\sqrt{2}\cdot\cos\frac{\alpha}{2}}{\sqrt{\pi\cdot(\pi-\alpha)}}$$

Substituting the given value of firing angle $\alpha = 60° = \frac{\pi}{3}$ rad:

$$g = \frac{2\sqrt{2}\cdot\cos\left(\frac{60°}{2}\right)}{\sqrt{\pi\cdot\left(\pi-\frac{\pi}{3}\right)}}$$

$$= \frac{3}{\pi} \approx 0.955$$

$$\boxed{0.955}$$

Key Points

- Due to the presence of freewheeling action in a semi-converter, the input source current remains zero for a duration of α in each half cycle, reducing the RMS current requirements.

Q2.24.2 NAT 2024 2M Ans: 114.58 ● ● ●

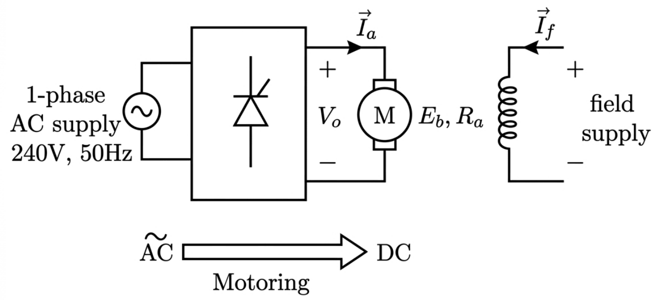

Fig. Solution Q2.24.2

Evaluation of the motor constants from nameplate ratings

Given nameplate ratings of the DC machine during motoring: Rated Voltage, $V_0 = 210$ V Rated Armature Current, $I_0 = 10$ A Rated Speed, $N = 1200$ rpm Armature Resistance, $R_a = 1\,\Omega$

Since the field current (I_f) is constant, the field flux (ϕ) is also constant. The back EMF (E_b) under rated motoring condition is given by:

$$E_b = K_m\omega = K_m \cdot \frac{2\pi \cdot N}{60}$$

Using the terminal voltage equation for motoring ($V_0 = E_b + I_0R_a$):

$$210 = K_m \cdot \frac{2\pi \cdot N}{60} + I_0R_a$$

$$210 = K_m \cdot \left(\frac{2\pi \cdot 1200}{60}\right) + (10 \times 1)$$

$$210 = K_m \cdot (40\pi) + 10$$

$$200 = K_m \cdot (40\pi) \implies K_m = \frac{200}{40\pi} \approx 1.5915\ \text{V} \cdot \text{s/rad}$$

Regenerative Braking

When the armature terminals are reversed for regenerative braking, the back EMF (E_b) opposes the polarity of the converter output voltage, transitioning the mode to inversion. The governing equation is:

$$V_0 = -E_b + I_0R_a$$

For a single-phase full-converter feeding a DC motor, the average output voltage is given by:

$$\frac{2V_m}{\pi}\cos\alpha = -E_b + I_0R_a$$

$$\frac{2V_m}{\pi}\cos\alpha = -K_m \cdot \frac{2\pi N}{60} + I_0R_a$$

Given parameters during braking:
AC Source Supply Voltage (RMS), $V_s = 240$ V $\implies V_m = 240\sqrt{2}$ V
Braking Speed, $N = 600$ rpm
Armature Current is maintained at rated value, $I_0 = 10$ A

Substituting these values into the expression:

$$\frac{2 \cdot 240\sqrt{2}}{\pi} \cdot \cos\alpha = -1.5915 \times \left(\frac{2\pi \cdot 600}{60}\right) + 10$$

$$\frac{480\sqrt{2}}{\pi} \cdot \cos\alpha = -1.5915 \times (20\pi) + 10$$

$$216.08 \cdot \cos\alpha = -100 + 10$$
$$216.08 \cdot \cos\alpha = -90$$
$$\cos\alpha = \frac{-90}{216.08} \approx -0.4165$$
$$\alpha = \cos^{-1}(-0.4165) \approx 114.61^\circ$$

$$\boxed{114.58^\circ}$$

Key Points

- During regenerative braking, the machine acts as a generator. Reversing the armature connection makes E_b negative relative to the converter terminal voltage reference.
- Regenerative braking via a line-commutated converter requires an inversion mode operation where the firing angle satisfies $90^\circ < \alpha < 180^\circ$.

Mistakes to be avoided

- Do not use $V_m = 240\,\text{V}$ directly. Always convert the standard RMS supply voltage to its peak component value $V_m = V_s\sqrt{2}$ when analyzing converter output formulas.
- Ensure that the negative sign for E_b is explicitly included when setting up the terminal voltage loop equation for the regenerative braking configuration.

Q2.23.1 NAT 2023 2M Ans: 39.78

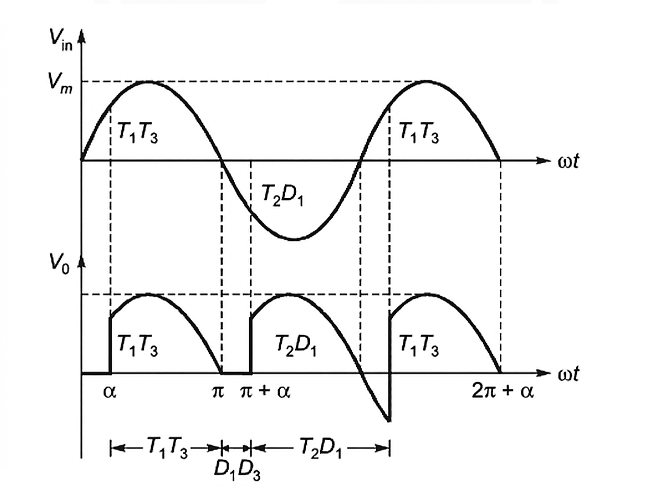

Fig. Solution Q2.23.1

The average output voltage V_o is given by integrating the voltage waveform over its periodic interval:

$$V_o = \frac{1}{2\pi}\left[\int_{\alpha}^{\pi} V_m \sin\omega t\, d(\omega t) + \int_{\pi+\alpha}^{2\pi+\alpha} -V_m \sin\omega t\, d(\omega t)\right]$$

Evaluating the integrals simplifies to the following expression:

$$V_o = \frac{V_m}{2\pi}[1 + 3\cos\alpha]$$

Given parameters from the problem context:

- Peak voltage, $V_m = 100\,\text{V}$

- Firing angle, $\alpha = 60°$

Substituting these values into the average voltage equation:

$$V_o = \frac{100}{2\pi}[1 + 3\cos 60°] = \frac{100}{2\pi}[1 + 3(0.5)] = \frac{250}{2\pi} \approx 39.78\,\text{V}$$

$$\boxed{39.78}$$

Key Points

- From π to $\pi + \alpha$, T_3 and D_1 will perform freewheeling action of the load current, hence output voltage will be 0.

Q2.22.1 NAT 2022 1M Ans: 800 ● ○ ○

For a 1-ϕ full-bridge diode rectifier connected to a resistive load:
Given data:

- Load resistance, $R = 50\,\Omega$
- 1-ϕ Active power supply voltage, $V_s = 200$ V

The average output power $P_{o(\text{avg})}$ delivered to a purely resistive load depends on the root-mean-square (RMS) value of the output voltage:

$$P_{o(\text{avg})} = \frac{V^2_{o(\text{rms})}}{R}$$

For a 1-ϕ full-bridge rectifier, the magnitude of the output voltage waveform matches the input supply voltage waveform at all times (except during ideal diode transitions). Therefore, the RMS value of the output voltage V_{or} is equal to the RMS value of the input supply voltage $V_{s(\text{rms})}$:

$$V_{or} = V_{s(\text{rms})} = 200\,\text{V}$$

Substituting the values into the power equation:

$$P_{o(\text{avg})} = \frac{200^2}{50} = \frac{40000}{50} = 800\,\text{W}$$

$$\boxed{800}$$

Q2.22.2 NAT 2022 1M Ans: 0.447 ● ○ ○

The distortion factor (or current modification factor) g is defined as the ratio of the RMS value of the fundamental component of the input current (I_{s1}) to the total RMS input current (I_{sr}):

$$g = \frac{I_{s1}}{I_{sr}} = \frac{10/\sqrt{2}}{\sqrt{\left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{4}{\sqrt{2}}\right)^2 + \left(\frac{3}{\sqrt{2}}\right)^2}}$$

Evaluating the expression gives:

$$g = \frac{10}{\sqrt{10^2 + 4^2 + 3^2}} = \frac{10}{\sqrt{100 + 16 + 9}} = \frac{10}{\sqrt{125}} \approx 0.8944$$

The total input power factor (PF) is the product of the distortion factor (g) and the displacement factor (FDF):

$$\text{PF} = g \cdot \text{FDF}$$

Given that the displacement angle is $\frac{\pi}{3}$:

$$PF = g \cdot \cos\left(\frac{\pi}{3}\right)$$

Substituting the calculated value of g:

$$PF = 0.8944 \times \cos(60^\circ) = 0.8944 \times 0.5 \approx 0.447$$

$$\boxed{0.447}$$

Q2.22.3 NAT 2022 2M Ans: 17.65

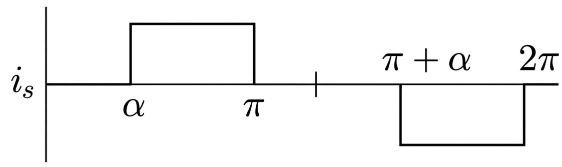

Fig. Solution Q2.22.3

$$i_s = \sum_{n=1,3,5}^{\infty} \frac{4I_o}{n\pi} \cos\left(\frac{n\alpha}{2}\right) \sin n\left(\omega t - \frac{\alpha}{2}\right)$$

The amplitude of the n^{th} harmonic component is:

$$\hat{I}_{sn} = \frac{4I_o}{n\pi} \cos\left(\frac{n\alpha}{2}\right)$$

For the fundamental component ($n = 1$):

$$\hat{I}_{s1} = \frac{4 \times 15}{\pi} \cos(22.5^\circ) = 17.65\text{ A}$$

Hence, the answer is

$$\boxed{17.65}$$

Q2.21.1 NAT 2021 1M Ans: 390.32

The worst-case peak diode voltage occurs at maximum input voltage:

$$(V_D)_{\text{peak}} = (230 + 20\%)\sqrt{2} = \left(230 + 230 \times \frac{20}{100}\right)\sqrt{2}$$

$$= 276\sqrt{2} \approx 390.32\text{ V}$$

Hence, the answer is

$$\boxed{390.32}$$

Q2.20.1 MCQ 2020 1M Ans: C ● ○ ○

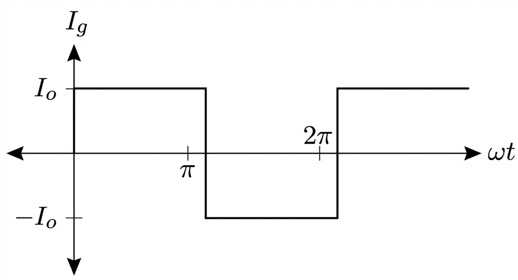

Fig. Solution Q2.20.1

$$i_g(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4I_o}{n\pi} \sin(n\omega t)$$

The source current contains only odd harmonic components due to half-wave symmetry. Hence, the frequency components in the source current are:

$$50\,\text{Hz},\ 150\,\text{Hz},\ 250\,\text{Hz},\ \ldots$$

$$\boxed{C}$$

Q2.20.2 NAT 2020 2M Ans: 4.80 ● ● ○

Given data for a 1-ϕ SCR bridge rectifier:

- Firing angle, $\alpha = 45°$
- Load resistance, $R = 10\,\Omega$
- Source voltage, $V_{\text{rms}} = 230\,\text{V}$
- Supply frequency, $f = 50\,\text{Hz}$
- Source inductance, $L_s = 2.28\,\text{mH}$

The average load current is obtained from:

$$I_0 R = \frac{2V_m}{\pi} \cos\alpha - 4fL_s I_0$$

where

$$V_m = 230\sqrt{2}\,\text{V}$$

Substituting values:

$$I_0 \times 10 = \frac{2 \times 230\sqrt{2}}{\pi} \cos 45° - 4 \times 50 \times 2.28 \times 10^{-3} I_0$$

$$I_0 = \frac{146.49}{10.456} = 14.00\,\text{A}$$

The overlap angle μ is found using:

$$\frac{V_m}{\pi} [\cos\alpha - \cos(\alpha + \mu)] = 4fL_s I_0$$

$$4 \times 50 \times 2.28 \times 10^{-3} \times 14 = 6.384$$

$$\frac{230\sqrt{2}}{\pi}[\cos 45^\circ - \cos(45^\circ + \mu)] = 6.384$$

$$\cos 45^\circ - \cos(45^\circ + \mu) = 0.061659$$

$$\cos(45^\circ + \mu) = 0.707107 - 0.061659 = 0.645448$$

$$45^\circ + \mu = \cos^{-1}(0.645448) = 49.80^\circ$$

$$\mu = 49.80^\circ - 45^\circ = 4.80^\circ$$

$$\boxed{4.80}$$

Key Points

- Source inductance causes an overlap period (μ) during which multiple thyristors conduct simultaneously, resulting in a reduction of the average output voltage.

Mistakes to be avoided

- **RMS vs Peak Voltage:** Ensure you use $V_m = V_{\text{rms}}\sqrt{2}$ in the formulas rather than using the raw 230 V directly.

Q2.19.1 MCQ 2019 1M Ans: C ● ○ ○

$$i_s = \sum_{6k\pm1}^{\infty} \frac{4I_o}{n\pi} \sin\left(\frac{n\pi}{3}\right) \sin n(\omega t - \alpha)$$

Triplen harmonics do not exist in the supply current, so the minimum order harmonic is the 5^{th} harmonic:

$$5 \times 50\,\text{Hz} = 250\,\text{Hz}$$

$$\boxed{C}$$

Q2.19.2 NAT 2019 2M Ans: 81.65 ● ○ ○

$$\text{AC line current rms} = (I_s)_{\text{rms}} = I_0\sqrt{\frac{2}{3}} = 100 \times \sqrt{\frac{2}{3}}$$
$$= 81.65\,\text{A}$$

$$\boxed{81.65}$$

Q2.19.3 NAT 2019 2M Ans: 0.78 ● ○ ○

For a single-phase fully controlled converter, input and output power are related by:

$$V_{sr} I_{sr} \cos\phi = V_0 I_0$$

Since the source and load currents are equal:

$$I_0 = I_{sr} = 10\,\text{A}$$

Substituting into the power equation and cancelling current:

$$V_{sr} \cos \phi = V_0$$

Hence,

$$\cos \phi = \frac{V_0}{V_{sr}} = \frac{180}{230} = 0.78$$

$$\boxed{0.78}$$

Q2.18.1 MCQ 2018 1M Ans: C

Case 1: Resistive load (R-load)

Since the load is purely resistive, there is no energy storage and hence no freewheeling action. Therefore, the freewheeling diode does not conduct.

Thus, from the waveform:

$$V_o(t) \geq 0, \qquad i_o(t) \geq 0$$

Case 2: Inductive load (R–L load)

For a highly inductive load, the load current remains continuous:

$$i_o(t) \geq 0$$

Since the freewheeling diode provides a path for current during the negative half-cycle, the output voltage also remains non-negative:

$$V_o(t) \geq 0$$

$$\boxed{C}$$

Q2.18.2 NAT 2018 1M Ans: 17.39

The RMS value of the fundamental source current is:

$$\begin{aligned} i_{s1} &= \frac{4I_0}{\pi} \cos \frac{\alpha}{2} \\ I_{s1(\text{rms})} &= \frac{2\sqrt{2}}{\pi} I_0 \cos \frac{\alpha}{2} \\ &= \frac{2\sqrt{2}}{\pi} \times 20 \times \cos\left(\frac{30^\circ}{2}\right) \\ &= 17.39 \text{ A} \end{aligned}$$

$$\boxed{17.39}$$

Q2.18.3 NAT 2018 2M Ans: 90

For the converter,

$$V_0 = \frac{V_m}{2\pi}(3 + \cos \alpha)$$

Given motor power:

$$E_b I_0 = 1600 \text{ W}$$

$$I_0 = \frac{1600}{80} = 20\,\text{A}$$

Using the armature voltage equation:

$$V_0 = E_b + I_0 R_a$$

$$V_0 = 80 + (20 \times 2) = 120\,\text{V}$$

Substituting into the converter equation:

$$\frac{V_m}{2\pi}(3 + \cos\alpha) = 120$$

Given $V_m = 80\pi$:

$$\frac{80\pi}{2\pi}(3 + \cos\alpha) = 120$$

$$40(3 + \cos\alpha) = 120$$

$$3 + \cos\alpha = 3$$

$$\cos\alpha = 0$$

$$\alpha = 90^\circ$$

$$\boxed{90}$$

Q2.17.1 MCQ 2017 1M Ans: A ● ○ ○

The inductance given in the problem is the source inductance due to which there is overlap between the two diodes. Hence, each diode conducts for

$$(180^\circ + \mu)$$

where μ is the overlap angle.

$$\boxed{A}$$

Key Points

- Source inductance causes commutation overlap in rectifier circuits.
- Due to overlap, each diode conducts for an angle of $(180^\circ + \mu)$ instead of 180°.
- μ is known as the overlap (commutation) angle.

Mistakes to be avoided

- Do not assume each diode always conducts for exactly 180° when source inductance is present.
- Do not confuse the overlap angle μ with the firing angle α.

Q2.17.2 NAT 2017 2M Ans: 224.17

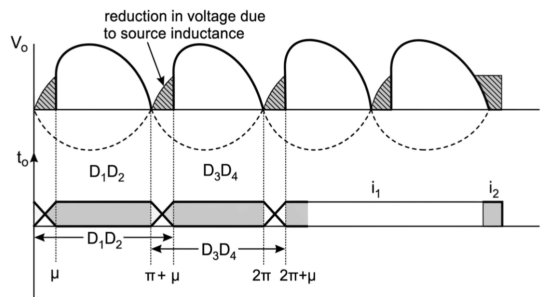

Fig. Solution Q2.17.2

Average Output Voltage

Without source inductance (L_s),

$$V_{o(\text{avg})} = \frac{2V_m}{\pi}$$

With source inductance (L_s),

$$V'_{o(\text{avg})} = \frac{1}{\pi}\int_{\mu}^{\pi} V_m \sin\omega t \, d\omega t = \frac{V_m}{\pi}(1+\cos\mu)$$

Reduction in average output voltage:

$$\Delta V_{do} = V_{o(\text{avg})} - V'_{o(\text{avg})} = \frac{V_m}{\pi}(1-\cos\mu)$$

Also,

$$\Delta V_{do} = 4fL_sI_o$$

On equating,

$$\frac{V_m}{\pi}(1-\cos\mu) = 4fL_sI_o$$

Substituting the given values,

$$\frac{222\sqrt{2}}{\pi}(1-\cos\mu) = 4\times 50\times 10\times 10^{-3}\times 14$$

$$\mu = 44.17^\circ$$

Conduction angle of each diode,

$$\gamma_D = 180^\circ + \mu$$

$$= 180^\circ + 44.17^\circ = 224.17^\circ$$

$$\boxed{224.17^\circ}$$

Q2.17.3 NAT 2017 1M Ans: 0 ● ○ ○

Since the load is purely resistive, the conduction does not go beyond 180°.

Hence, the freewheeling diode does not come into conduction.

Therefore, the RMS current through diode D_3 is

$$I_{D_3,\text{RMS}} = 0$$

$$\boxed{0}$$

Key Points

- For a purely resistive load, current becomes zero when the supply voltage becomes zero.
- Freewheeling diode conducts only when stored inductive energy forces current to continue.
- Since there is no inductance in a purely resistive load, freewheeling action does not occur.

Q2.17.4 MCQ 2017 1M Ans: C ● ○ ○

In a single-phase full-bridge rectifier, the output voltage waveform has two pulses in every supply cycle.

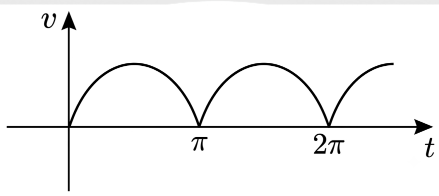

Fig. Solution Q2.17.4

Hence, the ripple frequency is

$$f_o = 2f_s$$

For a supply frequency of 50 Hz,

$$f_o = 2 \times 50 = 100\,\text{Hz}$$

Hence, the correct option is

$$\boxed{C}$$

Q2.17.5 NAT 2017 2M Ans: 0.71 ● ○ ○

The circuit is a half-wave rectifier whose output voltage waveform is shown below.

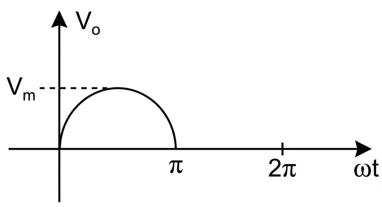

Fig. Solution Q2.17.5

The RMS value of the output voltage is

$$V_{o,\text{RMS}} = \left[\frac{1}{2\pi}\int_0^{\pi} (V_m \sin\omega t)^2 \, d(\omega t)\right]^{1/2}$$

$$= V_m \left[\frac{1}{2\pi}\int_0^{\pi} \frac{1-\cos 2\omega t}{2} \, d(\omega t)\right]^{1/2} = \frac{V_m}{2}$$

The input power factor is

$$\text{IPF} = \frac{P_o}{V_{s,\text{RMS}} I_{s,\text{RMS}}} = \frac{I_{o,\text{RMS}}^2 R}{V_{s,\text{RMS}} I_{s,\text{RMS}}}$$

Since,

$$I_{s,\text{RMS}} = I_{o,\text{RMS}}$$

$$\text{IPF} = \frac{V_{o,\text{RMS}}}{V_{s,\text{RMS}}} = \frac{V_m/2}{V_m/\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.707$$

$$\boxed{0.71}$$

Key Points

- For a half-wave rectifier,
$$V_{o,\text{RMS}} = \frac{V_m}{2}.$$
- Source RMS voltage is
$$V_{s,\text{RMS}} = \frac{V_m}{\sqrt{2}}.$$
- For a resistive load,
$$\text{IPF} = \frac{V_{o,\text{RMS}}}{V_{s,\text{RMS}}}.$$

Mistakes to be avoided

- Do not use the full-wave RMS value for a half-wave rectifier.
- Use the integration limits from 0 to π only.
- Remember that the source RMS voltage is calculated over the complete cycle.

Q2.16.1 NAT 2016 2M Ans: C ● ○ ○

The input power factor at the AC mains for a controlled rectifier configuration is evaluated as the product of the Current Distortion Factor (C.D.F.) and the Displacement Factor (D.F.):

$$\text{Input power factor} = \text{power factor at ac mains}$$

$$= \text{C.D.F.} \times \text{D.F.} = g\cos\alpha$$

For the given converter configuration, the distortion factor (g) is given by $\frac{2\sqrt{2}}{\pi}$. Therefore, the expression for the input power factor becomes:

$$= \frac{2\sqrt{2}}{\pi}\cos\alpha$$

Given that the firing angle delay is $\alpha = 30°$, substituting this value into the equation yields:

$$= \frac{2\sqrt{2}}{\pi}\cos 30°$$

$$= \frac{2\sqrt{2}}{\pi} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{\pi} \approx 0.78$$

$$\boxed{C}$$

Key Points

- Total Power Factor (PF) in non-linear power electronic circuits is given by PF = $g \times$ DPF, where g is the distortion factor and DPF = $\cos\alpha$ is the displacement power factor.
- The current distortion factor (g) represents the ratio of the RMS fundamental component of the input current to the total RMS input current ($g = \frac{I_{s1}}{I_s}$).

Q2.16.2 NAT 2016 2M Ans: 9.21 ● ● ○

The input power is

$$P = V_s I_s \cos\phi$$

$$5 \times 10^3 = 220 \times I_s \times 1$$

$$I_s = 22.72\text{ A}$$

Applying KVL,

$$V_s = V_{c1} + jI_sX_s$$

$$V_{c1} = V_s - jI_sX_s$$

$$V_{c1} = 220\angle 0^\circ - 22.72\angle 0^\circ \left(2\pi \times 50 \times 5 \times 10^{-3}\right) \angle 90^\circ$$

$$= 220 - 35.68\angle 90^\circ$$

$$V_{c1} = 222.87\angle -9.21^\circ$$

Therefore, the phase shift between V_s and V_{c1} is

$$\boxed{9.21^\circ}$$

Q2.16.3 NAT 2016 2M Ans: 6 ● ● ○

Since the converter operates in continuous conduction mode,

$$v_m(t) = 200\pi \sin(100\pi t)\text{ V}$$

The average output dc voltage is

$$V_o = \frac{2V_m}{\pi} \cos\alpha$$

$$= \frac{2 \times 200\pi}{\pi} \cos 120^\circ = 400\left(-\frac{1}{2}\right) = -200\text{ V}$$

Also,

$$V_o = I_o R + E$$

$$-200 = I_o \times 20 - 800$$

$$I_o = \frac{600}{20} = 30\,\text{A}$$

The real power fed back to the source is

$$P = |V_o| I_o = 200 \times 30 = 6000\,\text{W} = 6\,\text{kW}$$

$$\boxed{6}$$

Key Points

- For a single-phase full converter in continuous conduction,

$$V_o = \frac{2V_m}{\pi}\cos\alpha.$$

- In inverter mode ($\alpha > 90°$), the average output voltage is negative.
- Real power fed back to the AC source is

$$P = |V_o| I_o.$$

Q2.16.4 | NAT | 2016 | 1M | Ans: 57.7

The load is highly inductive, so the output current is constant and continuous.

$$I_o = 100\,\text{A}$$

Each diode conducts for 120° out of 360°.
Hence, the RMS current through each diode is

$$(I_D)_{\text{RMS}} = I_o\sqrt{\frac{120}{360}} = \frac{I_o}{\sqrt{3}} = \frac{100}{\sqrt{3}} = 57.70\,\text{A}$$

$$\boxed{57.70\,\text{A}}$$

Q2.15.1 | MCQ | 2015 | 1M | Ans: B

For an HVDC link,

$$P = V \cdot I$$

In an LCC-HVDC system, current direction cannot be reversed due to thyristor unidirectionality. Hence, power reversal is achieved by reversing the polarity of DC voltages.

For power flow from system 2 to system 1,

$$V_1 = -485\,\text{kV}, \qquad V_2 = -500\,\text{kV}, \qquad I = 1.5\,\text{kA}$$

Since the voltage polarities are reversed while current remains in the same direction, the required power flow condition is satisfied.

$$\boxed{B}$$

Q2.15.2 MCQ 2015 1M Ans: C ● ○ ○

Given,

$$100\pi RC = 50$$

For the capacitor discharge path,

$$100\pi \cdot \frac{RC}{2} = 25$$

Thus,

$$\omega\tau = 25$$

Since

$$\omega T = 2\pi \approx 6.28$$

we have

$$\omega\tau \gg \omega T$$

Hence, capacitor discharge is negligible, so each capacitor charges to the peak input voltage:

$$V_{C1} \approx V_m = 100\,\text{V}$$

$$V_{C2} \approx V_m = 100\,\text{V}$$

Therefore, output voltage is

$$V_o = V_{C1} + V_{C2} = 2V_m = 200\,\text{V}$$

$$\boxed{C}$$

Key Points

- In a full-wave voltage doubler, each capacitor charges to approximately V_m.
- For $RC \gg T$, ripple is negligible and $V_o \approx 2V_m$.

Mistakes to be avoided

- Do not treat the circuit as a conventional rectifier; output voltage is doubled.

Q2.15.3 NAT 2015 2M Ans: 61.53 ● ● ○

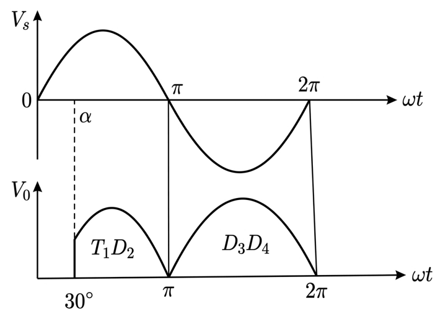

Fig. Solution Q2.15.3

For the given semi-converter with resistive load, average output voltage is

$$V_o = \frac{1}{2\pi}\left[\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) - \int_{\pi}^{2\pi} V_m \sin(\omega t)\, d(\omega t)\right]$$

$$= \frac{V_m}{2\pi}\left[(1+\cos\alpha)+2\right]$$

$$= \frac{V_m}{2\pi}(3+\cos\alpha)$$

Given,

$$V_m = 100\,\text{V}, \qquad \alpha = 30^\circ$$

$$V_o = \frac{100}{2\pi}(3+\cos 30^\circ)$$

$$= \frac{100}{2\pi}(3+0.866) = 61.53\,\text{V}$$

$$\boxed{61.53}$$

Key Points

- For a resistive load, current follows the output voltage and becomes zero at natural commutation.
- Average output voltage is obtained by integrating over both conducting intervals.

Mistakes to be avoided

- Do not use the standard symmetric semi-converter formula; this circuit has an asymmetric conduction pattern.

Q2.14.1 NAT 2014 1M Ans: 23 ● ○ ○

$$I_{o,\text{rms}} = \frac{V_s}{R}$$

$$R = \frac{V_s}{I_{o,\text{rms}}} = \frac{230}{10} = 23\,\Omega$$

Q2.14.2 NAT 2014 2M Ans: 69.31 ● ○ ○

For a single-phase semi-converter,

$$V_o = \frac{V_m}{2\pi}(1+\cos\alpha)$$

Given,

$$V_o = 70\,\text{V}, \qquad V_m = 325\,\text{V}$$

$$70 = \frac{325}{2\pi}(1+\cos\alpha)$$

$$\frac{70\times 2\pi}{325} - 1 = \cos\alpha$$

$$\alpha = \cos^{-1}\left(\frac{70 \times 2\pi}{325} - 1\right)$$

$$= \cos^{-1}(0.353) = 69.3°$$

$$\boxed{69.3°}$$

Mistakes to be avoided

- Use peak voltage V_m, not RMS voltage, in the formula.

Q2.14.3 NAT 2014 2M Ans: 0.78 ● ○ ○

Input power factor is

$$\text{P.F.} = \frac{I_{s1}}{I_s}\cos\phi$$

For this converter,

$$\text{P.F.} = \frac{2\sqrt{2}}{\pi}\cos\alpha$$

Given,

$$\alpha = 30°$$

$$\text{P.F.} = \frac{2\sqrt{2}}{\pi}\cos 30°$$

$$= \frac{2 \times 1.4142}{3.1416} \times 0.866$$

$$= 0.7797 \approx 0.78$$

$$\boxed{0.78}$$

Key Points

- Input power factor is the product of distortion factor and displacement factor.
- For controlled rectifiers, power factor decreases with increasing firing angle.

Mistakes to be avoided

- Do not forget to include both harmonic distortion and phase displacement effects.

Q2.14.4 NAT 2014 2M Ans: 17.07 ● ● ○

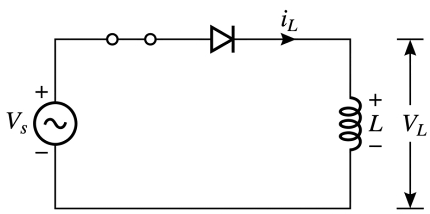

Fig. Solution Q2.14.4

Applying KVL,

$$V_s = L\frac{di}{dt}$$

Given,

$$V_s = V_m \sin(\omega t)$$

Hence,

$$\frac{V_m}{\omega L}\int \sin(\omega t)\, d(\omega t) = \int di$$

$$i(t) = \frac{V_m}{\omega L}[-\cos(\omega t)] + K$$

Substituting,

$$\frac{V_m}{\omega L} = \frac{100}{100\pi \times 31.83 \times 10^{-3}} = 10$$

So,

$$i(t) = 10[-\cos(\omega t)] + K$$

Given $i = 0$ at $t = 0.0025$ s,

$$0 = 10[-\cos(100\pi \times 0.0025)] + K$$

$$K = 7.07$$

Thus,

$$i(t) = 10[-\cos(\omega t)] + 7.07$$

Maximum current occurs at

$$\omega t = \pi$$

$$I_{\max} = 10[-\cos\pi] + 7.07$$

$$= 10(1) + 7.07$$

$$= 17.07\ \text{A}$$

$$\boxed{17.07}$$

Key Points

- The integration constant determines the DC offset in current.

Q2.14.5 MCQ 2014 2M Ans: B ● ● ●

Assume all secondary windings are identical and have the same resistance.

When phase R of the primary has the maximum voltage, phase R of the secondary also has the maximum voltage. Hence, lines a and c conduct. The current enters through terminal a and leaves through terminal c.

The current in phase R of the secondary is

$$(I_R)_{\text{sec}} = \frac{2R}{R+2R} \times 1 = \frac{2}{3}$$

Using the transformer turns ratio $1 : K$,

$$(I_R)_{\text{pri}} = K(I_R)_{\text{sec}} = \frac{2K}{3}$$

Similarly, when phase Y has the maximum voltage, lines b and c conduct. Applying current division,

$$(I_R)_{\text{sec}} = \frac{R}{R+2R} \times 1 = \frac{1}{3}$$

$$(I_R)_{\text{pri}} = K(I_R)_{\text{sec}} = \frac{K}{3}$$

Hence, the correct option is

$$\boxed{B}$$

Key Points

- During six-pulse rectifier operation, only two secondary phases conduct at any instant.
- Secondary winding currents are obtained using current division.
- Primary current is related to secondary current by the transformer turns ratio.

Q2.13.1 MCQ 2013 1M Ans: A ● ○ ○

Given,

$$v(t) = 30\sin(100t) + 10\cos(300t) + 6\sin\left(500t + \frac{\pi}{4}\right)$$

The angular frequencies present are

$$100, \quad 300, \quad 500 \text{ rad/s}$$

The fundamental angular frequency is the greatest common divisor:

$$\omega_0 = \gcd(100, 300, 500) = 100 \text{ rad/s}$$

$$\boxed{A}$$

Q2.13.2 MCQ 2013 1M Ans: B ● ○ ○

For the given diode network, conduction occurs only during the positive half-cycle.
When

$$v_{YZ} = 1000\sin\omega t > 0$$

the diodes conduct, so

$$v_{WX} = 1000\sin\omega t$$

When

$$v_{YZ} = 1000 \sin \omega t < 0$$

the diodes are reverse-biased, hence

$$v_{WX} = 0$$

Therefore,

$$v_{WX} = \begin{cases} 1000 \sin \omega t, & \sin \omega t > 0 \\ 0, & \sin \omega t < 0 \end{cases}$$

which can be written as

$$v_{WX} = 1000 \left(\frac{|\sin \omega t| + \sin \omega t}{2} \right)$$

$$\boxed{B}$$

Key Points

- Ideal diodes conduct only when forward biased.
- The circuit behaves as a positive half-wave rectifier.

Mistakes to be avoided

- Check diode orientation carefully before deciding the conducting interval.

Q2.13.3 MCQ 2013 2M Ans: C ● ● ●

Since the initial conditions are zero, the current in the LC oscillator is

$$i(t) = V\sqrt{\frac{C}{L}} \sin \omega_o t,$$

where

$$\omega_o = \frac{1}{\sqrt{LC}}.$$

After $\omega_o t = \pi$, the current reverses its direction. Since the thyristor is a unidirectional device, it cannot conduct reverse current and turns OFF.

Hence, the thyristor conducts only up to

$$\omega_o t = \pi$$

or

$$t = \frac{\pi}{\omega_o} = \pi\sqrt{LC}$$

$$= \pi\sqrt{\left(\frac{100}{\pi} \times 10^{-6}\right)\left(\frac{100}{\pi} \times 10^{-6}\right)} = 100\,\mu\text{s}$$

$$\boxed{100\,\mu\text{s}}$$

Key Points

- Natural resonant frequency:
$$\omega_o = \frac{1}{\sqrt{LC}}.$$
- The thyristor turns OFF when the current attempts to reverse.
- Conduction time of the thyristor equals one-half of the resonant period:
$$t = \pi\sqrt{LC}.$$

Q2.12.1 MCQ 2012 1M Ans: D ● ○ ○

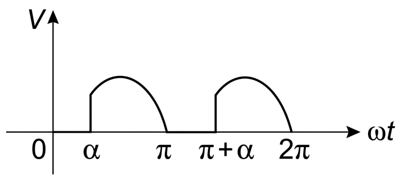

Fig. Solution Q2.12.1

The freewheeling diode conducts during the intervals when the output voltage is zero:

$$0 \rightarrow \alpha \quad \text{and} \quad \pi \rightarrow \pi + \alpha$$

Total conduction angle in one complete cycle:

$$\alpha + \alpha = 2\alpha$$

Hence, the fraction of the complete cycle is

$$\frac{2\alpha}{2\pi} = \frac{\alpha}{\pi}$$

$$\boxed{D}$$

Q2.11.1 MCQ 2011 2M Ans: B ● ● ○

The converter is used to feed power from the DC side to the AC side. Hence, the converter operates as a line-commutated inverter.

For a three-phase fully controlled bridge converter,

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha$$

For inverter operation,

$$\alpha > 90^\circ$$

and the average output voltage must be negative.

The current through the battery is

$$I_o = \frac{V_o - (-400)}{R} = \frac{V_o + 400}{R}$$

For maximum current, V_o should have the minimum negative value. This occurs when

$$\alpha = 90^\circ$$

At $\alpha = 90°$,

$$V_o = 0$$

Hence,

$$(I_o)_{\max} = \frac{400}{10} = 40\text{ A}$$

$$\boxed{B}$$

Key Points

- In line-commutated inverter operation, the converter transfers power from DC side to AC side.
- For inversion, the firing angle must satisfy $\alpha > 90°$.
- Thyristor current direction cannot reverse, so voltage polarity must reverse for power reversal.

Q2.11.2 | MCQ | 2011 | 2M | Ans: C | ● ○ ○

RMS value of supply current in case of a three-phase bridge converter is

$$I_s = I_o\sqrt{\frac{2}{3}}$$

Given,

$$I_o = 40\text{ A}$$

Therefore,

$$I_s = 40\sqrt{\frac{2}{3}}$$

$$I_s = 32.66\text{ A}$$

The kVA rating of the input transformer is

$$S = \sqrt{3}V_s I_s$$

$$S = \sqrt{3} \times 400 \times 32.66 \times 10^{-3}\text{ kVA}$$

$$S = 22.6\text{ kVA}$$

$$\boxed{C}$$

Key Points

- For a three-phase bridge converter with constant DC current, the supply current rms value is $I_s = I_o\sqrt{\frac{2}{3}}$.
- Three-phase transformer rating is $S = \sqrt{3}V_s I_s$.
- While calculating kVA, use line voltage and line current on the AC side.

Q2.11.3 | MCQ | 2011 | 2M | Ans: B | ● ○ ○

The active power is contributed only by the fundamental components of voltage and current. Since the supply voltage contains only the fundamental component, the active power is

$$P = V_{\text{rms}} I_{1,\text{rms}} \cos\phi$$

$$P = 100 \times 10 \times \cos\left(\frac{\pi}{3}\right) = 500\text{ W}$$

The total RMS current is

$$I_{rms} = \sqrt{10^2 + 5^2 + 2^2} = \sqrt{129}\text{ A}$$

The RMS value of the supply voltage is

$$V_{rms} = 100\text{ V}$$

Hence, the input power factor is

$$\text{Power Factor} = \frac{P}{V_{rms} I_{rms}} = \frac{500}{100 \times \sqrt{129}} = 0.44$$

$$\boxed{B}$$

Key Points

- Only the fundamental component of current contributes to active power when the supply voltage is sinusoidal.
- Total RMS current is obtained by the root-sum-square (RSS) of all harmonic RMS components.
- Input power factor is given by

$$\text{PF} = \frac{P}{V_{rms} I_{rms}}.$$

Mistakes to be avoided

- Do not use only the fundamental current while calculating apparent power.
- Harmonic currents increase RMS current but do not contribute to active power.
- Do not add harmonic currents arithmetically; use the RSS method.

Q2.11.4 MCQ 2011 2M Ans: B ● ○ ○

The active power drawn by the converter is contributed only by the fundamental component of the current.

$$P = V_{rms} I_{1,rms} \cos\phi$$

Given,

$$V_{rms} = 100\text{ V}, \quad I_{1,rms} = 10\text{ A}, \quad \phi = \frac{\pi}{3}$$

Hence,

$$P = 100 \times 10 \times \cos\left(\frac{\pi}{3}\right)$$

$$P = 100 \times 10 \times \frac{1}{2} = 500\text{ W}$$

$$\boxed{B}$$

Q2.10.1 MCQ 2010 1M Ans: A ● ○ ○

The average output voltage of a single-phase full-wave controlled rectifier is

$$V_o = \frac{2V_m}{\pi} \cos\alpha$$

When $\alpha = 0°$,

$$V_o = \frac{2V_m}{\pi} \cos 0° = 300\text{ V}$$

Hence,

$$\frac{2V_m}{\pi} = 300$$

When $\alpha = 60°$,

$$V_o = \frac{2V_m}{\pi}\cos 60°$$

Substituting $\frac{2V_m}{\pi} = 300$,

$$V_o = 300\cos 60°$$

$$V_o = 300 \times \frac{1}{2} = 150\text{ V}$$

$$\boxed{A}$$

Q2.09.1 MCQ | 2009 | 2M | Ans: C | ● ● ●

Given,

$$f = 50\text{ Hz}$$

Hence, the time period of the source voltage is

$$T = \frac{1}{f} = \frac{1}{50} = 20\text{ ms}$$

During the positive half cycle of the source voltage,

$$0 < t < \frac{T}{2},$$

energy is stored in the inductor and the current increases.
During the negative half cycle of the source voltage,

$$\frac{T}{2} \leq t \leq T,$$

the current decreases and the energy stored in the inductor is delivered back to the source.
The current through a purely inductive load is

$$i(t) = \frac{V_m}{\omega L}(1 - \cos\omega t)$$

Hence, the maximum current is

$$I_{\max} = \frac{2V_m}{\omega L}$$

Substituting the given values,

$$I_{\max} = \frac{2 \times 10}{100\pi \times \left(\frac{0.1}{\pi}\right)}$$

$$I_{\max} = 2\text{ A}$$

$$\boxed{C}$$

Key Points

- For a purely inductive load, current lags the applied voltage by 90°.
- The current expression is

$$i(t) = \frac{V_m}{\omega L}(1 - \cos \omega t) .$$

- The maximum current occurs at $\omega t = \pi$, giving

$$I_{\max} = \frac{2V_m}{\omega L} .$$

- An ideal inductor stores energy during one half cycle and returns it to the source during the next half cycle.

Q2.08.1 MCQ 2008 2M Ans: A ● ● ○

If the firing pulses are suddenly removed while the load current is continuous, the thyristor that is conducting at that instant remains latched because the load current is greater than the holding current.

The diodes continue to commutate naturally in every half cycle depending on which one is forward biased.

If the diode belonging to the same bridge leg as the conducting thyristor conducts, the output voltage becomes zero.

If the diode from the opposite bridge leg conducts, the supply voltage appears directly across the load. Hence, the output voltage follows the input voltage during that half cycle.

Therefore, the output waveform is as shown below.

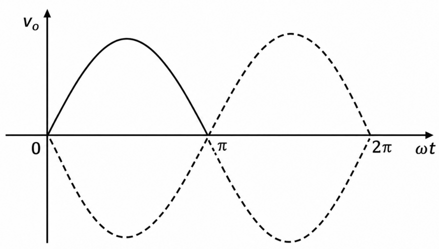

Fig. Solution Q2.08.1

$\boxed{A}$

Key Points

- An SCR cannot be turned OFF by removing the gate pulse alone; it continues to conduct until its current falls below the holding current.
- With continuous load current, the conducting SCR remains latched after gate pulse removal.
- Natural commutation of the diodes continues every half cycle.
- Depending on the conducting diode, the output is either 0 or equal to the instantaneous input voltage.

Mistakes to be avoided

- Do not assume that removing the gate pulse immediately turns OFF the SCR.
- Do not neglect the effect of continuous load current in maintaining SCR conduction.
- Do not confuse diode natural commutation with forced commutation of the SCR.
- Remember that the output alternates between zero and the input waveform depending on the conducting diode path.

Q2.08.2 MCQ | 2008 | 2M | Ans: C

The emf of the motor is shown with opposite polarity, so the dc output voltage is negative. This results in negative output power. Hence, power flows from the dc side to the ac side during regenerative braking.

$$\frac{2V_m}{\pi}\cos\alpha = -150 + 20$$

$$\frac{2\times 230\times\sqrt{2}}{\pi}\cos\alpha = -130$$

$$\cos\alpha = -0.62$$

$$\alpha = 129^\circ$$

$$\boxed{C}$$

Key Points

- For a single-phase fully controlled bridge converter,

$$V_{dc} = \frac{2V_m}{\pi}\cos\alpha$$

- In regenerative braking, power flows from the dc side to the ac side.
- Regeneration requires negative average dc output voltage with positive current direction.

Mistakes to be avoided

- Do not take the motor emf polarity as positive when it is shown opposite to the converter output.
- Do not use rms voltage directly as V_m; use $V_m = \sqrt{2}V_{rms}$.
- Do not ignore the resistance drop in the dc circuit.

Q2.08.3 MCQ | 2008 | 2M | Ans: B

Load current is

$$I_o = 10\text{ A}$$

Each phase of the source conducts for 240° out of 360°. Hence, the rms value of the source current is

$$I_s = I_o\sqrt{\frac{240}{360}}$$

$$I_s = 10\sqrt{\frac{2}{3}} = 8.165\text{ A}$$

The rms value of the fundamental component of the source current is

$$I_{s1} = \frac{4I_o}{\sqrt{2}\pi}\sin\frac{\pi}{3}$$

$$I_{s1} = \frac{4\times 10}{\sqrt{2}\pi}\times\frac{\sqrt{3}}{2} = 7.8\text{ A}$$

The total harmonic distortion is

$$\text{THD} = \sqrt{\left(\frac{I_s}{I_{s1}}\right)^2 - 1} \times 100\%$$

$$= \sqrt{\left(\frac{8.165}{7.8}\right)^2 - 1} \times 100\% = 31\%$$

$\boxed{B}$

Q2.08.4 MCQ 2008 2M Ans: B

The input power factor (IPF) is given by

$$\text{IPF} = g \cos\alpha$$

where g is the distortion factor.
Substituting the given values,

$$\text{IPF} = \frac{2\sqrt{2}}{\pi} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{6}}{\pi}$$

$$= \frac{\sqrt{6}}{\pi} = 0.78$$

$\boxed{B}$

Q2.07.1 MCQ 2007 1M Ans: C

The fundamental input power factor is given by

$$\text{Fundamental Power Factor} = \cos\left(\alpha + \frac{\mu}{2}\right)$$

Substituting the given values,

$$= \cos\left(25^\circ + \frac{10^\circ}{2}\right)$$

$$= \cos 30^\circ$$

$$= 0.866$$

$\boxed{C}$

Q2.07.2 MCQ 2007 1M Ans: C ● ○ ○

Let the dc link current be I_d.

The dc voltage applied to the inverter is

$$V_d = 420\,\text{V}$$

Power fed to the inverter is

$$P = V_d I_d = 50\,\text{kW}$$

$$420 I_d = 50 \times 10^3$$

$$I_d = 119.05\,\text{A}$$

Each thyristor conducts for a period of

$$\frac{2\pi}{3}.$$

Therefore, the rms current of each thyristor is

$$(I_{th})_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi/3} I_d^2\, d(\omega t)}$$

$$(I_{th})_{rms} = \frac{I_d}{\sqrt{3}} = \frac{119.05}{\sqrt{3}} = 68.73\,\text{A}$$

$$\boxed{C}$$

Q2.07.3 MCQ 2007 1M Ans: C ● ○ ○

A free-wheeling diode is required to prevent short-circuiting, which can occur if the current from one set of thyristors is not completely transferred before the other set of thyristors starts conducting.

$$\boxed{C}$$

Key Points

- A free-wheeling diode provides an alternate path for the load current during commutation.
- It prevents simultaneous conduction of two thyristor groups, thereby avoiding source short-circuit.
- It improves commutation reliability and reduces voltage spikes across the thyristors.

Q2.07.4 MCQ 2007 2M Ans: A ● ● ○

Back emf of the motor is

$$E_a = V_o - I_a R_a$$

Here, V_o is the average output voltage of the converter. Since losses are neglected,

$$R_a = 0$$

Therefore,

$$E_a = V_o$$

Also,

$$V_o = E_a = K\phi N$$

At 440 V dc voltage, speed is 1500 rpm. At half the rated speed,

$$N = 750\,\text{rpm}$$

Hence,

$$V_o = 220\,\text{V}$$

If I_a is the average armature current, then rms value of supply current is

$$I_s = I_a\sqrt{\frac{2}{3}}$$

Power delivered to the motor is

$$P = V_o I_a$$

Input volt-ampere is

$$\text{Input VA} = \sqrt{3}V_s I_s$$

Therefore, input power factor is

$$\text{Input power factor} = \frac{V_o I_a}{\sqrt{3}V_s I_s}$$

$$= \frac{220 I_a}{\sqrt{3} \times 440 \times I_a\sqrt{\frac{2}{3}}}$$

$$= 0.354$$

$\boxed{A}$

Key Points

- With $R_a = 0$, motor back emf is equal to the converter average output voltage.
- For constant flux,
$$E_a \propto N.$$
- For this converter,
$$I_s = I_a\sqrt{\frac{2}{3}}.$$
- Input power factor is
$$\text{IPF} = \frac{\text{Output power}}{\text{Input VA}}.$$

Q2.07.5 MCQ 2007 2M Ans: D ● ● ○

The input to the converter is

$$V_s = \frac{230}{4} = 57.5\,\text{V}$$

The diode conducts when

$$V_s \geq E.$$

Hence, the diode starts conducting at θ_1 such that

$$V_m \sin\theta_1 = E$$

$$57.5\sqrt{2}\sin\theta_1 = 12$$

$$\theta_1 = 8.486^\circ = 0.148\,\text{rad}$$

The average output current is

$$I_o = \frac{1}{2\pi}\int_{\theta_1}^{\pi-\theta_1}\left(\frac{V_m \sin\omega t - E}{R}\right) d(\omega t)$$

$$I_o = \frac{1}{2\pi R}\left[2V_m \cos\theta_1 - E(\pi - 2\theta_1)\right]$$

$$= \frac{1}{2\pi \times 19.04}\left[2 \times 57.5\sqrt{2} \times \cos 8.486^\circ - 12\,(\pi - 2 \times 0.148)\right]$$

$$I_o = 1.06\,\text{A} \approx 1\,\text{A}$$

$$\boxed{D}$$

Key Points

- Diode conduction starts when the instantaneous source voltage equals the opposing emf.
- Conduction angle is from θ_1 to $\pi - \theta_1$.
- The average output current is obtained by integrating the load current over the conduction interval.

Mistakes to be avoided

- Do not assume conduction starts at 0°; it starts when $V_m \sin\theta_1 = E$.
- Use θ_1 in radians while evaluating the integral term $(\pi - 2\theta_1)$.
- Do not forget to divide the integral by 2π to obtain the average current.

Q2.07.6 MCQ 2007 2M Ans: D ● ○ ○

For a pure inductive load, the load current is

$$I = \frac{V_m}{\omega L}(1 - \cos\omega t)$$

The diode stops conducting when the current through it becomes zero, i.e.,

$$I = 0$$

$$1 - \cos \omega t = 0$$

$$\cos \omega t = 1$$

$$\omega t = 360^\circ$$

Hence, during the positive half cycle the diode current rises from zero to a maximum value, and during the negative half cycle it falls back to zero.

$$\boxed{D}$$

Key Points

- The diode continues conducting until the stored energy in the inductor becomes zero.
- Conduction ceases when the current through the diode becomes zero.

Mistakes to be avoided

- Do not assume diode conduction ends at 180°; for a pure inductive load it continues until 360°.
- Set the current equal to zero, not the voltage, to determine the extinction angle.

Q2.06.1 MCQ 2006 1M Ans: D ● ● ○

For

$$0 \leq \omega t \leq \pi,$$

diode D_1 conducts and diode D_2 is reverse biased.
For

$$\pi < \omega t < 2\pi,$$

diode D_1 becomes reverse biased and diode D_2 conducts.
Hence, during

$$\pi < \omega t < 2\pi,$$

the load current freewheels through D_2. Therefore, the diode D_2 current remains zero during

$$0 \leq \omega t \leq \pi,$$

and is equal to the load current during

$$\pi < \omega t < 2\pi.$$

Hence, the diode current waveform is as shown below.

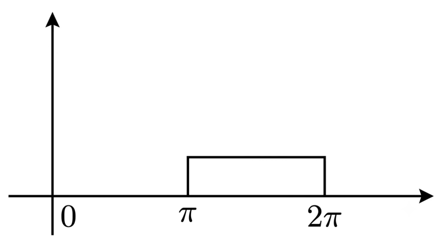

Fig. Solution Q2.06.1

$$\boxed{D}$$

Key Points

- D_1 conducts during the positive half cycle, while D_2 remains reverse biased.
- During the negative half cycle, the load current freewheels through D_2.
- The current through the freewheeling diode is equal to the load current during the freewheeling interval.

Mistakes to be avoided

- Do not assume both diodes conduct simultaneously.
- Do not forget that the freewheeling diode conducts only after the source voltage becomes negative.
- Distinguish between load current and diode current while sketching the waveform.

Q2.06.2 MCQ 2006 2M Ans: C ● ○ ○

For a three-phase fully controlled bridge converter, the input displacement factor is

$$\cos\alpha$$

Given,

$$\alpha = 60^\circ$$

$$\text{Input displacement factor} = \cos\alpha = \cos 60^\circ = 0.5$$

The input power factor is

$$\text{Input power factor} = \text{Distortion factor} \times \cos\alpha$$

For a three-phase fully controlled bridge converter, the distortion factor is

$$\frac{3}{\pi}$$

Therefore,

$$\text{Input power factor} = \frac{3}{\pi} \times 0.5$$

$$= 0.478$$

$$\boxed{C}$$

Key Points

- Input power factor is the product of distortion factor and displacement factor.
- For a three-phase fully controlled bridge converter,

$$\text{Displacement factor} = \cos\alpha.$$

- Distortion factor for this converter is

$$\frac{3}{\pi}.$$

Q2.06.3 MCQ | 2006 | 2M | Ans: A | ● ● ○

Given,

$$E = 350\,\text{V}$$

$$I_{dc} = 20\,\text{A}$$

$$R_{cell} = 0.5\,\Omega$$

Since the power flow is reversed, the power must come out of the source. Hence, the converter operates in inversion mode. The thyristor cannot carry negative current, therefore the output voltage must be negative. Consequently, the firing angle must be greater than 90°.

The output voltage is

$$V_o = -(E - I_{dc}R_{cell}) = -(350 - 20 \times 0.5) = -340\,\text{V}$$

For a three-phase full converter,

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha$$

Substituting,

$$\frac{3V_{ml}}{\pi}\cos\alpha = -340 = \frac{3 \times 440 \times \sqrt{2} \times \cos\alpha}{\pi}$$

Hence,

$$\alpha = 125^\circ$$

Since

$$\alpha > 60^\circ,$$

the thyristor will be reverse biased for a duration of

$$180^\circ - 125^\circ = 55^\circ.$$

$\boxed{A}$

Key Points

- In inversion mode, the average DC output voltage is negative while the current remains positive.
- For a three-phase fully controlled converter,

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha.$$

- Inversion operation requires $\alpha > 90^\circ$.

Mistakes to be avoided

- Do not assume the DC current becomes negative during inversion; only the average output voltage is negative.
- Include the internal cell resistance drop $I_{dc}R_{cell}$ while calculating the effective DC voltage.

Q2.06.4 MCQ 2006 2M Ans: C ● ● ●

Given,

$$\theta_1 = \sin^{-1}\left(\frac{E}{V_m}\right) = \sin^{-1}\left(\frac{200}{230\sqrt{2}}\right)$$

Therefore,

$$\theta_1 = 38^\circ \text{ or } 0.66 \text{ radian}$$

For a RE Load,

$$I_{avg} = \frac{1}{2\pi R_L}\left[2V_m \cos\theta_1 - E\,(\pi - 2\theta_1)\right]$$

Substituting the given values,

$$I_{avg} = \frac{1}{2\pi \times 2}\left[2\sqrt{2} \times 230 \cos 38^\circ - 200\,(\pi - 2 \times 0.66)\right]$$

$$I_{avg} = 11.9 \text{ A}$$

$$\boxed{C}$$

Key Points

- For an RE load, conduction starts when the source voltage exceeds the back EMF, i.e., $\theta_1 = \sin^{-1}(E/V_m)$.
- The average load current is obtained from the average output voltage divided by the load resistance R_L.

Q2.05.1 MCQ 2005 1M Ans: B ● ○ ○

Maximum value of input line voltage is

$$V_m = 400\sqrt{2}\text{ V}$$

If V_{AB} is taken as reference, then for diode bridge rectifier,

$$V_o = V_{AB} \qquad \text{for } 60^\circ \leq \omega t \leq 120^\circ$$

Since,

$$V_{AB} = 400\sqrt{2}\sin\omega t$$

Hence, maximum output voltage occurs at

$$\omega t = 90^\circ$$

Therefore, peak instantaneous output voltage is

$$V_m = 400\sqrt{2}\text{ V}$$

$$\boxed{\text{B}}$$

Key Points

- In a three-phase diode bridge rectifier, the output voltage is equal to the maximum instantaneous line-to-line voltage.

Q2.05.2 MCQ | 2005 | 2M | Ans: B | ● ● ○

Given,

$$V_{\text{rms}} = 230\,\text{V}$$

Therefore,

$$V_m = 230\sqrt{2}\,\text{V}$$

For $\alpha > 90°$, the peak output voltage is

$$V_{o,\text{peak}} = V_m \sin\alpha$$

Given,

$$V_{o,\text{peak}} = 230\,\text{V}$$

$$230 = 230\sqrt{2}\sin\alpha$$

$$\sin\alpha = \frac{1}{\sqrt{2}}$$

Since firing angle is in the second quadrant,

$$\alpha = 135°$$

$$\boxed{\text{B}}$$

Key Points

- For a half-wave controlled rectifier, if $\alpha < 90°$, the peak output is V_m.
- If $\alpha > 90°$, the peak output occurs at firing instant and equals $V_m \sin\alpha$.

Mistakes to be avoided

- Do not take 230 V as the peak value of the input voltage; it is RMS.
- Do not choose $\alpha = 45°$ because firing angle must be greater than 90° for reduced peak output.

Q2.04.2 MCQ | 2004 | 1M | Ans: C | ● ○ ○

Ripple frequency is given by

$$f_r = (\text{Number of pulses}) \times (\text{Supply frequency})$$

A three-phase half-wave rectifier produces 3 pulses in every cycle.
Hence,

$$f_r = 3 \times 400$$

$$f_r = 1200\,\text{Hz}$$

$$\boxed{\text{C}}$$

Q2.03.1 MCQ 2003 1M Ans: A

If V_{AB} is chosen as reference, then it conducts from 60° to 120° for zero firing angle delay.

$$V_{AB} = V_m \sin \omega t$$

For a firing angle delay of 30°, conduction occurs from

$$90^\circ \leq \omega t \leq 150^\circ$$

The maximum voltage occurs at

$$\omega t = 90^\circ$$

and the minimum voltage occurs at

$$\omega t = 150^\circ$$

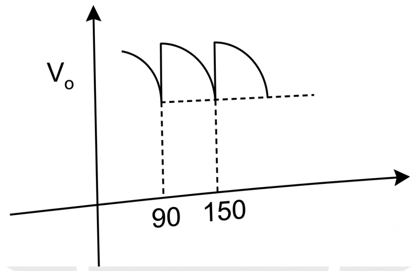

Fig. Solution Q2.03.1

Hence,

$$\frac{\text{Peak-to-peak ripple voltage}}{\text{Peak output dc voltage}} = \frac{V_m - V_m \sin 150^\circ}{V_m}$$

$$= \frac{1 - \frac{1}{2}}{1} = 0.5$$

$\boxed{\text{A}}$

Key Points

- For a three-phase controlled rectifier, firing delay shifts the conduction interval by angle α.
- With V_{AB} as reference, conduction shifts from 60°–120° to 90°–150° when $\alpha = 30^\circ$.
- Peak-to-peak ripple is the difference between maximum and minimum output voltage in the conduction interval.

Mistakes to be avoided

- Do not take the conduction interval as 60° to 120° after applying firing delay.

Q2.03.2 MCQ 2003 2M Ans: A

As current source is connected as load, the load current is constant and it continuously flows through pair T_1, T_2 or T_3, T_4.

This is a case of continuous conduction.

T_1 and T_2 are fired at

$$\alpha = 30^\circ.$$

These SCRs will get turned ON as the firing angle is 30°. Pair T_3, T_4 will be fired at

$$(180^\circ + \alpha).$$

So, T_1 and T_2 conduct for

$$\alpha < \omega t < \pi + \alpha$$

and

$$V_{dc} = V_{ac}.$$

Similarly, T_3 and T_4 conduct for

$$\pi + \alpha < \omega t < 2\pi + \alpha$$

and

$$V_{dc} = -V_{ac}.$$

Thus, the waveform is as shown.

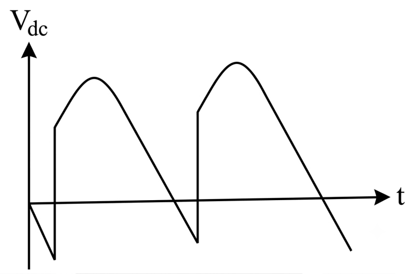

Fig. Solution Q2.03.2

$\boxed{\text{A}}$

Key Points

- A current-source load maintains continuous load current, resulting in continuous conduction.
- Each SCR pair conducts for 180° before natural commutation transfers current to the next pair.
- During conduction of T_1, T_2, the output voltage equals the input voltage, whereas during conduction of T_3, T_4, the output voltage is its negative.

Q2.02.2 MCQ 2002 1M Ans: C ● ○ ○

Triplen harmonics are not present in the AC source line current.

Hence, the lowest frequency harmonic component is the fifth harmonic.

$$f_h = 5f$$

Given,

$$f = 50\,\text{Hz}$$

Therefore,

$$f_h = 5 \times 50$$

$$f_h = 250\,\text{Hz}$$

$\boxed{\text{C}}$

Q2.02.3 MCQ 2002 2M Ans: B ● ● ○

The displacement power factor is given by

$$\cos\alpha$$

Hence, the reactive power is

$$Q \propto \tan\alpha$$

As the firing angle α increases, the reactive power drawn by the converter increases.
Therefore, the reactive power is maximum when

$$\alpha = 30^\circ$$

Hence, the correct option is

$$\boxed{\text{B}}$$

Key Points

- The displacement power factor of a line-commutated converter is $\cos\alpha$.
- Reactive power is proportional to $\tan\alpha$.
- Higher firing angle gives higher reactive power demand.

Q2.02.4 MCQ 2002 2M Ans: C ● ○ ○

For a single-phase diode bridge rectifier with resistive load, the output voltage is the full-wave rectified form of the input voltage.

Given,

$$v = 200\sin(\omega t)$$

Therefore,

$$V_m = 200\text{ V}$$

The RMS value of the load voltage is

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

Power dissipated in the load resistor is

$$P = \frac{V_{\text{rms}}^2}{R}$$

$$P = \frac{\left(\frac{200}{\sqrt{2}}\right)^2}{50}$$

$$P = \frac{20000}{50}$$

$$P = 400\text{ W}$$

$$\boxed{\text{C}}$$

Q2.01.1 MCQ | 2001 | 2M | Ans: A | ● ● ○

When the thyristor is ON,

$$V_m \sin \omega t = L\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{V_m}{L}\sin \omega t$$

Integrating from firing angle α to ωt,

$$\int_0^i di = \frac{V_m}{\omega L}\int_\alpha^{\omega t} \sin(\omega t)\, d(\omega t)$$

$$i = -\frac{V_m}{\omega L}\cos \omega t\Big|_\alpha^{\omega t}$$

$$i = \frac{V_m}{\omega L}(\cos\alpha - \cos\omega t)$$

For extinction angle β,

$$i = 0$$

$$0 = \frac{V_m}{\omega L}(\cos\alpha - \cos\beta)$$

$$\cos\alpha = \cos\beta$$

$$\beta = 2n\pi \pm \alpha = 360^\circ n \pm \alpha$$

For

$$\alpha = 120^\circ,$$

$$\beta = 240^\circ,\ 480^\circ$$

Since β cannot exceed 360°,

$$\boxed{A}$$

Key Points

- During SCR conduction with a pure inductive load, the source voltage appears entirely across the inductor.
- The load current is obtained by integrating the inductor voltage equation

$$V = L\frac{di}{dt}.$$

- Extinction occurs when the inductor current becomes zero, i.e., $i(\beta) = 0$.

Q2.01.2 MCQ 2001 1M Ans: A ● ○ ○

For a circulating current dual converter, both converters operate simultaneously.

To avoid short circuit and to maintain equal average DC output voltages with opposite polarities,

$$V_{dc1} + V_{dc2} = 0$$

For converter 1,

$$V_{dc1} = V_{d0} \cos \alpha_1$$

For converter 2,

$$V_{dc2} = V_{d0} \cos \alpha_2$$

Hence,

$$V_{d0} \cos \alpha_1 + V_{d0} \cos \alpha_2 = 0$$

$$\cos \alpha_1 + \cos \alpha_2 = 0$$

$$\cos \alpha_1 = -\cos \alpha_2$$

$$\alpha_1 + \alpha_2 = 180^\circ$$

$\boxed{\text{A}}$

Key Points

- In a circulating current dual converter, both converters conduct simultaneously.
- Average output voltages of the two converters must be equal and opposite.
- The firing angle relation is
$$\alpha_1 + \alpha_2 = 180^\circ.$$

Q2.00.1 MCQ 2000 1M Ans: B ● ○ ○

If the current drops to zero at any instant, the converter operates in discontinuous conduction mode.

Applying KVL,

$$V = E_b + I_a R_a$$

At the instant when the armature current becomes zero,

$$I_a = 0$$

Hence,

$$V = E_b$$

Therefore, the supply voltage is equal to the back EMF.

$\boxed{\text{B}}$

Q2.00.2 MCQ 2000 1M Ans: C

For a three-phase fully controlled converter feeding a constant current load,

$$I_{dc} = I_d = \text{constant}$$

Each thyristor conducts for 120°, and the input line current is a rectangular waveform having magnitude

$$\pm I_d.$$

Hence, the peak value of the AC input line current is

$$I_{\text{line,peak}} = I_d.$$

Hence, the correct option is

$$\boxed{\text{C}}$$

Key Points

- In a three-phase fully controlled bridge with continuous current, each line current is a rectangular waveform of amplitude $\pm I_d$.
- Each thyristor conducts for 120°.
- The peak value of the AC line current is equal to the DC load current.
- The RMS value of the line current is

$$I_{\text{line,rms}} = \sqrt{\frac{2}{3}}\, I_d,$$

and the fundamental RMS component is

$$I_{1,\text{rms}} = \frac{\sqrt{6}}{\pi} I_d.$$

Mistakes to be avoided

- Do not assume the RMS value of the line current equals the DC current.
- Do not confuse the peak value with the average value of the AC line current.
- Remember that the average value of the AC line current over one cycle is zero because it contains equal positive and negative intervals.

Q2.00.3 NAT 2000 5M Ans: *

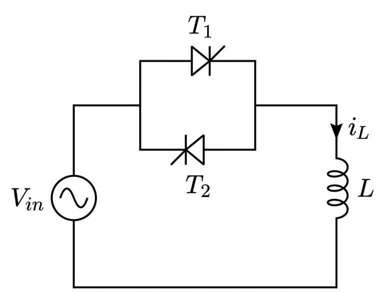

Fig. Solution Q2.00.3

When the positive thyristor is fired at $\theta = \alpha$,

$$V_m \sin \omega t = L \frac{di}{dt}$$

$$di = \frac{V_m}{\omega L} \sin \omega t \, d\omega t$$

$$\int_0^i di = \frac{V_m}{\omega L} \int_\alpha^{\omega t} \sin \theta \, d\theta$$

$$i = \frac{V_m}{\omega L} (\cos \alpha - \cos \theta)$$

The current becomes zero at extinction angle β.

$$0 = \frac{V_m}{\omega L} (\cos \alpha - \cos \beta)$$

$$\cos \alpha = \cos \beta$$

Since $\alpha \geq \frac{\pi}{2}$,

$$\beta = 2\pi - \alpha$$

Hence, for the positive thyristor conduction,

$$i_L = \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t), \qquad \alpha \leq \omega t \leq 2\pi - \alpha$$

For the reverse thyristor conduction,

$$i_L = \frac{V_m}{\omega L} [-\cos \alpha - \cos \omega t], \qquad \pi + \alpha \leq \omega t \leq 3\pi - \alpha$$

Thus,

$$\boxed{i_L = \begin{cases} \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t), & \alpha \leq \omega t \leq 2\pi - \alpha \\ 0, & 2\pi - \alpha < \omega t < \pi + \alpha \\ \frac{V_m}{\omega L} (-\cos \alpha - \cos \omega t), & \pi + \alpha \leq \omega t \leq 3\pi - \alpha \end{cases}}$$

Key Points

- For a pure inductive load,
$$v_L = L \frac{di_L}{dt}.$$
- The inductor current is obtained by integrating the applied sinusoidal voltage.
- For $\alpha \geq \pi/2$, conduction is discontinuous.
- Extinction occurs when $i_L = 0$.

Mistakes to be avoided

- Do not assume continuous conduction for $\alpha \geq \pi/2$.
- Do not forget the factor ω while integrating with respect to ωt.
- Do not take the extinction angle as $\pi + \alpha$; it is $2\pi - \alpha$ for positive conduction.

Q2.99.1 MCQ 1999 1M Ans: B ● ○ ○

Resonant converters employ zero-voltage switching (ZVS) or zero-current switching (ZCS) techniques.

Under these conditions, the voltage across or the current through the switching device becomes zero during switching.

Hence, the switching loss is greatly reduced. Hence, the correct option is

$$\boxed{\text{B}}$$

Q2.99.2 NAT 1999 5M Ans: * ● ● ●

For a three-phase diode bridge rectifier,

(a) Average output voltage

$$(V_o)_{\text{avg}} = \frac{3V_{ml}}{\pi} = \frac{3 \times 400 \times \sqrt{2}}{\pi} = 540.19\,\text{V}$$

(b) Average output current

$$(I_o)_{\text{avg}} = \frac{(V_o)_{\text{avg}}}{R} = \frac{540.19}{10} = 54.01\,\text{A}$$

Each diode conducts for 120° out of 360°.

$$(I_D)_{\text{RMS}} = \sqrt{\frac{120}{360}}\,I_o = \frac{54.01}{\sqrt{3}} = 31.18\,\text{A}$$

(c) RMS source phase current

Each source phase conducts for 240° out of 360°.

$$(I_s)_{\text{RMS}} = \sqrt{\frac{240}{360}}\,I_o = \sqrt{\frac{2}{3}} \times 54.01 = 44.10\,\text{A}$$

(d) Output power

$$P = V_o I_o = \frac{(540.19)^2}{10} = 29.179\,\text{kW}$$

Apparent power drawn from mains is

$$S = \sqrt{3}\,V_{\text{rms}} I_{\text{rms}} = \sqrt{3} \times 400 \times 44.10 = 30.557\,\text{kVA}$$

These values can be used to determine the input power factor.

$$\boxed{540.19\,\text{V} \quad 31.18\,\text{A} \quad 44.10\,\text{A} \quad 30.557\,\text{kVA}}$$

Key Points

- For a three-phase diode bridge rectifier,

$$(V_o)_{\text{avg}} = \frac{3V_{LL,\text{peak}}}{\pi}.$$

- Each diode conducts for 120° in one cycle.
- Each source phase carries current for 240° in one cycle.
- Apparent input power is

$$S = \sqrt{3}\,V_{LL} I_L.$$

Q2.98.1 MCQ 1998 2M Ans: D

For a three-phase fully controlled converter operating with continuous load current, each thyristor conducts for

$$120^\circ$$

out of every

$$360^\circ.$$

Hence, the RMS current through each thyristor is

$$I_{T,\text{RMS}} = \sqrt{\frac{120}{360}}\, I_d$$

Given,

$$I_d = 150\,\text{A}$$

Therefore,

$$I_{T,\text{RMS}} = \sqrt{\frac{1}{3}} \times 150 = \frac{150}{\sqrt{3}} = 86.6\,\text{A}$$

$$\boxed{\text{D}}$$

Q2.98.2 MCQ 1998 2M Ans: D

For a single-phase fully controlled rectifier operating with continuous current, the displacement power factor is

$$\cos\phi = \cos\alpha$$

Given,

$$\alpha = 30^\circ$$

Hence,

$$\cos\phi = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Hence, the correct option is

$$\boxed{\text{D}}$$

Key Points

- For a single-phase fully controlled converter with continuous load current, the input current lags the supply voltage by the firing angle α.
- Therefore, the displacement angle is $\phi = \alpha$.
- The displacement power factor is
$$\cos\phi = \cos\alpha.$$

Q2.97.1 MCQ | 1997 | 1M | Ans: C | ● ○ ○

In a circulating current dual converter, both converters remain active simultaneously.

An interconnecting reactor limits the circulating current between the converters.

Since both converters are always ready to conduct, the direction of load current can be reversed smoothly without any dead time.

Hence, the response of the drive is faster.

Hence, the correct option is

$$\boxed{\text{C}}$$

Q2.96.1 MCQ | 1996 | 1M | Ans: A | ● ○ ○

The average output voltage of a three-phase fully controlled converter considering source inductance (overlap angle μ) is

$$(V_o)_{\text{avg}} = \frac{3V_m}{2\pi}\left[\cos\alpha + \cos(\alpha+\mu)\right]$$

As the overlap angle μ increases,

$$\alpha + \mu \uparrow$$

which results in

$$\cos(\alpha+\mu)\downarrow .$$

Hence,

$$(V_o)_{\text{avg}} = \frac{3V_m}{2\pi}\left[\cos\alpha + \cos(\alpha+\mu)\right]$$

decreases with increase in overlap angle.

Therefore, the correct option is

$$\boxed{\text{A}}$$

Q2.95.1 NAT | 1995 | 5M | Ans: 7.406,3.504 | ● ● ○

For a purely resistive load, the converter operates in discontinuous conduction.

The maximum input voltage is

$$V_m = 200\sqrt{2}\text{ V}$$

The extinction angle is

$$\beta = 180^\circ - \sin^{-1}\left(\frac{E}{V_m}\right) = 180^\circ - \sin^{-1}\left(\frac{100}{200\sqrt{2}}\right) = 159.295^\circ$$

The average output current is

$$I = \frac{1}{\pi R}\int_{60^\circ}^{159.295^\circ}(V_m \sin\omega t - E)\ d(\omega t)$$

$$= \frac{1}{\pi R}\left[V_m(\cos 60^\circ - \cos 159.295^\circ) - E(159.295^\circ - 60^\circ)\frac{\pi}{180}\right]$$

$$= \frac{1}{10\pi}\left[200\sqrt{2}(\cos 60^\circ - \cos 159.295^\circ) - 100(159.295 - 60)\frac{\pi}{180}\right]$$

I=7.406 A

(ii)If a large inductor is connected, conduction becomes continuous and the diode provides freewheeling action after 180°.

The average output voltage is

$$(V_o)_{\text{avg}} = \frac{V_m}{\pi}(1 + \cos\alpha)$$

$$= \frac{200\sqrt{2}}{\pi}(1 + \cos 60^\circ) = 135.047\ \text{V}$$

Hence,

$$I = \frac{(V_o)_{\text{avg}} - E}{R} = \frac{135.047 - 100}{10} = 3.504\ \text{A}$$

$$\boxed{7.406\ \text{A} \qquad 3.504\ \text{A}}$$

Key Points

- Pure resistive load results in discontinuous conduction.
- Extinction angle is obtained from
$$V_m \sin\beta = E.$$
- With a large inductor, current becomes continuous due to freewheeling.
- For continuous conduction,
$$(V_o)_{\text{avg}} = \frac{V_m}{\pi}(1 + \cos\alpha).$$

Q2.95.2 NAT 1995 1M Ans: 594.12

For a three-phase diode bridge rectifier,

$$(V_o)_{\text{avg}} = \frac{3V_m}{\pi}$$

where

$$V_m = \sqrt{2}\,V_{LL}$$

Given,

$$V_{LL} = 440\ \text{V}$$

Hence,

$$V_m = 440\sqrt{2} = 622.25\ \text{V}$$

Therefore,

$$(V_o)_{\text{avg}} = \frac{3 \times 440\sqrt{2}}{\pi} = 594.12\ \text{V}$$

$$\boxed{594.12}$$

Q2.95.3 MCQ 1995 1M Ans: C

For a single-phase diode bridge rectifier feeding a highly inductive load,

$$I_o = \text{constant}$$

since the large inductance makes the load current ripple-free.
At any instant, two diodes conduct, and the supply current is equal to the load current in magnitude.
Hence, the input current alternates between

$$+i_o \quad \text{and} \quad -i_o,$$

forming a square waveform.
Hence, the correct option is

$$\boxed{\text{C}}$$

Q2.94.1 NAT 1994 1M Ans: False

The given statement is **False**.
A six-pulse double star rectifier is *not* the same as a three-phase full-wave rectifier.
For a six-pulse double star rectifier, the average output voltage is

$$V_{d1} = \frac{3V_m}{\pi}\cos\alpha$$

For a three-phase half-wave rectifier, the average output voltage is

$$V_{o2} = \frac{3V_m}{2\pi}\cos\alpha$$

Clearly,

$$\frac{3V_m}{\pi}\cos\alpha \neq \frac{3V_m}{2\pi}\cos\alpha$$

Hence, the output voltage of a six-pulse double star rectifier is different from that of a three-phase half-wave rectifier.
Therefore, the statement is

$$\boxed{\text{False}}$$

Q2.94.2 NAT 1994 1M Ans: True

The given statement is **True**.
A line-commutated inverter converts DC power into AC power.
It operates with a firing angle

$$\alpha > 90^\circ.$$

Hence, the average DC output voltage becomes negative while the DC current remains positive.
Therefore,

$$P = V_{dc}I_{dc} < 0,$$

which indicates that power flows from the DC side to the AC side.
Hence, the statement is

$$\boxed{\text{True}}$$

Q2.94.3 MCQ 1994 1M Ans: C ● ○ ○

In the output voltage waveform, a constant voltage appears for a finite interval after the extinction angle when the current becomes zero.

This characteristic is observed only with an RLE load (such as a DC motor) operating in discontinuous conduction.

The constant voltage interval is due to the presence of the back EMF of the DC motor after the current ceases.

Hence, the correct option is

$$\boxed{\text{C}}$$

Q2.93.1 MCQ 1993 1M Ans: C ● ○ ○

When a line-commutated converter operates in the inverter mode, it transfers real power from the DC side to the AC supply.

However, due to the firing angle delay and harmonic currents, the converter requires reactive power.

Since the DC side cannot supply reactive power, the required reactive power is drawn from the AC side.

Hence, the correct option is

$$\boxed{C}$$

DEBUG INFO

Chapter II

AC to DC Converters/Phase Controlled Rectifiers

Total Questions : 87

MCQ : 55

MSQ : 0

NAT : 32

SOLUTIONS

III

DC to DC Converters-Choppers

Topics

1. Buck Converter or Step Down Chopper with R/RL/RLE Load
2. Step Up Chopper or Boost Converter
3. Buck-Boost Converter
4. Fourier Series Analysis and Design of DC-DC Converters
5. Commutation Circuits

Q3.26.1 MCQ | 2026 | 2M | Ans: C | ● ● ○

Given that the inductor current rises for 10 μs and falls for 10 μs over a complete switching cycle, the duty cycle (D) of the converter is calculated as:

$$D = \frac{T_{\text{on}}}{T_{\text{on}} + T_{\text{off}}} = \frac{10\,\mu\text{s}}{10\,\mu\text{s} + 10\,\mu\text{s}} = 0.5$$

For an ideal boost converter operating in continuous conduction mode (CCM), the relationship between input voltage (V_{in}) and output voltage (V_{out}) is expressed as:

$$V_{\text{out}} = \frac{V_{\text{in}}}{1 - D} = \frac{V_{\text{in}}}{1 - 0.5} = 2V_{\text{in}}$$

Average inductor current is 3 A

$$I_{\text{out}} = \frac{V_{\text{out}}}{10}$$

Applying the input-to-output power balance equation ($P_{\text{in}} = P_{\text{out}}$) for an ideal lossless converter:

$$V_{\text{in}} \times I_{\text{L(avg)}} = \frac{V_{\text{out}}^2}{10}$$

Substituting $I_{\text{L(avg)}} = 3$ A and $V_{\text{out}} = 2V_{\text{in}}$ into the power balance equation:

$$V_{\text{in}} \times 3 = \frac{(2V_{\text{in}})^2}{10}$$

$$3V_{\text{in}} = \frac{4V_{\text{in}}^2}{10}$$

Assuming $V_{\text{in}} \neq 0$, we can divide both sides by V_{in}:

$$3 = \frac{4V_{\text{in}}}{10}$$

$$4V_{\text{in}} = 30 \implies V_{\text{in}} = 7.5\text{ V}$$

$$\boxed{7.5}$$

Key Points

- Boost Converter Voltage Ratio: In critical conduction mode, a boost converter steps up voltage according to the conversion ratio $V_{\text{out}} = \frac{V_{\text{in}}}{1-D}$.
- Power Conservation Principle: For any ideal DC-DC converter, average input power equals average output power ($V_{\text{in}} I_{\text{in}} = V_{\text{out}} I_{\text{out}}$).

Q3.25.1 NAT | 2025 | 1M | Ans: 90 | ● ○ ○

For a Boost Converter operating under continuous conduction mode (CCM), the relationship between the average output voltage V_0 and the input source voltage V_s is given as:

$$V_0 = \frac{V_s}{1 - \alpha}$$

Assuming an ideal lossless converter, the input power equals the output power ($P_s = P_0$), which gives the relation between voltage and current ratios:

$$\frac{V_0}{V_s} = \frac{I_s}{I_0} = \frac{1}{1 - \alpha}$$

From this, the average output current I_0 can be expressed in terms of the source current I_s and the duty cycle α:

$$I_0 = (1-\alpha)I_s$$

Given the operating parameters $\alpha = 0.4$ and $I_s = 10\,\text{A}$:

$$I_0 = (1-0.4)\times 10 = 6\,\text{A}$$

Given that the output voltage $V_0 = 15\,\text{V}$, the total output power P_0 delivered to the load is:

$$P_0 = V_0 I_0 = 15\times 6 = 90\,\text{W}$$

$$\boxed{90\,\text{W}}$$

Q3.25.2 NAT 2025 2M Ans: 1.8 ●●○

For a DC-DC Buck Converter, the given parameters yield the following step-by-step analysis:

The peak-to-peak inductor ripple current ΔI_L and capacitor ripple current ΔI_C are given as:

$$\Delta I_L = \Delta I_C = 0.2\,\text{A}$$

The switching frequency f is calculated from the total time period $T = 50\times 10^{-6}$ s:

$$f = \frac{1}{T} = \frac{1}{50\times 10^{-6}} = 20\,\text{kHz}$$

The duty cycle α is found using the turn-on time $T_{\text{ON}} = 30\,\mu$s and total time period $T = 50\,\mu$s:

$$\alpha = \frac{T_{\text{ON}}}{T} = \frac{30}{50} = 0.6$$

The fundamental formula for the peak-to-peak inductor ripple current ΔI_L in a buck converter is:

$$\Delta I_L = \frac{\alpha(1-\alpha)V_s}{fL}$$

Rearranging the equation to solve for the filter inductance L and substituting the given values ($V_s = 30\,\text{V}$):

$$L = \frac{\alpha(1-\alpha)V_s}{\Delta I_L\times f} = \frac{0.6(1-0.6)\times 30}{0.2\times 20\times 10^3}$$

$$L = 1.8\,\text{mH}$$

$$\boxed{1.8\,\text{mH}}$$

Q3.24.1 NAT 2024 1M Ans: 50 ●○○

Voltage Commutated Chopper

$$t_{\text{cm}} = \frac{C}{I_0}V_s$$

$$= \frac{10\cdot 10^{-6}}{10}\cdot 50$$

$$= 50\,\mu\text{s}$$

$$\boxed{50}$$

Q3.24.2 NAT 2024 2M Ans: 13 ● ● ○

For a DC-DC Boost Converter operating under continuous conduction mode (CCM), the average output voltage V_0 is related to the input source voltage V_s and the duty cycle α by:

$$V_0 = \frac{V_s}{1-\alpha}$$

Given parameters: Input voltage, $V_s = 20\,\text{V}$; Output voltage, $V_0 = 40\,\text{V}$; Load resistance, $R = 10\,\Omega$; Switching frequency, $f = 500\,\text{Hz}$

Substituting the voltages to determine the operating duty cycle α:

$$40 = \frac{20}{1-\alpha} \implies 1-\alpha = \frac{20}{40} = 0.5$$

$$\alpha = 0.5$$

The average output current I_0 is:

$$I_0 = \frac{V_0}{R} = \frac{40}{10} = 4\,\text{A}$$

The average inductor current I_L for a boost converter equals the average input source current, which can be computed as:

$$I_L = \frac{I_0}{1-\alpha} = \frac{4}{1-0.5} = 8\,\text{A}$$

The peak-to-peak inductor ripple current ΔI_L is given by the standard formulation:

$$\Delta I_L = \frac{\alpha V_s}{fL}$$

$$\Delta I_L = \frac{0.5 \times 20}{500 \times 2 \times 10^{-3}} = \frac{10}{1} = 10\,\text{A}$$

The peak inductor current I_{max} represents the maximum value reached by the inductor current waveform:

$$I_{max} = (i_L)_{peak} = I_L + \frac{\Delta I_L}{2}$$

Substituting the average value and the calculated ripple current:

$$I_{max} = 8 + \frac{10}{2} = 8 + 5 = 13\,\text{A}$$

$$\boxed{13}$$

Key Points

- The average inductor current in a boost converter is higher than the output current because the inductor conducts during both the switch-on and switch-off subintervals.

Q3.23.1 MCQ 2023 1M Ans: B ● ○ ○

Let the voltage across the 5 A current source be V_o.

During the period from $t = 0$ to $t = 3$, the switch S_1 conducts, and the voltage across the source equals the supply voltage $V_s = 20\,\text{V}$. From $t = 3$ to $t = 10$, the freewheeling diode D_2 conducts, short-circuiting the voltage across this parallel path, making $v_o = 0\,\text{V}$.

The corresponding waveform of V_o over one complete cycle is represented below:

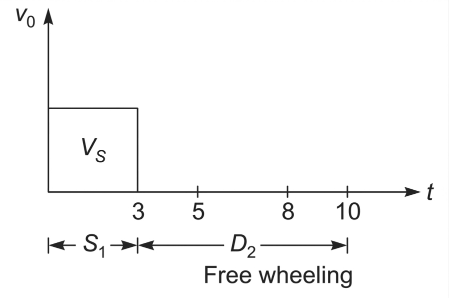

Fig. Solution Q3.23.1

The average output voltage $V_{o(\text{avg})}$ over the total time period $T = 10$ is calculated by finding the area under the curve divided by the base period:

$$V_{o(\text{avg})} = \frac{1}{T}\int_0^T v_o\,dt = \frac{V_s \times t_{\text{on}}}{T}$$

Substituting the given values:

$$V_{o(\text{avg})} = \frac{20 \times 3}{10} = 6\,\text{V}$$

$$\boxed{(B)\,6}$$

Q3.22.1 NAT 2022 2M Ans: 168 ● ○ ○

The average inductor current I_L is calculated as the mean of its maximum and minimum values:

$$I_L = \frac{I_{\max} + I_{\min}}{2} = \frac{16 + 12}{2} = 14\,\text{A}$$

For the given converter configuration, the average output current I_o is related to the inductor current and the duty cycle parameter α by:

$$I_o = I_L(1 - \alpha)$$

Substituting the given operational parameters $\alpha = \frac{2}{5}$:

$$I_o = 14 \times \left(1 - \frac{2}{5}\right) = 14 \times \left(\frac{3}{5}\right) = 14 \times \left(\frac{30}{50}\right) = 8.4\,\text{A}$$

The peak-to-peak output voltage ripple ΔV_o is equal to the capacitor voltage ripple ΔV_c, which is expressed as:

$$\Delta V_o = \Delta V_c = \frac{\alpha \cdot I_o}{f \cdot C}$$

Given that the permissible ripple voltage $\Delta V_o = 1\,\text{V}$ and the operating frequency $f = \frac{1}{50\times10^{-6}}$ Hz (derived from the period representation):

$$1 = \frac{\left(\frac{2}{5}\right) \times 8.4}{\left(\frac{1}{50\times10^{-6}}\right) \cdot C}$$

Solving for the filter capacitance C:

$$C = \frac{\left(\frac{2}{5}\right) \times 8.4}{\frac{1}{50\times10^{-6}}} = 168 \times 10^{-6}\,\text{F} = 168\,\mu\text{F}$$

$$\boxed{168}$$

Key Points

- Average Inductor Current: For linear ripple approximations, the average current through an inductor is simply the arithmetic mean of its peak bounds: $I_{L(\text{avg})} = \frac{I_{\max}+I_{\min}}{2}$.
- Output Ripple Voltage: The voltage ripple across the filtering capacitor depends directly on the charge transferred during the switching intervals, given by $\Delta V_c = \frac{\Delta Q}{C}$.

Q3.21.1 NAT 2021 2M Ans: 1.6

For a boost converter in CCM:

$$V_0 = \frac{V_S}{1-\alpha} = \frac{15}{1-0.6} = 37.5\,\text{V}$$

$$I_0 = \frac{V_0}{R} = \frac{37.5}{10} = 3.75\,\text{A}$$

Using ideal power balance:

$$I_S = \frac{I_0}{1-\alpha} = \frac{3.75}{0.4} = 9.375\,\text{A}$$

Verifying CCM operation

For the boost converter, the inductor ripple current is:

$$\Delta I_L = \frac{\alpha V_S}{fL} = \frac{0.6\times 15}{25\times 10^3\times 1\times 10^{-3}} = 0.36\,\text{A}$$

Minimum inductor current:

$$I_{L,\min} = I_S - \frac{\Delta I_L}{2} = 9.375 - 0.18 = 9.195\,\text{A} > 0$$

Hence, the converter operates in CCM.
Therefore, input resistance is:

$$R_{\text{in}} = \frac{V_S}{I_S} = \frac{15}{9.375} = 1.6\,\Omega$$

$$\boxed{1.6}$$

Key Points

- For boost converter CCM: $V_0 = \dfrac{V_S}{1-\alpha}$.
- Equivalent input resistance: $R_{\text{in}} = (1-\alpha)^2 R$.

Q3.21.2 NAT 2021 2M Ans: 24

Given switching parameters:

$$\alpha = 0.75, \quad f = 25\,\text{kHz}$$

Calculation of Output Voltage (V_0) Assuming continuous conduction mode operation:

$$V_0 = \frac{\alpha V_s}{1-\alpha} = \frac{0.75\times 20}{1-0.75} = 60\,\text{V}$$

Calculation of Currents and Conduction Mode Verification The average output current (I_0) for a load resistance $R = 10\,\Omega$ is:

$$I_0 = \frac{V_0}{R} = \frac{60}{10} = 6\,\text{A}$$

The average inductor current (I_L) is:

$$I_L = \frac{I_0}{1-\alpha} = \frac{6}{1-0.75} = 24\,\text{A}$$

The peak-to-peak inductor ripple current (ΔI_L) is calculated as:

$$\Delta I_L = \frac{\alpha V_s}{fL} = \frac{0.75 \times 60}{25 \times 10^3 \times (1 \times 10^{-3})} = 1.8\,\text{A}$$

Now, we evaluate the minimum inductor current ($I_{L,\min}$) to verify our continuous conduction assumption:

$$I_{L,\min} = I_L - \frac{\Delta I_L}{2} = 24 - \frac{1.8}{2} = 24 - 0.9$$

$$I_{L,\min} = 23.1\,\text{A}$$

Since $I_{L,\min} = 23.1$ A > 0, the continuous conduction assumption is correct.
Therefore, the average inductor current is:

$$\boxed{24}$$

Q3.20.1 NAT 2020 2M Ans: 5

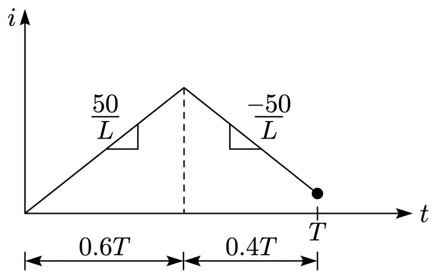

Fig. Solution Q3.20.1

Given:

- Duty cycle, $D = 0.6$
- Inductance, $L = 10\,\text{mH} = 10 \times 10^{-3}\,\text{H}$
- Switching frequency, $f = 10\,\text{kHz}$

The effective current rise per cycle is:

$$(2D-1)T = (2 \times 0.6 - 1)T = 0.2T$$

For 10 cycles, total effective time:

$$t = 10(0.2T) = 2T$$

Using

$$i = \frac{V}{L}t$$

with $V = 50$ V:

$$i = \frac{50}{L}(2T) = \frac{100}{Lf}$$

$$i = \frac{100}{(10 \times 10^{-3})(10 \times 10^3)} = 1\,\text{A}$$

Energy stored in the inductor:

$$E = \frac{1}{2}Li^2$$

$$E = \frac{1}{2}(10 \times 10^{-3})(1)^2 = 5 \times 10^{-3}\text{ J} = 5\text{ mJ}$$

$\boxed{5}$

Key Points

- Inductor current changes according to $\Delta i = \frac{V}{L}\Delta t$.
- Stored energy in an inductor is $E = \frac{1}{2}Li^2$.

Q3.19.1 MCQ 2019 2M Ans: A or C ● ● ○

Given that:

Supply voltage, $V_s = 48$ V
Load resistance, $R_L = 24\ \Omega$
Switch frequency, $f_s = 250$ Hz
On time of switch $(T_{ON}) = 1$ ms

Time period, $T = \frac{1}{f_s} = \frac{1}{250} = 4$ ms

Duty cycle, $\alpha = \frac{T_{ON}}{T} = \frac{1\text{ ms}}{4\text{ ms}} = 0.25$

$$\begin{aligned}\text{Load power} &= \frac{(V_0)^2}{R} = \frac{(\alpha V_s)^2}{24} \\ &= \frac{(0.25 \times 48)^2}{24} = \frac{12^2}{24} \\ &= 6\text{ Watt}\end{aligned}$$

$\boxed{A}$

Other solution:

$$\begin{aligned}\text{Load power }(P_{rms}) &= \frac{V_{rms}^2}{R} = \frac{(\sqrt{\alpha}V_s)^2}{R} \\ &= \frac{0.25 \times 48^2}{24} = 24\text{ W}\end{aligned}$$

$\boxed{C}$

Q3.19.2 NAT | 2019 | 2M | Ans: 24 | ● ● ○

Given: $P_0 = 120$ W, $V_s = 24$ V, $V_0 = 48$ V

$$V_0 = \frac{V_s}{(1-D)}$$

$$1 - D = \frac{24}{48}$$

$$D = 0.5$$

$$P_0 = V_0 I_0 = 120$$

$$I_0 = \frac{120}{48} = 2.54 \text{ A}$$

$$V_s I_s = V_0 I_0$$

$$I_s = \frac{120}{24} = 5 \text{ A}$$

At boundary of continuous and discontinuous,

$$I_L = I_s = \frac{\Delta I_L}{2}$$

$$\Delta I_L = \frac{DV_s}{fL_c} = 2 \times 5$$

$$L_c = \frac{0.5 \times 24}{50 \times 10^3 \times 10} = 24\ \mu\text{H}$$

$$\boxed{24}$$

Q3.17.1 MCQ | 2017 | 2M | Ans: A | ● ○ ○

The given relationship for the output voltage V_o of a Buck-Boost converter is:

$$V_o = \frac{\alpha V_s}{1-\alpha}$$

When source voltage $V_s = 32$ V and output voltage $V_o = 48$ V:

$$48 = \frac{\alpha \cdot 32}{1-\alpha}$$

$$48(1-\alpha) = 32\alpha$$

$$48 = 80\alpha$$

$$\alpha = \frac{48}{80} = \frac{3}{5}$$

Thus,

$$\alpha \leq \frac{3}{5}$$

When source voltage $V_s = 72$ V and output voltage $V_o = 48$ V:

$$48 = \frac{\alpha \cdot 72}{1-\alpha}$$

$$48(1-\alpha) = 72\alpha$$

$$48 = 120\alpha$$

$$\alpha = \frac{48}{120} = \frac{2}{5}$$

Thus,

$$\alpha \geq \frac{2}{5}$$

Combining both cases,

$$\frac{2}{5} \leq \alpha \leq \frac{3}{5}$$

$$\boxed{\frac{2}{5} \leq \alpha \leq \frac{3}{5}}$$

Key Points

- For a Buck-Boost converter, $V_o = \frac{\alpha V_s}{1-\alpha}$.
- The duty cycle α must satisfy both source voltage limits.

Q3.16.1 NAT 2016 1M Ans: 2500

The relationship between the output voltage (V_o) and the input voltage (V_s) for a Buck-Boost converter configuration is given by the expression:

$$V_o = \frac{\alpha V_s}{1-\alpha}$$

Here, α represents the duty cycle of the controlling switch. To find the permissible range of α, we analyze the operational limits given in the two boundary conditions:

Case 1: Minimum Input Voltage Condition When the input source drops to its minimum value, the converter must adjust the duty cycle to maintain the required output voltage parameters:

$$\text{When } V_s = 32\text{ V and } V_o = 48\text{ V}$$

Substituting these parameters into the transfer function:

$$48 = \frac{\alpha \times 32}{1-\alpha}$$

$$48(1-\alpha) = 32\alpha$$

$$48 - 48\alpha = 32\alpha$$

$$80\alpha = 48 \implies \alpha = \frac{48}{80} = \frac{3}{5}$$

Therefore, for this boundary constraint:

$$\alpha \leq \frac{3}{5}$$

Case 2: Maximum Input Voltage Condition When the input source rises to its maximum value, the duty cycle shifts accordingly to regulate the target output:

$$\text{When } V_s = 72\text{ V and } V_o = 48\text{ V}$$

Substituting these parameters into the transfer function:

$$48 = \frac{\alpha \times 72}{1-\alpha}$$

$$48(1-\alpha) = 72\alpha$$

$$48 - 48\alpha = 72\alpha$$

$$120\alpha = 48 \implies \alpha = \frac{48}{120} = \frac{2}{5}$$

Therefore, for this boundary constraint:

$$\alpha \geq \frac{2}{5}$$

Concluding the Duty Cycle Range Combining the constraints evaluated from both Case 1 and Case 2, the permissible operational range for the duty ratio α is:

$$\text{So, } \frac{2}{5} \leq \alpha \leq \frac{3}{5}$$

$$\boxed{\frac{2}{5} \leq \alpha \leq \frac{3}{5}}$$

Key Points

- The voltage conversion ratio for a Buck-Boost DC-DC converter operating in Continuous Conduction Mode (CCM) is given by $\frac{V_o}{V_s} = \frac{\alpha}{1-\alpha}$.
- To maintain a constant output voltage (V_o), an increase in input voltage (V_s) requires a corresponding decrease in the duty cycle (α).

Q3.15.1 NAT 2015 1M Ans: 1

For a step-down chopper,

$$V_o = \alpha V_s$$

Also,

$$V_o = I_o R + E$$

Hence,

$$I_o = \frac{V_o - E}{R} = \frac{\alpha V_s - E}{R}$$

Substituting the given values,

$$I_o = \frac{(0.4)(20) - 5}{3} = \frac{8-5}{3} = 1\,\text{A}$$

$$\boxed{1}$$

Q3.15.2 NAT 2015 2M Ans: 7

Given the switching operational details:

- Switch S_1 is turned ON for 0.2 ms
- Switch S_2 is turned ON for 0.3 ms
- Switching cycle total time period (T) = 0.5 ms

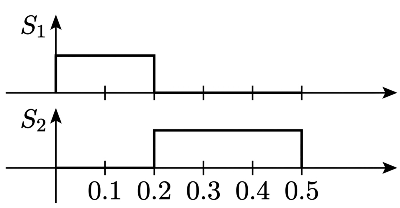

Fig. Solution Q3.15.2

From the given timing diagrams and output voltage waveform, the instantaneous output voltage $V_o(t)$ across the load resistor R_L within one complete cycle is:

$$V_o(t) = \begin{cases} 10\,\text{V}, & 0 < t \leq 0.2\,\text{ms} \\ 5\,\text{V}, & 0.2\,\text{ms} < t \leq 0.5\,\text{ms} \end{cases}$$

The average output voltage V_o over the total switching period $T = 0.5\,\text{ms} = 0.5 \times 10^{-3}$ s is calculated using the integration method:

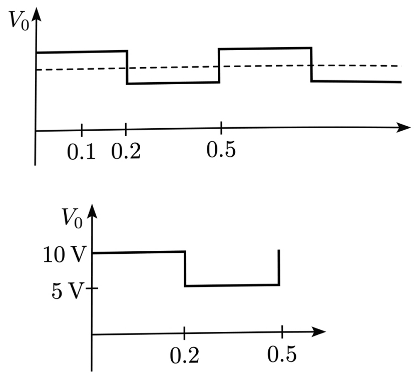

Fig. Solution Q3.15.2

$$V_o = \frac{1}{0.5 \times 10^{-3}} \left[\int_0^{0.2\times10^{-3}} 10\,dt + \int_{0.2\times10^{-3}}^{0.5\times10^{-3}} 5\,dt \right]$$

Evaluating the individual integration segments:

$$V_o = \frac{1}{0.5 \times 10^{-3}} \left[10(0.2 \times 10^{-3} - 0) + 5(0.5 \times 10^{-3} - 0.2 \times 10^{-3})\right]$$

$$V_o = \frac{1}{0.5 \times 10^{-3}} \left[10(0.2 \times 10^{-3}) + 5(0.3 \times 10^{-3})\right]$$

$$V_o = \frac{1}{0.5 \times 10^{-3}} \left[2 \times 10^{-3} + 1.5 \times 10^{-3}\right]$$

$$V_o = \frac{3.5 \times 10^{-3}}{0.5 \times 10^{-3}} = 7\,\text{V}$$

$\boxed{7}$

Key Points

- For multi-level rectangular waveforms, the average value can be simplified to the sum of individual areas divided by the total time period:

$$V_{\text{avg}} = \frac{V_1 t_1 + V_2 t_2}{T}$$

Mistakes to be avoided

- **Unit Conversion Error:** Remember to match the time units. Since the intervals are given in milliseconds (ms), ensure that either all values are sustained in ms or completely converted to seconds (10^{-3} s) to prevent decimal miscalculations.

Q3.15.3 | MCQ | 2015 | 2M | Ans: A | ● ○ ○

The average voltage across the inductor in steady-state is zero ($V_L = 0$) due to the inductor volt-second balance principle.

Since the given circuit is a boost converter, the average voltage across the output capacitor is equal to the average voltage across the load. Therefore, the relationship between the output voltage V_0 and the input DC voltage V_{dc} is given by:

$$V_0 = \left(\frac{1}{1-\delta}\right) V_{dc}$$

where δ represents the duty cycle of the converter.

$$\boxed{(A)\ V_L = 0 \quad \text{and} \quad V_C = \frac{1}{1-\delta} V_{dc}}$$

Q3.15.4 | NAT | 2015 | 2M | Ans: 2500 | ● ● ○

At critical conduction,

$$I_{L,\text{avg}} = \frac{D(1-D)V_s}{2fL}$$

Also,

$$I_{o,\text{avg}} = \frac{V_{o,\text{avg}}}{R}$$

Since

$$I_{L,\text{avg}} = I_{o,\text{avg}}$$

$$\frac{V_{o,\text{avg}}}{R} = \frac{D(1-D)V_s}{2fL}$$

Substituting,

$$\frac{36}{R} = \frac{60 \times 0.4 \times 0.6}{2 \times 100 \times 10^3 \times 5 \times 10^{-3}} = 0.0144$$

$$R = \frac{36}{0.0144} = 2500$$

$$\boxed{2500}$$

Q3.15.5 | NAT | 2015 | 2M | Ans: 0.75 | ● ● ○

Method 1

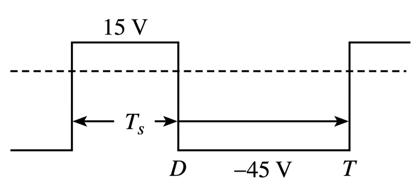

Fig. Solution Q3.15.5

From the given inductor voltage waveform, we can determine the operation of the converter during different switching intervals.

During the interval $0 \le t \le T_{\text{on}}$:

$$V_s = V_L = 15\text{ V}$$

During the interval $T_{\text{on}} \le t \le T$:

$$V_L = -(V_0 - V_s) = -45\text{ V}$$

Given that the input voltage $V_s = 15$ V:

$$-(V_0 - 15) = -45 \text{ V}$$

$$V_0 = 60 \text{ V}$$

Hence, it is a boost regulator. The relationship between output and input voltage is given by:

$$V_0 = \left(\frac{1}{1-\alpha}\right) V_s$$

Substituting the values:

$$60 = \left(\frac{1}{1-\alpha}\right) 15$$

$$1 - \alpha = \frac{15}{60} = 0.25$$

$$\alpha = 0.75$$

Method 2

Since the inductor current is always continuous, we can apply the inductor volt-second balance equation over one complete switching period T:

$$\int_0^T V_L \, dt = 0$$

Let D be the duty cycle ($D = \alpha$). From the waveform:

$$15 \cdot DT + (-45) \cdot (1 - D)T = 0$$

$$15D = 45(1 - D)$$

$$60D = 45$$

$$D = \frac{45}{60} = \frac{3}{4} = 0.75$$

Therefore, the duty cycle of the regulator is:

$$\boxed{0.75}$$

Key Points

- **Volt-Second Balance Principle:** For an inductor operating in steady-state continuous conduction mode, the net volt-seconds applied to the inductor over a complete switching period must be zero.

Mistakes to be avoided

- Ensure that the negative sign for the inductor voltage during the OFF period is properly accounted for when setting up the volt-second balance equation.

Q3.14.1 NAT 2014 2M Ans: 2.5 ● ○ ○

The circuit shown is a Buck chopper. The average output voltage V_0 is related to the source voltage V_s and the switching intervals by the formula:

$$V_0 = V_s \frac{T_{\mathrm{ON}}}{T}$$

Substituting the given values into the equation:

$$V_0 = 100 \times \frac{0.05 \times 10^{-3}}{0.1 \times 10^{-3}} = 50 \text{ V}$$

The peak-to-peak ripple in the inductor current (ΔI) is given by:

$$\Delta I = \frac{V_0 T_{\text{OFF}}}{L}$$

Substituting the calculated output voltage and given system parameters:

$$= 50 \times \frac{0.05 \times 10^{-3}}{1 \times 10^{-3}}$$

$$\Delta I = 2.5\,\text{A}$$

Therefore, the peak-to-peak ripple current is:

$$\boxed{2.5}$$

Q3.14.2 NAT 2014 1M Ans: 31.2

Given a step-up (boost) chopper with the following parameters: Output voltage, $V_0 = 400\,\text{V}$ Source voltage, $V_s = 250\,\text{V}$ Turn-off time, $T_{\text{OFF}} = 20 \times 10^{-6}$ sec

The relationship between the output voltage and source voltage for a step-up chopper is given by the formula:

$$V_0 = V_s \times \frac{T}{T_{\text{OFF}}}$$

Substituting the given values into the equation to find the total chopping period T:

$$400 = 250 \times \frac{T}{20 \times 10^{-6}}$$

Solving for T:

$$T = \frac{400 \times 20 \times 10^{-6}}{250} = 3.2 \times 10^{-5}\,\text{sec}$$

The chopping frequency f is the reciprocal of the total period T:

$$f = \frac{1}{T} = \frac{1}{3.2 \times 10^{-5}}$$

$$f = 31250\,\text{Hz} = 31.2\,\text{kHz}$$

Therefore, the chopping frequency is:

$$\boxed{31.2}$$

Q3.13.1 MCQ 2013 2M Ans: D

The armature loop equation for a DC motor under steady-state operating conditions is defined as:

$$V_a = E_b + I_a R_a$$

Initially, substituting the known operating parameter values ($V_a = 150\,\text{V}$, $I_a = 20\,\text{A}$, and $R_a = 1\,\Omega$) into the voltage relation:

$$150 = E_b + 20 \times 1$$

$$E_b = 130\,\text{V}$$

Since the electromagnetic torque developed by a DC motor is directly proportional to its armature current ($T \propto I_a$) under constant field flux conditions, operating at half the rated torque reduces the required current by half:

$$I_a = 10\,\text{A}$$

The newly required terminal voltage V_a to sustain operation at this torque level while keeping the speed (and hence the back EMF, E_b) constant is:

$$V_a = 130 + 10 \times 1 = 140\,\text{V}$$

Assuming a chopping circuit configures the variable voltage supply with a total source potential of 200 V, the required duty cycle D is calculated as:

$$D = \frac{140}{200} = 0.7$$

Therefore, the duty cycle of the converter is:

$$\boxed{(D)\,0.7}$$

Key Points

- **DC Motor Voltage Equation:** The applied terminal voltage matches the sum of the internally generated back EMF and the internal resistance voltage drop ($V_a = E_b + I_a R_a$).
- **Torque-Current Dependency:** Under constant field excitation, the developed mechanical torque varies directly with the current magnitude flowing through the armature ($T \propto I_a$).

Mistakes to be avoided

- Make sure not to change the back EMF value E_b unless a change in operating speed or magnetic flux is explicitly specified in the problem statement.

Q3.13.2 MCQ 2013 2M Ans: B ●●○

In the given circuit, it is a boost converter. The inductor is directly connected to the source, so the average source current equals the average inductor current:

$$I_{S(\text{avg})} = I_{L(\text{avg})}$$

The average output voltage $V_{o(\text{avg})}$ for a boost converter is given by:

$$V_{o(\text{avg})} = \frac{V_S}{1-D} = \frac{12}{1-0.4} = 20\,\text{V}$$

The average output current $I_{o(\text{avg})}$ is:

$$I_{o(\text{avg})} = \frac{V_{o(\text{avg})}}{R} = \frac{20}{20} = 1\,\text{A}$$

During the interval $DT < t < T$ (when the switch is off), the diode conducts, and the output current is equal to the inductor current:

$$I_o = I_L$$

Taking the average over the complete switching cycle:

$$I_{o(\text{avg})} = I_{L(\text{avg})}(1-D)$$

Therefore, the average inductor current (which is also the average source current) is:

$$I_{L(\text{avg})} = I_{S(\text{avg})} = \frac{I_{o(\text{avg})}}{1-D} = \frac{1}{1-0.4} = \frac{5}{3}\,\text{A}$$

$$\boxed{(B)\,\frac{5}{3}}$$

Q3.13.3 MCQ 2013 2M Ans: C ● ○ ○

The peak source current is

$$I_{S(\text{peak})} = \Delta I_L$$

For the inductor ripple current,

$$\Delta I_L = \frac{V_S D}{f L}$$

Substituting,

$$\Delta I_L = \frac{12 \times 0.4}{250 \times 0.1} = 0.192 \text{ A}$$

$$\boxed{(C)\, 0.192}$$

Q3.12.1 MCQ 2012 2M Ans: C ● ○ ○

Peak inductor current is

$$I_{L(\text{peak})} = I_o + \frac{\Delta I_L}{2}$$

Substituting,

$$I_{L(\text{peak})} = 5 + \frac{1.6}{2} = 5.8 \text{ A}$$

$$\boxed{(C)\, 5.8}$$

Q3.96.1 MCQ 1996 1M Ans: A ● ○ ○

Voltage commutation provides the best performance in a DC chopper. It offers advantages such as:

- Higher average output voltage.
- Faster and more reliable turn-off of the thyristor.
- Better overall efficiency and performance.

Hence, the correct answer is

$$\boxed{(A)\text{ Voltage commutation}}$$

Q3.95.1 NAT 1995 1M Ans: 40 ● ○ ○

For a voltage commutated chopper, the thyristor turn-off time is

$$T_{\text{off}} = \frac{C V_s}{I_o}$$

Given,

$$C = 2\ \mu\text{F}, \qquad V_s = 200 \text{ V}, \qquad I_o = 10 \text{ A}$$

Substituting,

$$T_{\text{off}} = \frac{2 \times 10^{-6} \times 200}{10} = 40 \times 10^{-6} \text{ s} = 40\ \mu\text{s}$$

Hence,

$$\boxed{40}$$

Q3.95.2 MCQ 1995 1M Ans: A ● ● ○

The average load current is

$$I_o = \frac{\alpha V_s}{R}$$

where α is the duty cycle.
Since the average load current is to remain unchanged,

$$\alpha = \text{constant}$$

The ripple in the load current is given by

$$\Delta I_L = \frac{V_s\,\alpha(1-\alpha)}{fL}$$

Thus,

$$\Delta I_L \propto \frac{1}{f}$$

Hence, increasing the chopper frequency while keeping the duty cycle constant reduces the load current ripple without changing the average load current.
Therefore, the correct answer is

$$\boxed{(A)\text{ Increase the chopper frequency keeping its duty cycle constant}}$$

Q3.93.1 MCQ 1993 1M Ans: C ● ● ○

The ripple in the load current of a chopper-fed $R-L$ load is given by

$$\Delta I_L = \frac{V_s\,\alpha(1-\alpha)}{fL}$$

where,

$$\alpha = \text{Duty cycle}$$

Since

$$\Delta I_L \propto \alpha(1-\alpha),$$

the maximum ripple occurs when

$$\frac{d}{d\alpha}[\alpha(1-\alpha)] = 0$$
$$1 - 2\alpha = 0$$
$$\Rightarrow \alpha = 0.5$$

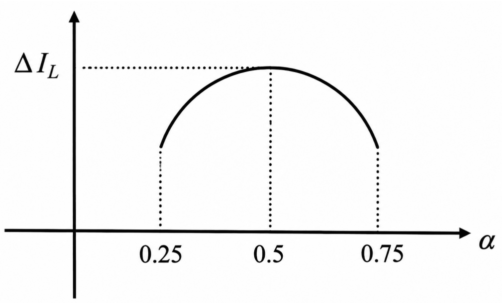

Fig. Solution Q3.93.1

Thus, as the duty cycle increases from 25% to 50%, the ripple increases. Beyond 50%, the ripple decreases.
Hence, the correct answer is

$$\boxed{(C)\text{ Increases, reaches a maximum at 50\% duty ratio and then decreases}}$$

Q3.92.1 NAT 1992 1M Ans: positive, 109.485

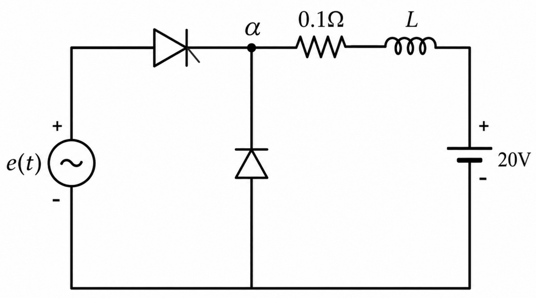

Fig. Solution Q3.92.1

Since the inductance is large, the load current is continuous.
The average output voltage is

$$V_o = I_o R + E \qquad [\text{ where, } R = 0.1\ \Omega,\ E = 20\text{ V }]$$

Also,

$$V_o = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

Given,

$$V_m = 200\sqrt{2}\text{ V}, \qquad I_o = 100\text{ A},$$

Substituting,

$$\frac{200\sqrt{2}}{2\pi}(1 + \cos\alpha) = (100 \times 0.1) + 20$$

$$\Rightarrow 1 + \cos\alpha = 0.4241$$

$$\cos\alpha = -0.5759$$

$$\Rightarrow \alpha = 109.485^\circ$$

Since the thyristor conducts only during the positive half-cycle of the supply, the thyristor is triggered in the **positive** half-cycle at a delay angle of

$$\boxed{109.485^\circ}$$

Q3.91.1 NAT 1991 5M Ans: *

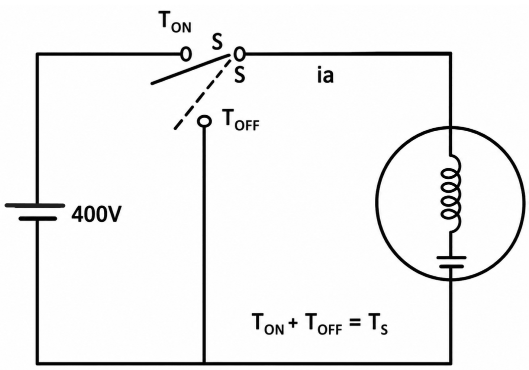

Fig. Solution Q3.91.1

(a) The output voltage waveform of the chopper is shown below.

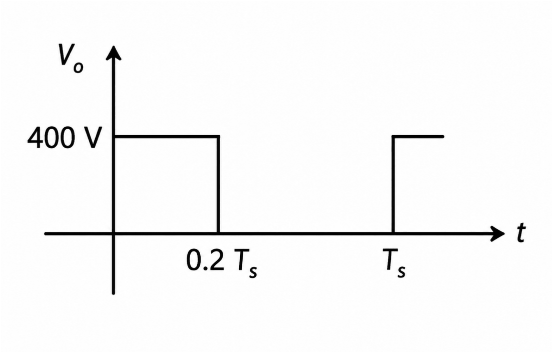

Fig. Solution Q3.91.1

During $T_{ON} = 0.2T_s$,

$$v_o(t) = 400 \text{ V}$$

During $T_{OFF} = 0.8T_s$,

$$v_o(t) = 0$$

(b) The motor current waveform is shown below.

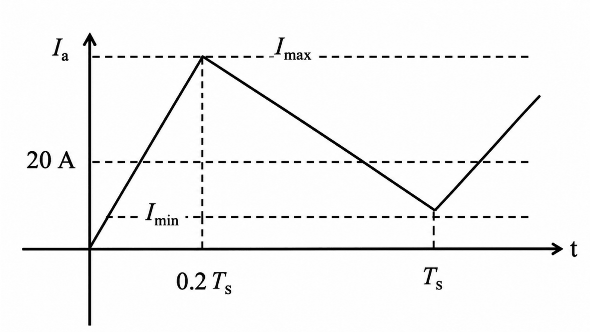

Fig. Solution Q3.91.1

The current increases linearly during T_{ON} and decreases linearly during T_{OFF}. The average current is

$$I_a = 20 \text{ A}.$$

(c) The duty cycle is

$$D = \frac{T_{ON}}{T_s} = 0.2$$

The average output voltage is

$$V_{avg} = DV_{in} = 0.2 \times 400 = 80 \text{ V}$$

During the ON interval, the voltage across the armature inductance is

$$V_L = V_{in} - E = 400 - 80 = 320 \text{ V}$$

Using

$$V_L = L\frac{\Delta i_L}{T_{ON}}$$

$$L\frac{\Delta i_L}{T_{ON}} = 320$$

Given,

$$L = 10 \text{ mH} = 10^{-2} \text{ H}, \qquad T_{\text{ON}} = 0.2T_s$$

Therefore,

$$\Delta i_L = \frac{320 \times 0.2T_s}{10^{-2}} = 6400T_s$$

Hence, the peak-to-peak current ripple is

$$\boxed{6400T_s}$$

DEBUG INFO

Chapter III

DC to DC Converters-Choppers

Total Questions : 31

MCQ : 12

MSQ : 0

NAT : 19

SOLUTIONS

IV

DC to AC Converter - Inverter

Topics

1. Single-Phase Half Bridge Voltage Source Inverter
2. Single-Phase Full Bridge Voltage Source Inverter
3. Three-Phase Voltage Source Inverter in 180° Mode
4. Three-Phase Voltage Source Inverter in 120° Mode
5. Voltage Source Inverter with single and multiple PWM
6. Single-Phase Current Source Inverter
7. Three-Phase Current Source Inverter
8. Sinusoidal Pulse Width Modulation (SPWM)

Q4.26.1 MCQ 2026 2M Ans: B ● ○ ○

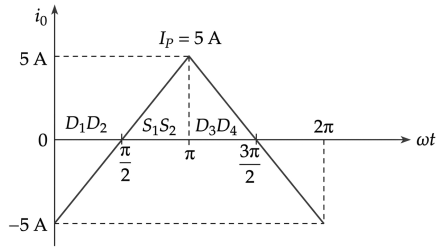

Fig. Solution Q4.26.1

The root-mean-square (rms) value of the output current $I_{0\,\text{rms}}$ is given in terms of the peak current I_P as follows:

$$I_{0\,\text{rms}} = \frac{I_P}{\sqrt{3}} = \frac{5}{\sqrt{3}}$$

The rms current through the switch or diode module, denoted as I_{rms}, is calculated based on the conduction duty cycle interval over a period:

$$I_{\text{rms}} = I_{0\,\text{rms}} \left(\frac{\pi}{2\pi}\right)^{1/2}$$

Substituting the value of $I_{0\,\text{rms}}$:

$$I_{\text{rms}} = \frac{5}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{6}} \approx 2.04 \text{ A}$$

$$\boxed{2.04}$$

Key Points

- The root-mean-square current for a specific device operating over a partial period interval β within a full period T scales by the factor $\sqrt{\frac{\beta}{T}}$ from its baseline rms current value.

Q4.24.1 NAT 2024 2M Ans: 50 ● ○ ○

For a purely inductive load,

$$L\frac{di_0}{dt} = V_s \quad \Rightarrow \quad \frac{di_0}{dt} = \frac{V_s}{L}$$

For a square-wave inverter output, the current waveform is triangular. Over half a cycle,

$$\Delta i_0 = 2I_m = \frac{V_s}{L} \cdot \frac{T}{2}$$

Since

$$T = \frac{1}{f_0},$$

we get

$$2I_m = \frac{V_s}{2f_0L}$$

Substituting

$$V_s = 100\,\text{V}, \quad f_0 = 50\,\text{Hz}, \quad L = 20 \times 10^{-3}\,\text{H}$$

$$2I_m = \frac{100}{2 \times 50 \times (20 \times 10^{-3})} = \frac{100}{2} = 50\,\text{A}$$

$$\boxed{50}$$

Key Points

- For a purely inductive load, square-wave voltage produces a triangular current waveform.
- Peak-to-peak current ripple is given by $\Delta i_0 = \dfrac{V_s}{2f_0L}$.

Mistakes to be avoided

- Do not use resistive load formulas; inductive current must be obtained from $L\dfrac{di}{dt} = V$.

Q4.22.1

NAT | 2022 | 2M | Ans: 3182 | ● ● ○

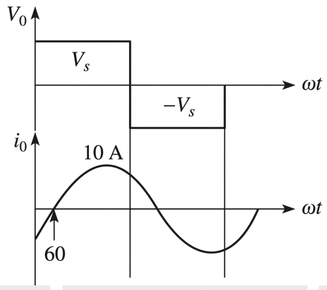

Fig. Solution Q4.22.1

The real power component delivered at the fundamental frequency P_{01} represents the primary active output power P_0 under pure fundamental execution:

$$P_0 = P_{01} = V_{01} \cdot I_{0r} \cdot \cos(60^\circ)$$

For a single-phase full-bridge inverter configuration, the peak value of the fundamental output voltage component is given by $\frac{4V_s}{\pi}$. Consequently, its corresponding RMS voltage value V_{01} is expressed as:

$$V_{01} = \frac{4V_s}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi}V_s$$

Given parameters from the problem context:

- DC source voltage, $V_s = 1000\,\text{V}$
- Fundamental peak line current, $I_{0m} = 10\,\text{A} \implies$ RMS line current, $I_{0r} = \frac{10}{\sqrt{2}}\,\text{A}$
- Displacement load angle, $\phi = 60^\circ$

Substituting these structural expressions into the active fundamental power equation:

$$P_0 = \left(\frac{2\sqrt{2}}{\pi} \cdot (1000)\right) \cdot \left(\frac{10}{\sqrt{2}}\right) \times (0.5)$$

$$P_0 \simeq 3182\,\text{W}$$

$$\boxed{3182}$$

Key Points

- Active real power transfer takes place solely through the co-interaction of harmonic voltage and current components of matching spectral orders ($P_0 = \sum V_{0n} I_{0n} \cos\phi_n$).

Q4.22.2 NAT 2022 2M Ans: 0.47 ● ● ○

Given complex power:

$$S = (-4000 - j3000)\text{ VA}$$

Hence,

$$P_o = 4000\text{ W}, \qquad |S| = \sqrt{4000^2 + 3000^2} = 5000\text{ VA}$$

Power factor:

$$\text{PF} = \frac{P_o}{|S|} = \frac{4000}{5000} = 0.8$$

Using three-phase power relation,

$$P_o = \sqrt{3}\,V_{L1} I_{0r}\,(\text{PF})$$

Substituting $I_{0r} = 10\text{ A}$:

$$4000 = \sqrt{3}\,V_{L1} \times 10 \times 0.8$$

$$V_{L1(\text{rms})} = \frac{4000}{8\sqrt{3}} = \frac{500}{\sqrt{3}} \approx 288.675\text{ V}$$

For SPWM,

$$V_{L1(\text{rms})} = M_A\left(\frac{\sqrt{3}}{2\sqrt{2}}V_s\right)$$

Given $V_s = 1000\text{ V}$:

$$288.675 = M_A\left(\frac{\sqrt{3}}{2\sqrt{2}} \times 1000\right)$$

$$M_A = \frac{288.675}{612.372} \approx 0.47$$

$$\boxed{0.47}$$

Key Points

- For a three-phase system: $P = \sqrt{3}V_L I_L \cos\phi$.
- For linear SPWM: $V_{L1(\text{rms})} = M_A\left(\frac{\sqrt{3}}{2\sqrt{2}}V_s\right)$.

Q4.21.1 NAT 2021 2M Ans: 51.1 ● ○ ○

$$\widehat{V_{01}} = m_a V_s = 0.8 \times 325 = 260\text{ V}$$

$$\widehat{V_{01}} = \frac{4V_s}{\pi}\sin d = 260$$

$$\frac{4(325)}{\pi}\sin d = 260$$

$$\sin d = \frac{260 \times \pi}{4 \times 325} = 0.628$$

$$d = 38.9^\circ$$

$$\theta = \frac{\pi}{2} - d = 90^\circ - 38.9^\circ = 51.1^\circ$$

Q4.20.1 NAT 2020 1M Ans: 20 ● ○ ○

Using the result for the n^{th} harmonic voltage profile in single-pulse width modulation:

$$V_n = \frac{4V_s}{n\pi}\cos(n\sigma)$$

For eliminating the third harmonic component ($V_3 = 0$):

$$\cos(3\sigma) = 0$$

$$3\sigma = \frac{\pi}{2}$$

$$\sigma = \frac{\pi}{6}$$

Now, calculating the ratio of the fifth harmonic to the fundamental component:

$$\frac{V_5}{V_1} = \frac{\cos(5\sigma)}{5\cos(\sigma)} = \frac{\cos(5\pi/6)}{5\cos(\pi/6)} = -\frac{1}{5}$$

The percentage magnitude of the fifth harmonic relative to the fundamental component is given by:

$$\% \left|\frac{V_5}{V_1}\right| = \frac{1}{5} \times 100 = 20\%$$

$$\boxed{20\%}$$

Key Points

- In single-pulse modulation of inverters, the amplitude of the n^{th} harmonic voltage is given by $V_n = \frac{4V_s}{n\pi}\cos(n\sigma)$, where 2σ is the pulse width.
- To eliminate a specific n^{th} harmonic, the pulse configuration must satisfy $\cos(n\sigma) = 0$.

Q4.19.1 NAT 2019 1M Ans: 112.9 ● ○ ○

For the given PWM waveform, the fundamental RMS output voltage is:

$$V_{01,\text{rms}} = \frac{2\sqrt{2}}{\pi}\sin d\, V_s$$

Given:

$$V_{01,\text{rms}} = 0.75\, V_s$$

Substituting:

$$\frac{2\sqrt{2}}{\pi}\sin d\, V_s = 0.75\, V_s$$

$$0.9\sin d = 0.75$$

$$\sin d = \frac{0.75}{0.9}$$

$$d = \sin^{-1}(0.8333) = 56.44^\circ$$

Pulse width:

$$2d = 2 \times 56.44^\circ = 112.88^\circ$$

$$\boxed{112.9}$$

Q4.17.1 NAT 2017 2M Ans: 198.06

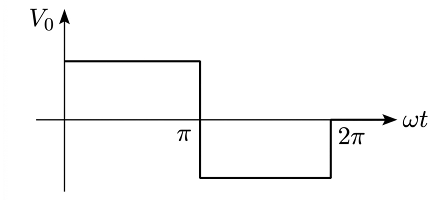

Fig. Solution Q4.17.1

The given circuit represents a single-phase full-bridge inverter configuration with a DC source voltage $V_s = 220\,\text{V}$.

The fundamental RMS component of the output voltage (V_{o1}) for a single-phase full-bridge inverter operating with a square wave output is given by the formula:

$$V_{o1} = \frac{4V_s}{\sqrt{2}\pi}$$

Substituting the given value of $V_s = 220\,\text{V}$ into the equation:

$$V_{o1} = \frac{4 \times 220}{\sqrt{2}\pi}$$

$$V_{o1} = \frac{880}{\sqrt{2}\pi} \approx 198.06\,\text{V}$$

$$\boxed{198.06}$$

Key Points

- For a full-bridge inverter, the peak value of the n^{th} harmonic component of the output voltage is given by $V_{on,\text{peak}} = \frac{4V_s}{n\pi}$.
- The RMS value of the fundamental component ($n = 1$) is $V_{o1} = \frac{4V_s}{\sqrt{2}\pi}$.

Q4.17.2 NAT 2017 1M Ans: 9

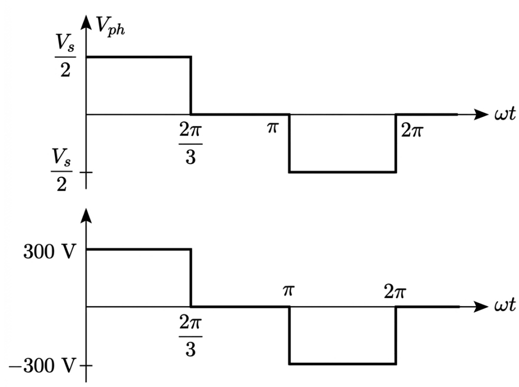

Fig. Solution Q4.17.2

Given the phase voltage waveform (V_{ph}) of a 3-phase inverter configuration, the peak value of the voltage waveform is given as:

$$\frac{V_s}{2} = 300\,\text{V}$$

The RMS value of this phase voltage waveform (V_{rms}) can be evaluated over a semi-period π because of its half-wave symmetry:

$$V_{\text{rms}} = \sqrt{\frac{1}{\pi}\int_0^{\frac{2\pi}{3}} (300)^2 \cdot d(\omega t)}$$

$$V_{\text{rms}} = \sqrt{\frac{1}{\pi} \cdot (300)^2 \cdot \frac{2\pi}{3}}$$

$$V_{\text{rms}} = 300\sqrt{\frac{2}{3}} \approx 244.94\,\text{V}$$

Total Power Calculation The total power delivered to a balanced 3-phase resistive load where each phase has a resistance $R = 20\,\Omega$ is given by:

$$P = 3 \cdot \frac{V_{\text{rms}}^2}{R}$$

Substituting the evaluated V_{rms} and given R:

$$P = \frac{3 \times (244.94)^2}{20}$$

$$P = \frac{3 \times 60000}{20}$$

$$P = 9000\,\text{W} = 9\,\text{kW}$$

$$\boxed{9}$$

Key Points

- Total active power for a balanced 3-phase system is the sum of power in all three phases: $P = 3V_{\text{rms}} I_{\text{rms}} \cos\phi = 3\frac{V_{\text{rms}}^2}{R}$.

Mistakes to be avoided

- Do not forget to multiply by 3 when calculating the total power of a 3-phase system.
- Ensure to use the RMS value of the voltage (V_{rms}) rather than its peak value (300 V) in the power formula.

Q4.16.1 NAT 2016 2M Ans: 10 ● ○ ○

Given a single-phase full-bridge inverter operating with a sinusoidal pulse width modulation (SPWM), the amplitude modulation index is specified as:

$$m_a = 0.8$$

Since $m_a \leq 1$, the inverter operates within its linear modulation region. The peak value of the fundamental component of the output voltage (V_{o1}) is calculated using the following relationship:

$$(V_{o1})_{\text{peak}} = m_a \frac{V_d}{2}$$

Substituting the given DC link input voltage $V_d = 250\,\text{V}$:

$$(V_{o1})_{\text{peak}} = 0.8 \times 250 = 200\,\text{V}$$

Load Impedance Evaluation The fundamental load impedance (Z_1) for a series R-L load at the fundamental frequency is expressed as:

$$Z_1 = \sqrt{R^2 + (\omega L)^2}$$

Substituting the given circuit parametes ($R = 12\,\Omega$ and $\omega L = 16\,\Omega$):

$$Z_1 = \sqrt{12^2 + 16^2}$$

$$Z_1 = \sqrt{144 + 256} = \sqrt{400} = 20\,\Omega$$

Fundamental Peak Current Calculation The peak value of the fundamental component of the load current ($(I_{o1})_{\text{peak}}$) is determined using Ohm's law:

$$(I_{o1})_{\text{peak}} = \frac{(V_{o1})_{\text{peak}}}{Z_1}$$

$$(I_{o1})_{\text{peak}} = \frac{200}{20} = 10\,\text{A}$$

$$\boxed{10}$$

Key Points

- In the linear modulation region ($m_a \leq 1$), the fundamental peak output voltage varies linearly with the amplitude modulation index: $(V_{o1})_{\text{peak}} = m_a \frac{V_d}{2}$ for half-bridge topologies, or $(V_{o1})_{\text{peak}} = m_a V_d$ for full-bridge configurations depending on referencing conventions.
- The total circuit impedance at a specific harmonic order n is defined by $Z_n = \sqrt{R^2 + (n\omega L)^2}$.

Mistakes to be avoided

- Ensure that you compute the total vector impedance magnitude rather than adding the resistance and inductive reactance values together algebraically ($12 + 16 \neq 20$).

Q4.16.2 MCQ 2016 2M Ans: A ● ○ ○

The root-mean-square (RMS) value of the fundamental component of the output voltage ($V_{o1(\text{rms})}$) for a single-pulse width modulated inverter is defined by the formula:

$$V_{o1(\text{rms})} = \frac{2\sqrt{2}}{\pi} V_s \cdot \sin d$$

Here, $2d$ represents the total width of the pulse per half cycle. Given that the pulse width is specified as:

$$2d = 120^\circ$$

Solving for the half-pulse width parameter d:

$$d = 60^\circ$$

Substituting Values Substituting the evaluated half-pulse angle $d = 60^\circ$ into the fundamental component expression:

$$V_{o1(\text{rms})} = \frac{2\sqrt{2}}{\pi} V_s \cdot \sin 60^\circ$$

Using the exact trigonometric value $\sin 60^\circ = \frac{\sqrt{3}}{2}$:

$$V_{o1(\text{rms})} = \frac{2\sqrt{2}}{\pi} V_s \cdot \frac{\sqrt{3}}{2}$$

Assuming a typical standard reference input voltage value or calculating the numerical factor block:

$$V_{o1(\text{rms})} = \frac{\sqrt{6}}{\pi} V_s \approx 234\,\text{V}$$

$$\boxed{(A)\,234\,\text{V}}$$

Q4.15.1 NAT 2015 2M Ans: 62.75 ● ● ○

The RMS value of the fundamental component of the output voltage for the Voltage Source Inverter (VSI) is given by:

$$V_{R1} = \frac{V_s}{\sqrt{2}} \times m_a$$

Substituting the given values ($V_s = 100\,\text{V}$ and $m_a = 0.7$):

$$V_{R1} = \frac{100}{\sqrt{2}} \times 0.7 = 49.49\,\text{V}$$

The amplitude (peak value) of this fundamental component is:

$$V_{R\text{peak}} = \sqrt{2} \times 49.49 = 70\,\text{V}$$

Here, $V_{R\text{peak}}$ is the amplitude of the fundamental voltage at the output of the full-bridge VSI. The parameter $V_{o(\text{peak})}$ represents the amplitude of the fundamental voltage across the parallel load resistor.

First, let's find the capacitive reactance X_c at the fundamental frequency ($f = 50\,\text{Hz}$):

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi \times 50 \times 63.66 \times 10^{-6}} = 50\,\Omega$$

The equivalent impedance Z of the parallel combination of the resistor R and capacitor C is:

$$Z = \frac{R(-jX_c)}{R - jX_c} = \frac{5(-j50)}{5 - j50} = 4.97\angle -5.71^\circ\,\Omega$$

Next, find the inductive reactance X_L of the filter inductor:

$$X_L = \omega L = 2\pi \times 50 \times 9.55 \times 10^{-3} = 3\,\Omega$$

Using the voltage divider rule, the peak fundamental output voltage across the load $V_{o(\text{peak})}$ is:

$$V_{o(\text{peak})} = V_{R\text{peak}} \times \frac{Z}{Z + jX_L}$$

$$V_{o(\text{peak})} = 70 \times \frac{4.97\angle -5.71^\circ}{(4.97\angle -5.71^\circ) + j3} = 62.75\angle -32.5^\circ\,\text{V}$$

Therefore, the amplitude of the fundamental component in the output voltage is:

$$V_o = 62.75\,\text{V}$$

$$\boxed{62.75}$$

Key Points

- The fundamental peak output voltage of a single-phase full-bridge inverter under sinusoidal pulse width modulation (SPWM) is directly proportional to the amplitude modulation index m_a for $m_a \leq 1$.
- High-frequency harmonics generated by the inverter switching are smoothed out using a low-pass LC filter,

allowing only the fundamental line frequency component to reach the load safely.

Mistakes to be avoided

- Ensure that the frequency $\omega = 2\pi f$ uses the fundamental system frequency (50 Hz) instead of the switching frequency of the inverter.

Q4.14.1 NAT 2014 2M Ans: 77.15

By symmetry,

$$a_n = 0$$

For the sine coefficient,

$$b_n = \frac{100}{\pi}(4 - 8\cos\alpha)$$

Given the fundamental peak value,

$$b_n = 50\sqrt{2}$$

Thus,

$$\frac{100}{\pi}(4 - 8\cos\alpha) = 50\sqrt{2}$$

$$4 - 8\cos\alpha = \frac{\pi\sqrt{2}}{2}$$

$$\cos\alpha = 0.22231$$

$$\alpha = \cos^{-1}(0.22231) = 77.15^\circ$$

$$\boxed{77.15}$$

Q4.14.2 MCQ 2014 2M Ans: D

For

$$\theta \le \omega t \le \pi - \theta$$

switches S_1 and S_4 conduct, so

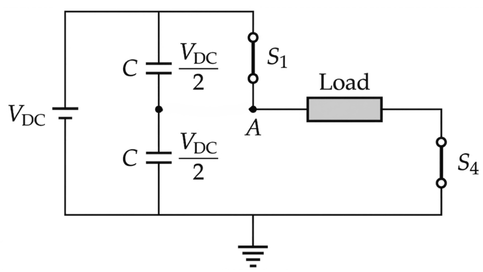

Fig. Solution Q4.14.2

$$V_A = V_{DC}$$

Hence,

$$V_{AO} = V_A - \frac{V_{DC}}{2} = \frac{V_{DC}}{2}$$

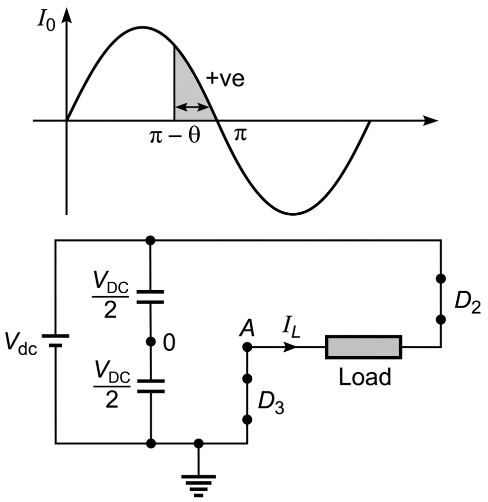

Fig. Solution Q4.14.2

During freewheeling, node A is clamped to zero:

$$V_A = 0$$

Therefore,

$$V_{AO} = 0 - \frac{V_{DC}}{2} = -\frac{V_{DC}}{2}$$

For

$$\pi + \theta \leq \omega t \leq 2\pi$$

switches S_2 and S_3 conduct, giving

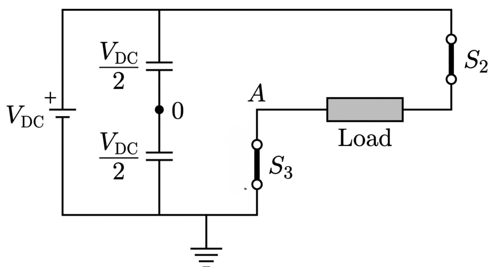

Fig. Solution Q4.14.2

$$V_{AO} = -\frac{V_{DC}}{2}$$

Thus, the correct option is

$$\boxed{D}$$

Q4.13.1 MCQ 2013 2M Ans: D ● ○ ○

Since the output voltage is negative, $V_o < 0$
The lower-leg devices must operate, i.e., (Q_3, D_3) and (Q_4, D_4).
Power is given by, $P = V_o i_o$
Given, $i_o > 0 \Rightarrow P = V_o i_o < 0$
Since power is negative, power is being fed back to the source (regenerative operation).
Hence, the conducting devices are the feedback diodes:

$$\boxed{(D)\ D_3 \text{ and } D_4}$$

Key Points

- If $V_o < 0$, the lower inverter leg determines the output polarity.
- If $P = V_o i_o < 0$, the converter operates in regenerative mode and power flows back to the source.

Mistakes to be avoided

- Negative power indicates diode feedback conduction, not active switch power delivery.

Q4.13.2 MCQ 2013 1M Ans: D ● ○ ○

For IGBTs, switching losses are minimized when the device is turned ON under zero-current conditions.
In Zero Current Switching (ZCS), the current through the device is zero at the switching instant, thereby reducing the overlap of voltage and current during turn-ON and minimizing the switching loss.
Hence, the appropriate switching technique is ZCS during turn-ON.
Therefore, the correct answer is

$$\boxed{(D)\ \text{ZCS during turn-ON}}$$

Q4.12.1 MCQ 2012 2M Ans: B ● ○ ○

The RMS value of the load line voltage is

$$V_L = \left[\frac{1}{\pi}\int_0^{2\pi/3} V_s^2\, d(\omega t)\right]^{1/2} = \sqrt{\frac{2}{3}}V_s$$

The load phase voltage is

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{\sqrt{2}}{3}V_s$$

Substituting $V_s = 300\,\text{V}$,

$$V_p = \frac{300\sqrt{2}}{3} = 100\sqrt{2} = 141.4\,\text{V}$$

$$\boxed{(B)\ 141.4}$$

Q4.12.2 MCQ 2012 2M Ans: D ● ○ ○

The power consumed by each phase resistor is

$$P_{\text{phase}} = \frac{V_p^2}{R} = \frac{(141.4)^2}{20} = 1000\,\text{W}$$

Hence, total three-phase power is

$$P_{\text{total}} = 3P_{\text{phase}} = 3 \times 1000 = 3000\,\text{W} = 3\,\text{kW}$$

$$\boxed{(D)\ 3}$$

Q4.11.1 MCQ 2011 1M Ans: A ● ● ○

In a current source inverter (CSI), each switch must be capable of blocking reverse voltage.
Since a power MOSFET cannot block reverse voltage because of its intrinsic body diode, it cannot be used alone in a CSI.
Therefore, a diode is connected in series with the MOSFET to provide reverse voltage blocking capability.
Our requirement is blocking bidirectional voltage and carrying unidirectional current, which is satisfied by option A.
Hence, the proper switch configuration is

$$\boxed{(A)}$$

Q4.09.1 MCQ 2009 2M Ans: B ● ● ○

The theoretical maximum output frequency of a current source inverter (CSI) is

$$f_{\max} = \frac{1}{4RC}$$

Given,

$$R = 10\ \Omega, \qquad C = 0.1\ \mu\text{F} = 0.1 \times 10^{-6}\ \text{F}$$

Substituting,

$$f_{\max} = \frac{1}{4 \times 10 \times 0.1 \times 10^{-6}}$$
$$= 250 \times 10^{3}\ \text{Hz} = 250\ \text{kHz}$$

Hence, the correct answer is

$$\boxed{(B)\ 250\ \text{kHz}}$$

Q4.08.1 MCQ 2008 1M Ans: B ● ● ○

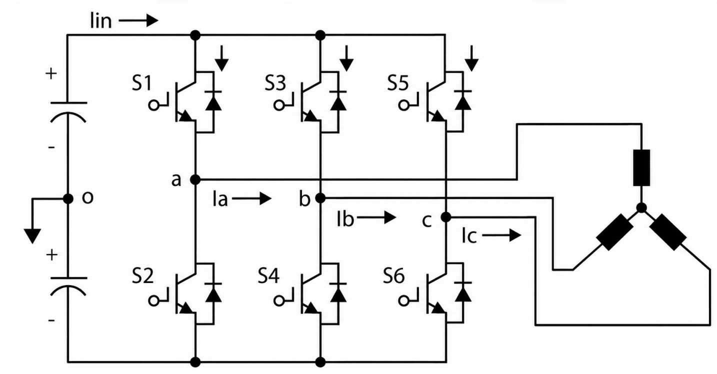

Fig. Solution Q4.08.1

The line voltage of a three-phase voltage source inverter is

$$V_{ab} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin\left(n\omega t + \frac{\pi}{6}\right)$$

For triplen harmonics, $n = 3, 9, 15, \ldots$

$$\because \cos\left(\frac{3\pi}{6}\right) = \cos\left(\frac{\pi}{2}\right) = 0,$$

all triplen harmonics are absent from the line voltage.

The pole voltage is given by;

$$V_{ao} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{2V_s}{n\pi} \sin(n\omega t),$$

which contains all odd harmonics, including the third harmonic.
Hence, the correct answer is

$\boxed{(B)\text{ Pole-voltage will have 3rd harmonic component but line-voltage will be free from 3rd harmonic}}$

Q4.08.2 MCQ 2008 2M Ans: B ● ● ○

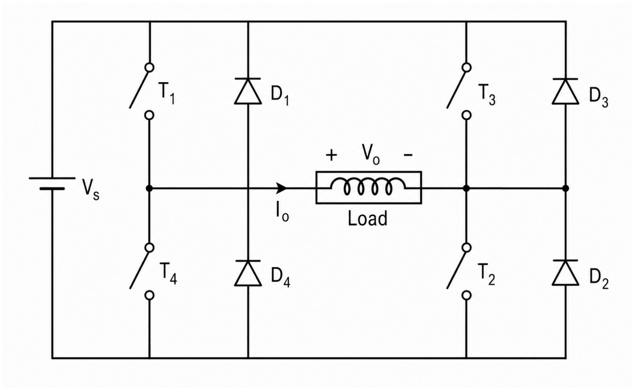

Fig. Solution Q4.08.2

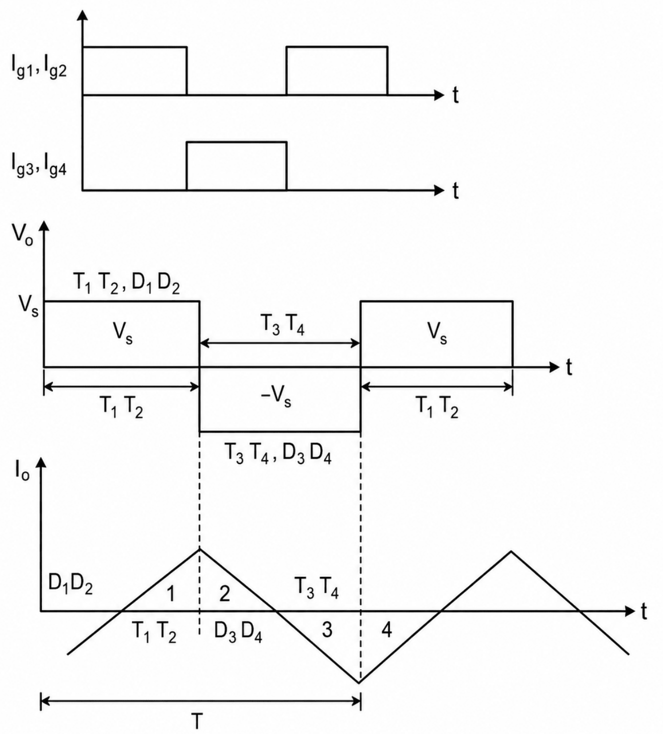

Fig. Solution Q4.08.2

For a purely inductive load,

$$v_L = L\frac{di_o}{dt}$$

The inverter operates in 180° square-wave mode at

$$f = 50 \text{ Hz}$$

Hence, the time period is

$$T = \frac{1}{f} = \frac{1}{50} = 20 \text{ ms}$$

Since the load current has no dc component, the current waveform is symmetrical about zero.
During one-quarter of the cycle, the inductor current rises from 0 to its peak value $I_{\max}$.
Therefore,

$$V_{dc} = L\frac{I_{\max}}{T/4}$$

or,

$$I_{\max} = \frac{V_{dc}T}{4L} = \frac{V_{dc}}{4fL}$$

Given,

$$V_{dc} = 200 \text{ V}, \qquad f = 50 \text{ Hz}, \qquad L = 0.1 \text{ H}$$

Substituting,

$$I_{\max} = \frac{200}{4 \times 50 \times 0.1} = 10 \text{ A}$$

Hence, the correct answer is

$$\boxed{(B)\ 10 \text{ A}}$$

Q4.07.1 MCQ | 2007 | 1M | Ans: B | ● ● ○

A Voltage Source Inverter (VSI) requires each switch to conduct current in both directions while blocking reverse voltage.
A power MOSFET has an inherent anti-parallel body diode, which provides the necessary reverse current path required in a VSI.
However, because of this body diode, a MOSFET cannot block reverse voltage and hence cannot be directly used in a Current Source Inverter (CSI).
Therefore, six MOSFETs connected in a bridge configuration (without any additional power devices) must be operated as a Voltage Source Inverter (VSI).
Hence, the correct answer is

$$\boxed{(B)\ \text{True, because MOSFETs have inherently anti-parallel diodes}}$$

Q4.07.2 MCQ | 2007 | 2M | Ans: C | ● ● ○

For single pulse-width modulation,

$$2d = 150°$$

Hence,

$$d = 75°$$

The RMS value of the output voltage is

$$V_{\text{rms}} = V_s\sqrt{\frac{2d}{\pi}} = V_s\sqrt{\frac{5}{6}} = 0.913V_s$$

The Fourier series of the output voltage is

$$v_o = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi}\sin\left(\frac{n\pi}{2}\right)\sin(nd)\sin(n\omega t)$$

The RMS value of the fundamental component is

$$V_1 = \frac{1}{\sqrt{2}}\left(\frac{4V_s}{\pi}\sin 75^\circ\right) = 0.87V_s$$

The total harmonic distortion is

$$\text{THD} = \frac{\sqrt{V_{\text{rms}}^2 - V_1^2}}{V_1} \times 100$$

Substituting,

$$\text{THD} = \frac{\sqrt{(0.913V_s)^2 - (0.87V_s)^2}}{0.87V_s} \times 100$$

$$= 31.83\%$$

Hence, the correct answer is

$$\boxed{(C)\ 31.83\%}$$

Q4.06.1 MCQ | 2006 | 2M | Ans: B

The Fourier series of the output voltage for single-pulse modulation is

$$v_o = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi}\sin(nd)\sin\left(\frac{n\pi}{2}\right)\sin(n\omega t)$$

The third harmonic component is

$$V_{o3} = \frac{4V_s}{3\pi}\sin 3d \sin\frac{3\pi}{2}$$

The maximum value of the fundamental component is

$$(V_{o1})_{\max} = \frac{4V_s}{\pi}\sin d \sin\frac{\pi}{2}$$

Given,

$$2d = 144^\circ$$

Hence,

$$d = 72^\circ$$

Therefore,

$$\frac{V_{o3}}{(V_{o1})_{\max}} = \frac{\frac{4V_s}{3\pi}\sin 3d \sin\frac{3\pi}{2}}{\frac{4V_s}{\pi}\sin d \sin\frac{\pi}{2}}$$

$$= \frac{\sin 216^\circ \times \sin 270^\circ}{3\sin 72^\circ \times \sin 90^\circ} = 0.206$$

Hence,

$$\frac{V_{o3}}{(V_{o1})_{\max}} = 20.6\%$$

Therefore, the correct answer is

$$\boxed{(B)\ 20.6\%}$$

Q4.05.1 MCQ | 2005 | 1M | Ans: C | ● ● ○

The Fourier series of the line voltage of a three-phase square-wave inverter operating in 180° conduction mode is

$$V_{ab} = \sum_{n=1,3,5,\dots}^{\infty} \frac{4V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin\left(n\omega t + \frac{\pi}{6}\right)$$

For triplen harmonics,

$$n = 3, 9, 15, \dots$$

Since,

$$\cos\left(\frac{3\pi}{6}\right) = \cos\left(\frac{\pi}{2}\right) = 0,$$

all triplen harmonics are absent from the line voltage.
The line-to-neutral (phase) voltage in 180° conduction mode is

$$V_{ao} = \sum_{n=6k\pm1}^{\infty} \frac{2V_s}{n\pi} \sin(n\omega t),$$

where

$$k = 0, 1, 2, \dots$$

Thus, the phase voltage contains only harmonics of the order

$$n = 6k \pm 1,$$

which are all odd harmonics.
Similarly, for 120° conduction mode,

$$V_{ao} = \sum_{n=1,3,5,\dots}^{\infty} \frac{2V_s}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin\left(n\omega t + \frac{\pi}{6}\right)$$

and the line voltage is

$$V_{ab} = \sum_{n=6k\pm1}^{\infty} \frac{3V_s}{n\pi} \sin\left(n\omega t + \frac{\pi}{3}\right),$$

where

$$k = 0, 1, 2, \dots$$

Hence, in both 180° and 120° conduction modes, the output voltage waveform contains only odd harmonics.
Therefore, the correct answer is

$$\boxed{(C)\ \text{Only odd harmonics}}$$

Q4.03.1 MCQ | 2003 | 2M | Ans: A | ● ● ○

For a single-pulse modulation inverter, the Fourier series of the output voltage is

$$v_o = \sum_{n=1,3,5,\dots}^{\infty} \frac{4V_s}{n\pi} \sin(nd) \sin\left(\frac{n\pi}{2}\right) \sin(n\omega t)$$

The conduction angle is given as

$$\alpha = 120^\circ$$

Since the pulse width is

$$2d = \alpha,$$

we have

$$d = \frac{\alpha}{2} = \frac{120^\circ}{2} = 60^\circ.$$

The RMS value of the fundamental component is

$$V_{1,\text{RMS}} = \frac{1}{\sqrt{2}}\left(\frac{4V_s}{\pi}\sin d\right)$$

Substituting $d = 60°$,

$$V_{1,\text{RMS}} = \frac{4V_s}{\sqrt{2}\pi}\sin 60°$$

$$= \frac{4V_s}{\sqrt{2}\pi}\left(\frac{\sqrt{3}}{2}\right) = 0.78V_s$$

Since the waveform amplitude is

$$V_s = 1\text{ V},$$

$$V_{1,\text{RMS}} = 0.78\text{ V}$$

Hence, the correct answer is

$$\boxed{(A)\ 0.78\text{ V}}$$

Q4.03.2 MCQ | 2003 | 2M | Ans: A | ● ● ○

For a single-pulse modulation inverter, the Fourier series of the output voltage is

$$v_o = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi}\sin(nd)\sin\left(\frac{n\pi}{2}\right)\sin(n\omega t)$$

The amplitude of the n^{th} harmonic is proportional to

$$\sin(nd)$$

For elimination of the fifth harmonic,

$$\sin(5d) = 0$$

Hence,

$$5d = 0, \pi, 2\pi, \ldots$$

$$\Rightarrow d = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \ldots$$

Since the pulse width is

$$\alpha = 2d,$$

the possible pulse widths are

$$\alpha = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \ldots$$

or,

$$\alpha = 0°, 72°, 144°, \ldots$$

Among the given options, only

$$\alpha = 72°$$

satisfies the condition.
Hence, the correct answer is

$$\boxed{(A)\ \alpha = 72°}$$

Q4.02.1 | MCQ | 2002 | 2M | Ans: B | ● ● ○

The pole-to-pole voltage is obtained by subtracting the two pole voltages.

$$V_{12} = V_{10} - V_{20}$$

The corresponding pole-to-pole voltage waveform is shown below.

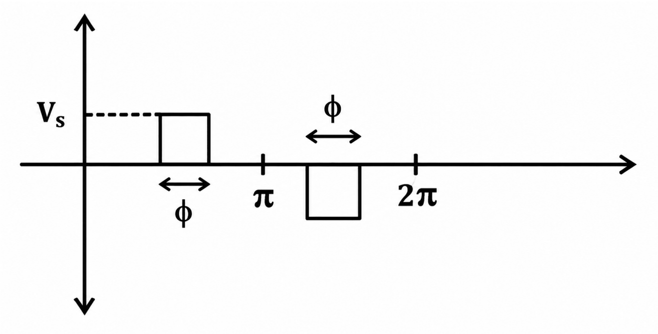

Fig. Solution Q4.02.1

The waveform has an amplitude of V_s and exists for a total duration of

$$2\phi$$

in one cycle of duration

$$2\pi.$$

Hence, the RMS value of the pole-to-pole voltage is

$$(V_{12})_{\text{RMS}} = V_s\sqrt{\frac{2\phi}{2\pi}}$$

$$= V_s\sqrt{\frac{\phi}{\pi}}$$

Therefore, the correct answer is

$$\boxed{(B)\ V_s\sqrt{\frac{\phi}{\pi}}}$$

Q4.01.1 | MCQ | 2001 | 2M | Ans: A | ● ● ○

The inverter operates in square-wave mode at

$$f = 50 \text{ Hz}$$

Hence, the time period is

$$T = \frac{1}{f} = \frac{1}{50} = 20 \text{ ms}$$

Since the load is purely inductive,

$$v_L = L\frac{di}{dt}$$

The output voltage is a square wave of amplitude $\pm V_s$. Therefore, the load current is a symmetrical triangular waveform. As the average load current is zero, the current changes from negative peak to positive peak during the positive half-cycle and vice versa during the negative half-cycle.
During the positive half-cycle,

- From 0 to $T/4$, the load current is negative while the output voltage is positive. Hence, energy flows from the load back to the source and the feedback diodes D_1 and D_2 conduct.
- From $T/4$ to $T/2$, the load current becomes positive and the transistors T_1 and T_2 conduct.

Similarly, during the negative half-cycle,

- From $T/2$ to $3T/4$, the feedback diodes D_3 and D_4 conduct.
- From $3T/4$ to T, the transistors T_3 and T_4 conduct.

Thus, each feedback diode conducts for

$$\frac{T}{4} = \frac{20}{4} = 5 \text{ ms}$$

Hence, the correct answer is

$$\boxed{(A)\ 5 \text{ ms}}$$

Q4.00.2 MCQ | 2000 | 2M | Ans: D | ● ● ○

Let the magnitude of the h^{th} harmonic component of the output voltage be

$$V_h = \alpha_h V_1,$$

where V_1 is the magnitude of the fundamental component and

$$\alpha_h < 1.$$

For a purely inductive load,

$$I_h = \frac{V_h}{h\omega L}$$

and the fundamental current is

$$I_1 = \frac{V_1}{\omega L}.$$

Therefore,

$$\frac{I_h}{I_1} = \frac{\frac{V_h}{h\omega L}}{\frac{V_1}{\omega L}} = \frac{V_h}{hV_1}$$

Since,

$$V_h = \alpha_h V_1,$$

we obtain

$$\frac{I_h}{I_1} = \frac{\alpha_h}{h}.$$

Hence, the h^{th} harmonic current has a magnitude equal to

$$\frac{\alpha_h}{h}$$

times the fundamental current component.
Therefore, the correct answer is

$$\boxed{(D)\ \frac{\alpha_h}{h} \text{ times the fundamental frequency component}}$$

Q4.00.2 MCQ | 2000 | 1M | Ans: D

Triangular PWM control shifts the dominant harmonics to higher frequencies.
Hence, it introduces very high-order harmonic voltages on the AC side of the inverter, making the lower-order harmonics negligible and easier to filter.
Therefore, the correct answer is

$$\boxed{(D)\ \text{Very high order harmonic voltages on the AC side}}$$

Q4.99.1 MCQ | 1999 | 1M | Ans: D

PWM switching in a three-phase inverter shifts the harmonic spectrum toward higher frequencies.
As a result, the lower-order harmonics are significantly reduced, while the higher-order harmonics become dominant and can be filtered more easily.
Hence, the correct answer is

$$\boxed{(D)\ \text{Reduce low order harmonics and increase high order harmonics}}$$

Q4.97.1 NAT | 1997 | 5M | Ans: *

The output voltage waveform of the single-phase bridge inverter is shown below.

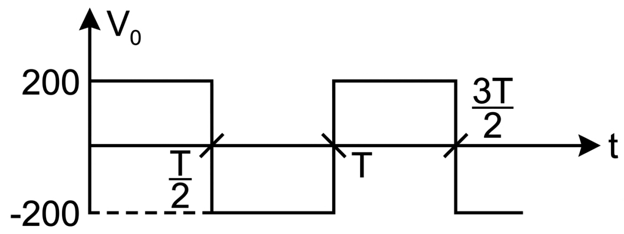

Fig. Solution Q4.97.1

During the positive half-cycle, the inductor stores energy, while during the negative half-cycle, it releases the stored energy.
Assume, $I_L = I_{\min}$, at the beginning of the positive half-cycle, and

$$I_L = I_{\max}$$

at the end of the positive half-cycle.
During the positive half-cycle,

$$i(\infty) = \frac{200}{20} = 10\text{ A}$$

The current response is

$$i(t) = 10 + (I_{\min} - 10)\, e^{-tR/L}$$

Since,

$$\frac{R}{L} = \frac{20}{0.2} = 100\text{ s}^{-1},$$

$$i(t) = 10 + (I_{\min} - 10)\, e^{-100t}$$

At

$$t = \frac{T}{2} = \frac{1}{2f} = \frac{1}{100}\text{ s},$$

the current reaches $I_{\max}$.

Hence,

$$I_{\max} = 10 + (I_{\min} - 10)\, e^{-1} \quad (1)$$

During the negative half-cycle,

$$i(\infty) = \frac{-200}{20} = -10 \text{ A}$$

The current response is

$$i(t) = -10 + (I_{\max} + 10)\, e^{-100t}$$

After another interval of

$$\Delta t = \frac{T}{2},$$

the current reaches $I_{\min}$.
Therefore,

$$I_{\min} = -10 + (I_{\max} + 10)\, e^{-1}$$

Substituting this expression for $I_{\min}$ into (1),

$$I_{\max} = 10 + \left[-10 + (I_{\max} + 10)\, e^{-1} - 10\right] e^{-1}$$

$$I_{\max} = 10 - 20e^{-1} + 10e^{-2} + I_{\max} e^{-2}$$

$$I_{\max} = \frac{10 - 20e^{-1} + 10e^{-2}}{1 - e^{-2}} = 4.621 \text{ A}$$

By symmetry,

$$I_{\min} = -4.621 \text{ A}$$

The steady-state load current waveform is shown below.

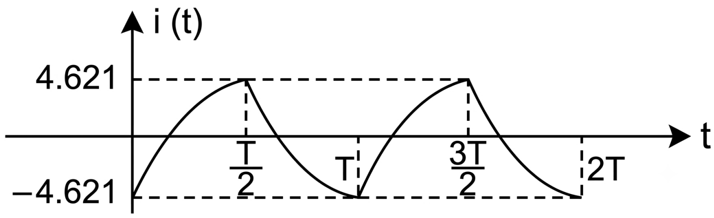

Fig. Solution Q4.97.1

Hence, the peak values of the load current are

$$\boxed{I_{\max} = 4.621 \text{ A}, \qquad I_{\min} = -4.621 \text{ A}}$$

Q4.95.1 NAT | 1995 | 1M | Ans: 20 | ● ○ ○

The output voltage of a single-phase square-wave inverter is

$$v_o = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi} \sin(n\omega t)$$

The RMS value of the fundamental component is

$$(V_1)_{\text{RMS}} = \frac{4V_s}{\pi\sqrt{2}}$$

The RMS value of the fifth harmonic is

$$(V_5)_{\text{RMS}} = \frac{4V_s}{5\pi\sqrt{2}}$$

Therefore,

$$\text{Percentage of fifth harmonic} = \frac{(V_5)_{\text{RMS}}}{(V_1)_{\text{RMS}}} \times 100\%$$

$$= \frac{\frac{4V_s}{5\pi\sqrt{2}}}{\frac{4V_s}{\pi\sqrt{2}}} \times 100\% = \frac{1}{5} \times 100\% = 20\%$$

Hence,

$$\boxed{20}$$

Q4.95.2 MCQ 1995 1M Ans: B ● ● ○

At starting, the slip is approximately unity. Since the stator resistance is negligible, the starting current is

$$I = \frac{V}{X}$$

As the reactance is proportional to frequency,

$$X \propto f$$

Hence,

$$I \propto \frac{V}{f}$$

To avoid excessive inrush current, the applied voltage should be reduced along with the frequency so that the V/f ratio remains constant.
Maintaining a constant V/f ratio also keeps the air-gap flux nearly constant, thereby preserving the starting torque while avoiding magnetic saturation.
Hence, the correct answer is

$$\boxed{(B)\ \text{Low voltage keeping the V/f ratio constant}}$$

DEBUG INFO

Chapter IV

DC to AC Converter - Inverter

Total Questions : 36

MCQ : 23

MSQ : 0

NAT : 13

SOLUTIONS

V

Miscellaneous Topics

Topics
1. AC Voltage Regulator

Q5.24.1 MCQ 2024 2M Ans: B ● ○ ○

The output voltage of an AC voltage controller feeding an RL load can be controlled only if the firing angle α is greater than the load impedance angle θ.

$$\alpha > \theta$$

Given,

$$f = 50\,\text{Hz}, \qquad R = 10\,\Omega, \qquad L = 18.37\,\text{mH}$$

$$\omega = 2\pi f = 2\pi \times 50 = 100\pi\ \text{rad/s}$$

The load impedance angle is

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\theta = \tan^{-1}\left(\frac{2\pi \times 50 \times 18.37 \times 10^{-3}}{10}\right)$$

$$\theta = \tan^{-1}(0.5771) \approx 30^\circ$$

Hence, the minimum triggering angle is

$$\boxed{\text{(B)}}$$

Key Points

- For an RL load, controllable output voltage is obtained only when $\alpha > \theta$.
- Load impedance angle:

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

- If $\alpha \leq \theta$, the load current is not properly controlled by the firing angle.

Q5.16.1 MCQ 2016 1M Ans: C ● ○ ○

For an induction motor, the slip at maximum torque is

$$s_{\max} = \frac{R_2}{X_2}$$

where R_2 is the rotor resistance and X_2 is the rotor reactance.
In a constant V/f drive,

$$\frac{V}{f} = \text{constant}$$

Since synchronous speed is proportional to frequency,

$$N_s \propto f$$

Also,

$$X_2 = 2\pi f L$$

Hence,

$$X_2 \propto f$$

Therefore,

$$s_{max} = \frac{R_2}{X_2} \propto \frac{1}{f}$$

Since

$$N_s \propto f$$

we get

$$s_{max} \propto \frac{1}{N_s}$$

Thus, the slip at maximum torque is inversely proportional to synchronous speed.

$$\boxed{\text{(C)}}$$

Key Points

- Slip at maximum torque:

$$s_{max} = \frac{R_2}{X_2}$$

- Rotor reactance varies with frequency:

$$X_2 \propto f$$

- Synchronous speed varies with frequency:

$$N_s \propto f$$

Q5.14.1 MCQ 2014 2M Ans: C ● ● ○

The maximum firing angle is approximately equal to the load impedance angle.

$$\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Substituting the given values,

$$\phi = \tan^{-1}\left(\frac{2\pi \times 50 \times 16 \times 10^{-3}}{5}\right)$$

$$\phi = 45.15^\circ \approx 45^\circ$$

The impedance is

$$Z = \sqrt{R^2 + (\omega L)^2}$$

$$Z = \sqrt{5^2 + \left(2\pi \times 50 \times 16 \times 10^{-3}\right)^2}$$

$$Z = 7.07\,\Omega$$

The rms current through SCR under this operating condition is

$$I_{T,\text{rms}} = \frac{V_m}{2Z}$$

where

$$V_m = 230\sqrt{2}\text{ V}$$

Therefore,

$$I_{T,\text{rms}} = \frac{230\sqrt{2}}{2 \times 7.07} = 22.98\text{ A} \approx 23\text{ A}$$

Hence, the correct option is

$$\boxed{\text{(C)}}$$

Key Points

- For an RL load, the critical firing angle is approximately the load angle:

$$\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

- Load impedance:

$$Z = \sqrt{R^2 + (\omega L)^2}$$

- Under the given condition:

$$I_{T,\text{rms}} = \frac{V_m}{2Z}$$

Mistakes to be avoided

- Do not use rms voltage directly in place of V_m.
- Do not omit the inductive reactance while calculating Z.

Q5.08.1 MCQ 2008 1M Ans: A ●●○

The load impedance is

$$Z = R + jX_L = 50 + j50\ \Omega$$

Hence, the load power factor angle is

$$\tan\phi = \frac{X_L}{R} = \frac{50}{50} = 1$$
$$\Rightarrow \phi = 45^\circ$$

For a single-phase AC voltage controller with an $R-L$ load, the output voltage is uncontrollable when the firing angle satisfies

$$0 < \alpha < \phi$$

Since,

$$\phi = 45^\circ,$$

the uncontrollable region is

$$0^\circ < \alpha < 45^\circ$$

Hence, the correct answer is

$$\boxed{(A)\ 0^\circ < \alpha < 45^\circ}$$

Q5.04.1 MCQ | 2004 | 2M | Ans: D

The instantaneous power dissipated in the resistive load is

$$p(t) = \frac{v_o^2(t)}{R}$$

Since the load is purely resistive, the instantaneous power is maximum when the output voltage reaches its peak value.
The supply voltage is

$$v_s(t) = 230\sqrt{2}\sin\omega t$$

Hence, the peak voltage is

$$V_m = 230\sqrt{2}\ \text{V}$$

The TRIAC is fired at

$$\alpha = \frac{\pi}{4} = 45^\circ.$$

Since the supply voltage reaches its maximum at

$$\omega t = \frac{\pi}{2} = 90^\circ,$$

which is after the firing instant, the TRIAC is conducting and the full peak supply voltage appears across the load.
Therefore,

$$P_{\text{peak}} = \frac{V_m^2}{R} = \frac{(230\sqrt{2})^2}{10}$$

$$= \frac{230^2 \times 2}{10} = 10580\ \text{W}$$

Hence, the correct answer is

$$\boxed{(D)\ 10580\text{W}}$$

DEBUG INFO

Chapter V

Miscellaneous Topics

Total Questions : 5

MCQ : 5

MSQ : 0

NAT : 0